

### Absolute Dependent Motion of Two Particles

The dependent motion of blocks that are suspended from pulleys and cables can be related by the geometry of the system. This is done by first establishing position coordinates, measured from a fixed origin to each block. Each coordinate must be directed along the line of motion of a block.

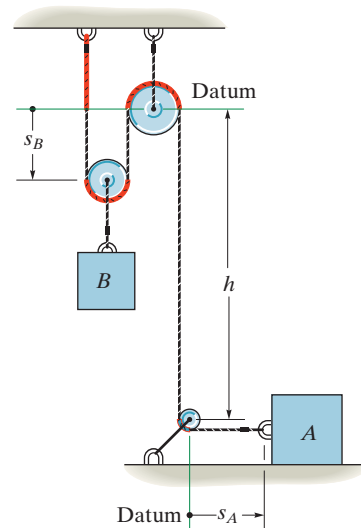
Using geometry and/or trigonometry, the coordinates are then related to the cable length in order to formulate a position coordinate equation.

The first time derivative of this equation gives a relationship between the velocities of the blocks, and a second time derivative gives the relation between their accelerations.

$$2s_B + h + s_A = l$$

$$2v_B = -v_A$$

$$2a_B = -a_A$$



### Relative-Motion Analysis Using Translating Axes

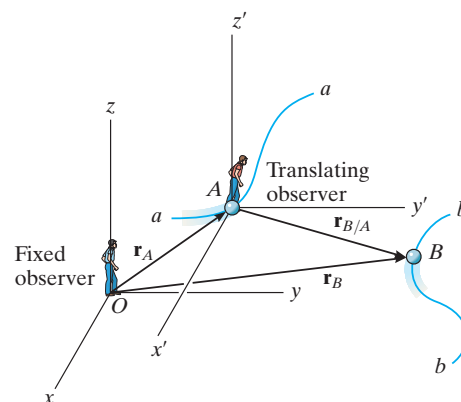
If two particles  $A$  and  $B$  undergo independent motions, then these motions can be related to their relative motion using a *translating set of axes* attached to one of the particles ( $A$ ).

For planar motion, each vector equation produces two scalar equations, one in the  $x$ , and the other in the  $y$  direction. For solution, the vectors can be expressed in Cartesian form, or the  $x$  and  $y$  scalar components can be written directly.

$$\mathbf{r}_B = \mathbf{r}_A + \mathbf{r}_{B/A}$$

$$\mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{B/A}$$

$$\mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{B/A}$$



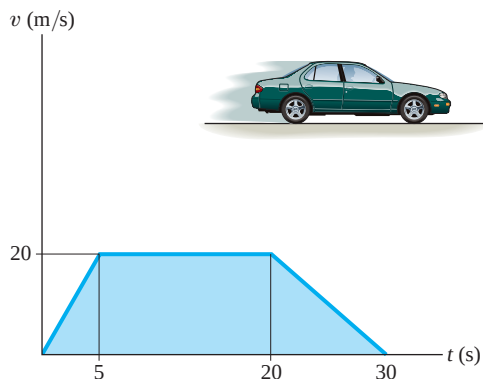
## REVIEW PROBLEMS

**R12-1.** The position of a particle along a straight line is given by  $s = (t^3 - 9t^2 + 15t)$  m, where  $t$  is in seconds. Determine its maximum acceleration and maximum velocity during the time interval  $0 \leq t \leq 10$  s.

**R12-2.** If a particle has an initial velocity  $v_0 = 12$  m/s to the right, and a constant acceleration of  $2$  m/s<sup>2</sup> to the left, determine the particle's displacement in 10 s. Originally  $s_0 = 0$ .

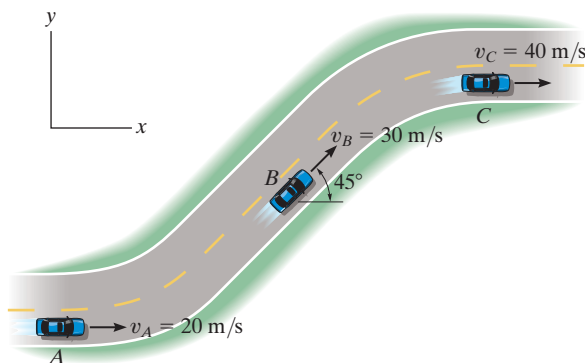
**R12-3.** A projectile, initially at the origin, moves along a straight-line path through a fluid medium such that its velocity is  $v = 1800(1 - e^{-0.3t})$  mm/s where  $t$  is in seconds. Determine the displacement of the projectile during the first 3 s.

**R12-4.** The  $v$ - $t$  graph of a car while traveling along a road is shown. Determine the acceleration when  $t = 2.5$  s, 10 s, and 25 s. Also if  $s = 0$  when  $t = 0$ , find the position when  $t = 5$  s, 20 s, and 30 s.



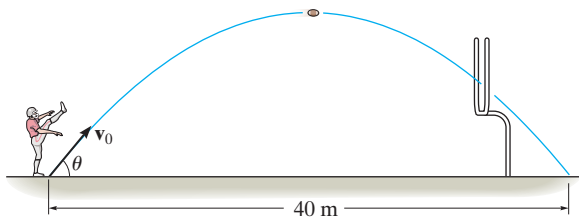
Prob. R12-4

**R12-5.** A car traveling along the straight portions of the road has the velocities indicated in the figure when it arrives at points  $A$ ,  $B$ , and  $C$ . If it takes 3 s to go from  $A$  to  $B$ , and then 5 s to go from  $B$  to  $C$ , determine the average acceleration between points  $A$  and  $B$  and between points  $A$  and  $C$ .



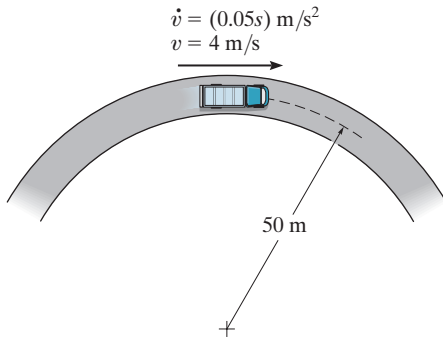
Prob. R12-5

**R12-6.** From a videotape, it was observed that a player kicked a football 40 m during a measured time of 3.6 seconds. Determine the initial speed of the ball and the angle  $\theta$  at which it was kicked.



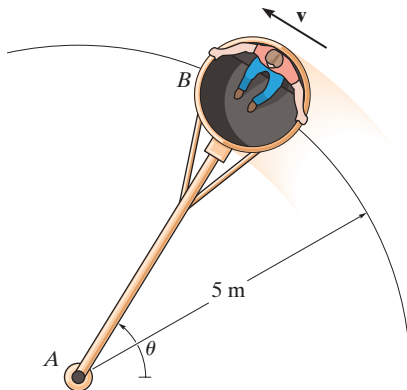
Prob. R12-6

**R12-7.** The truck travels in a circular path having a radius of 50 m at a speed of  $v = 4$  m/s. For a short distance from  $s = 0$ , its speed is increased by  $\dot{v} = (0.05s)$  m/s<sup>2</sup>, where  $s$  is in meters. Determine its speed and the magnitude of its acceleration when it has moved  $s = 10$  m.



**Prob. R12-7**

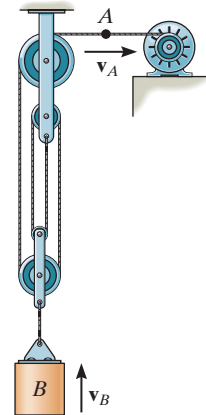
**R12-8.** Car  $B$  turns such that its speed is increased by  $(a_t)_B = (0.5e^t)$  m/s<sup>2</sup>, where  $t$  is in seconds. If the car starts from rest when  $\theta = 0^\circ$ , determine the magnitudes of its velocity and acceleration when  $t = 2$  s. Neglect the size of the car.



**Prob. R12-8**

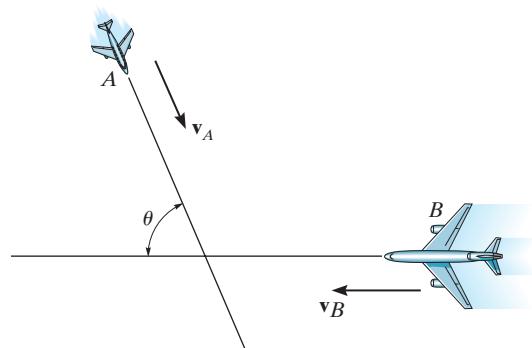
**R12-9.** A particle is moving along a circular path of 2-m radius such that its position as a function of time is given by  $\theta = (5t^2)$  rad, where  $t$  is in seconds. Determine the magnitude of the particle's acceleration when  $\theta = 30^\circ$ . The particle starts from rest when  $\theta = 0^\circ$ .

**R12-10.** Determine the time needed for the load at  $B$  to attain a speed of 8 m/s, starting from rest, if the cable is drawn into the motor with an acceleration of  $0.2$  m/s<sup>2</sup>.



**Prob. R12-10**

**R12-11.** Two planes,  $A$  and  $B$ , are flying at the same altitude. If their velocities are  $v_A = 600$  km/h and  $v_B = 500$  km/h such that the angle between their straight-line courses is  $\theta = 75^\circ$ , determine the velocity of plane  $B$  with respect to plane  $A$ .



**Prob. R12-11**

# Chapter 13



(© Migel/Shutterstock)

A car driving along this road will be subjected to forces that create both normal and tangential accelerations. In this chapter we will study how these forces are related to the accelerations they create.



# Kinetics of a Particle: Force and Acceleration

## CHAPTER OBJECTIVES

- To state Newton's Second Law of Motion and to define mass and weight.
- To analyze the accelerated motion of a particle using the equation of motion with different coordinate systems.
- To investigate central-force motion and apply it to problems in space mechanics.



Video Solutions are available for selected questions in this chapter.

## 13.1 Newton's Second Law of Motion

*Kinetics* is a branch of dynamics that deals with the relationship between the change in motion of a body and the forces that cause this change. The basis for kinetics is Newton's second law, which states that when an *unbalanced force* acts on a particle, the particle will *accelerate* in the direction of the force with a magnitude that is proportional to the force.

This law can be verified experimentally by applying a known unbalanced force  $\mathbf{F}$  to a particle, and then measuring the acceleration  $\mathbf{a}$ . Since the force and acceleration are directly proportional, the constant of proportionality,  $m$ , may be determined from the ratio  $m = F/a$ . This positive scalar  $m$  is called the *mass* of the particle. Being constant during any acceleration,  $m$  provides a quantitative measure of the resistance of the particle to a change in its velocity, that is its inertia.



The jeep leans backward due to its inertia, which resists its forward acceleration.

If the mass of the particle is  $m$ , Newton's second law of motion may be written in mathematical form as

$$\mathbf{F} = m\mathbf{a}$$

The above equation, which is referred to as the *equation of motion*, is one of the most important formulations in mechanics.\* As previously stated, its validity is based solely on *experimental evidence*. In 1905, however, Albert Einstein developed the theory of relativity and placed limitations on the use of Newton's second law for describing general particle motion. Through experiments it was proven that *time* is not an absolute quantity as assumed by Newton; and as a result, the equation of motion fails to predict the exact behavior of a particle, especially when the particle's speed approaches the speed of light (0.3 Gm/s). Developments of the theory of quantum mechanics by Erwin Schrödinger and others indicate further that conclusions drawn from using this equation are also invalid when particles are the size of an atom and move close to one another. For the most part, however, these requirements regarding particle speed and size are not encountered in engineering problems, so their effects will not be considered in this book.

**Newton's Law of Gravitational Attraction.** Shortly after formulating his three laws of motion, Newton postulated a law governing the mutual attraction between any two particles. In mathematical form this law can be expressed as

$$F = G \frac{m_1 m_2}{r^2} \quad (13-1)$$

where

$F$  = force of attraction between the two particles

$G$  = universal constant of gravitation; according to experimental evidence  $G = 66.73(10^{-12}) \text{ m}^3/(\text{kg} \cdot \text{s}^2)$

$m_1, m_2$  = mass of each of the two particles

$r$  = distance between the centers of the two particles

\*Since  $m$  is constant, we can also write  $\mathbf{F} = d(m\mathbf{v})/dt$ , where  $m\mathbf{v}$  is the particle's linear momentum. Here the unbalanced force acting on the particle is proportional to the time rate of change of the particle's linear momentum.

In the case of a particle located at or near the surface of the earth, the only gravitational force having any sizable magnitude is that between the earth and the particle. This force is termed the “weight” and, for our purpose, it will be the only gravitational force considered.

From Eq. 13–1, we can develop a general expression for finding the weight  $W$  of a particle having a mass  $m_1 = m$ . Let  $m_2 = M_e$  be the mass of the earth and  $r$  the distance between the earth's center and the particle. Then, if  $g = GM_e/r^2$ , we have

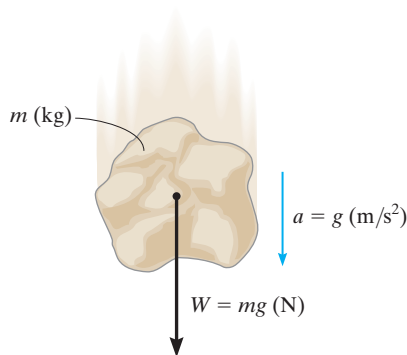
$$W = mg$$

By comparison with  $F = ma$ , we term  $g$  the acceleration due to gravity. For most engineering calculations  $g$  is measured at a point on the surface of the earth at sea level, and at a latitude of  $45^\circ$ , which is considered the “standard location.” Here the values  $g = 9.81 \text{ m/s}^2$  will be used for calculations.

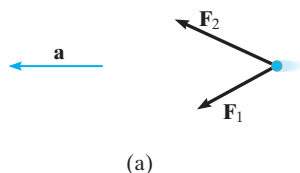
In the SI system the mass of the body is specified in kilograms, and the weight must be calculated using the above equation, Fig. 13–1. Thus,

$$W = mg \text{ (N)} \quad (g = 9.81 \text{ m/s}^2) \quad (13-2)$$

As a result, a body of mass 1 kg has a weight of 9.81 N; a 2-kg body weighs 19.62 N; and so on.



**Fig. 13–1**



## 13.2 The Equation of Motion

When more than one force acts on a particle, the resultant force is determined by a vector summation of all the forces; i.e.,  $\mathbf{F}_R = \Sigma \mathbf{F}$ . For this more general case, the equation of motion may be written as

$$\Sigma \mathbf{F} = m\mathbf{a} \quad (13-3)$$

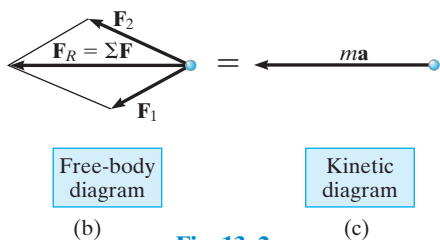


Fig. 13-2

To illustrate application of this equation, consider the particle shown in Fig. 13-2a, which has a mass  $m$  and is subjected to the action of two forces,  $\mathbf{F}_1$  and  $\mathbf{F}_2$ . We can graphically account for the magnitude and direction of each force acting on the particle by drawing the particle's *free-body diagram*, Fig. 13-2b. Since the *resultant* of these forces *produces* the vector  $m\mathbf{a}$ , its magnitude and direction can be represented graphically on the *kinetic diagram*, shown in Fig. 13-2c.\* The equal sign written between the diagrams symbolizes the *graphical* equivalency between the free-body diagram and the kinetic diagram; i.e.,  $\Sigma \mathbf{F} = m\mathbf{a}$ .† In particular, note that if  $\mathbf{F}_R = \Sigma \mathbf{F} = \mathbf{0}$ , then the acceleration is also zero, so that the particle will either remain at *rest* or move along a straight-line path with *constant velocity*. Such are the conditions of *static equilibrium*, Newton's first law of motion.

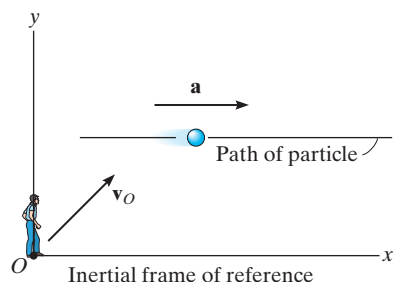


Fig. 13-3

**Inertial Reference Frame.** When applying the equation of motion, it is important that the acceleration of the particle be measured with respect to a reference frame that is *either fixed or translates with a constant velocity*. In this way, the observer will not accelerate and measurements of the particle's acceleration will be the *same* from *any* reference of this type. Such a frame of reference is commonly known as a *Newtonian* or *inertial reference frame*, Fig. 13-3.

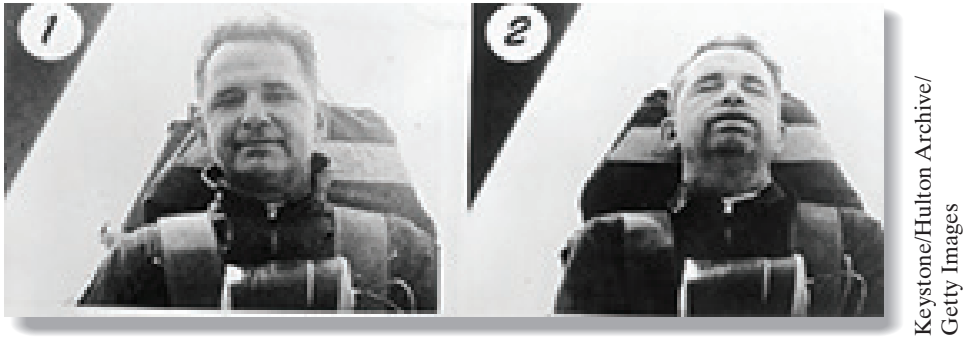
When studying the motions of rockets and satellites, it is justifiable to consider the inertial reference frame as fixed to the stars, whereas dynamics problems concerned with motions on or near the surface of the earth may be solved by using an inertial frame which is assumed fixed to the earth. Even though the earth both rotates about its own axis and revolves about the sun, the accelerations created by these rotations are relatively small and so they can be neglected for most applications.

\*Recall the free-body diagram considers the particle to be free of its surrounding supports and shows all the forces acting on the particle. The kinetic diagram pertains to the particle's motion as caused by the forces.

†The equation of motion can also be rewritten in the form  $\Sigma \mathbf{F} - m\mathbf{a} = \mathbf{0}$ . The vector  $-m\mathbf{a}$  is referred to as the *inertia force vector*. If it is treated in the same way as a "force vector," then the state of "equilibrium" created is referred to as *dynamic equilibrium*. This method of application, which will not be used in this text, is often referred to as the *D'Alembert principle*, named after the French mathematician Jean le Rond d'Alembert.

We are all familiar with the sensation one feels when sitting in a car that is subjected to a forward acceleration. Often people think this is caused by a “force” which acts on them and tends to push them back in their seats; however, this is not the case. Instead, this sensation occurs due to their inertia or the resistance of their mass to a change in velocity.

Consider the passenger who is strapped to the seat of a rocket sled. Provided the sled is at rest or is moving with constant velocity, then no force is exerted on his back as shown on his free-body diagram.



When the thrust of the rocket engine causes the sled to accelerate, then the seat upon which he is sitting exerts a force  $\mathbf{F}$  on him which pushes him forward with the sled. In the photo, notice that the inertia of his head resists this change in motion (acceleration), and so his head moves back against the seat and his face, which is nonrigid, tends to distort backward.



Upon deceleration the force of the seatbelt  $\mathbf{F}'$  tends to pull his body to a stop, but his head leaves contact with the back of the seat and his face distorts forward, again due to his inertia or tendency to continue to move forward. No force is pulling him forward, although this is the sensation he receives.





### 13.3 Equation of Motion for a System of Particles

The equation of motion will now be extended to include a system of particles isolated within an enclosed region in space, as shown in Fig. 13–4a. In particular, there is no restriction in the way the particles are connected, so the following analysis applies equally well to the motion of a solid, liquid, or gas system.

At the instant considered, the arbitrary  $i$ -th particle, having a mass  $m_i$ , is subjected to a system of internal forces and a resultant external force. The *internal force*, represented symbolically as  $\mathbf{f}_i$ , is the resultant of all the forces the other particles exert on the  $i$ th particle. The *resultant external force*  $\mathbf{F}_i$  represents, for example, the effect of gravitational, electrical, magnetic, or contact forces between the  $i$ th particle and adjacent bodies or particles *not* included within the system.

The free-body and kinetic diagrams for the  $i$ th particle are shown in Fig. 13–4b. Applying the equation of motion,

$$\Sigma \mathbf{F} = m\mathbf{a}; \quad \mathbf{F}_i + \mathbf{f}_i = m_i \mathbf{a}_i$$

When the equation of motion is applied to each of the other particles of the system, similar equations will result. And, if all these equations are added together *vectorially*, we obtain

$$\Sigma \mathbf{F}_i + \Sigma \mathbf{f}_i = \Sigma m_i \mathbf{a}_i$$

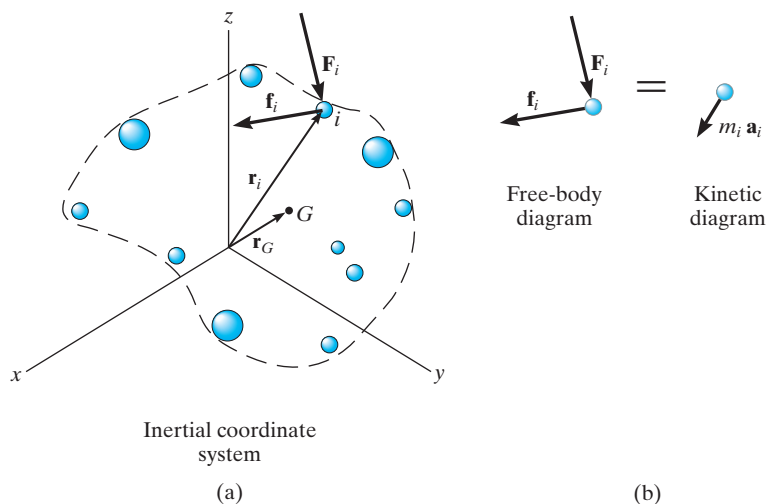


Fig. 13–4

The summation of the internal forces, if carried out, will equal zero, since internal forces between any two particles occur in equal but opposite collinear pairs. Consequently, only the sum of the external forces will remain, and therefore the equation of motion, written for the system of particles, becomes

$$\Sigma \mathbf{F}_i = \Sigma m_i \mathbf{a}_i \quad (13-4)$$

If  $\mathbf{r}_G$  is a position vector which locates the *center of mass*  $G$  of the particles, Fig. 13-4a, then by definition of the center of mass,  $m\mathbf{r}_G = \Sigma m_i \mathbf{r}_i$ , where  $m = \Sigma m_i$  is the total mass of all the particles. Differentiating this equation twice with respect to time, assuming that no mass is entering or leaving the system, yields

$$m\mathbf{a}_G = \Sigma m_i \mathbf{a}_i$$

Substituting this result into Eq. 13-4, we obtain

$$\Sigma \mathbf{F} = m\mathbf{a}_G \quad (13-5)$$

Hence, the sum of the external forces acting on the system of particles is equal to the total mass of the particles times the acceleration of its center of mass  $G$ . Since in reality all particles must have a finite size to possess mass, Eq. 13-5 justifies application of the equation of motion to a *body* that is represented as a single particle.

### Important Points

- The equation of motion is based on experimental evidence and is valid only when applied within an inertial frame of reference.
- The equation of motion states that the *unbalanced force* on a particle causes it to *accelerate*.
- An inertial frame of reference does not rotate, rather its axes either translate with constant velocity or are at rest.
- Mass is a property of matter that provides a quantitative measure of its resistance to a change in velocity. It is an absolute quantity and so it does not change from one location to another.
- Weight is a force that is caused by the earth's gravitation. It is not absolute; rather it depends on the altitude of the mass from the earth's surface.

## 13.4 Equations of Motion: Rectangular Coordinates

When a particle moves relative to an inertial  $x, y, z$  frame of reference, the forces acting on the particle, as well as its acceleration, can be expressed in terms of their  $\mathbf{i}, \mathbf{j}, \mathbf{k}$  components, Fig. 13–5. Applying the equation of motion, we have

$$\Sigma \mathbf{F} = m\mathbf{a}; \quad \Sigma F_x \mathbf{i} + \Sigma F_y \mathbf{j} + \Sigma F_z \mathbf{k} = m(a_x \mathbf{i} + a_y \mathbf{j} + a_z \mathbf{k})$$

For this equation to be satisfied, the respective  $\mathbf{i}, \mathbf{j}, \mathbf{k}$  components on the left side must equal the corresponding components on the right side. Consequently, we may write the following three scalar equations:

$$\begin{aligned} \Sigma F_x &= ma_x \\ \Sigma F_y &= ma_y \\ \Sigma F_z &= ma_z \end{aligned} \quad (13-6)$$

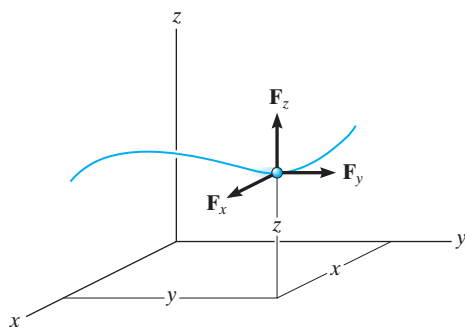


Fig. 13–5

In particular, if the particle is constrained to move only in the  $x$ – $y$  plane, then the first two of these equations are used to specify the motion.

### Procedure for Analysis

The equations of motion are used to solve problems which require a relationship between the forces acting on a particle and the accelerated motion they cause.

#### Free-Body Diagram.

- Select the inertial coordinate system. Most often, rectangular or  $x, y, z$  coordinates are chosen to analyze problems for which the particle has *rectilinear motion*.
- Once the coordinates are established, draw the particle's free-body diagram. Drawing this diagram is *very important* since it provides a graphical representation that accounts for *all the forces* ( $\Sigma \mathbf{F}$ ) which act on the particle, and thereby makes it possible to resolve these forces into their  $x, y, z$  components.
- The direction and sense of the particle's acceleration  $\mathbf{a}$  should also be established. If the sense is unknown, for mathematical convenience assume that the sense of each acceleration component acts in the *same direction* as its *positive* inertial coordinate axis.
- The acceleration may be represented as the  $m\mathbf{a}$  vector on the kinetic diagram.\*
- Identify the unknowns in the problem.

\*It is a convention in this text always to use the kinetic diagram as a graphical aid when developing the proofs and theory. The particle's acceleration or its components will be shown as blue colored vectors near the free-body diagram in the examples.

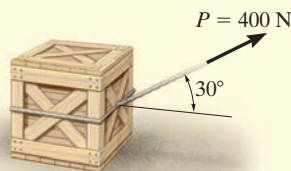
### Equations of Motion.

- If the forces can be resolved directly from the free-body diagram, apply the equations of motion in their scalar component form.
- If the geometry of the problem appears complicated, which often occurs in three dimensions, Cartesian vector analysis can be used for the solution.
- *Friction.* If a moving particle contacts a rough surface, it may be necessary to use the *frictional equation*, which relates the frictional and normal forces  $\mathbf{F}_f$  and  $\mathbf{N}$  acting at the surface of contact by using the coefficient of kinetic friction, i.e.,  $F_f = \mu_k N$ . Remember that  $\mathbf{F}_f$  always acts on the free-body diagram such that it opposes the motion of the particle relative to the surface it contacts. If the particle is *on the verge* of relative motion, then the coefficient of static friction should be used.
- *Spring.* If the particle is connected to an *elastic spring* having negligible mass, the spring force  $F_s$  can be related to the deformation of the spring by the equation  $F_s = ks$ . Here  $k$  is the spring's stiffness measured as a force per unit length, and  $s$  is the stretch or compression defined as the difference between the deformed length  $l$  and the undeformed length  $l_0$ , i.e.,  $s = l - l_0$ .

### Kinematics.

- If the velocity or position of the particle is to be found, it will be necessary to apply the necessary kinematic equations once the particle's acceleration is determined from  $\Sigma \mathbf{F} = m\mathbf{a}$ .
- If *acceleration* is a function of time, use  $a = dv/dt$  and  $v = ds/dt$  which, when integrated, yield the particle's velocity and position, respectively.
- If *acceleration* is a function of displacement, integrate  $a ds = v dv$  to obtain the velocity as a function of position.
- If *acceleration is constant*, use  $v = v_0 + a_c t$ ,  $s = s_0 + v_0 t + \frac{1}{2} a_c t^2$ ,  $v^2 = v_0^2 + 2a_c(s - s_0)$  to determine the velocity or position of the particle.
- If the problem involves the dependent motion of several particles, use the method outlined in Sec. 12.9 to relate their accelerations. In all cases, make sure the positive inertial coordinate directions used for writing the kinematic equations are the same as those used for writing the equations of motion; otherwise, simultaneous solution of the equations will result in errors.
- If the solution for an unknown vector component yields a negative scalar, it indicates that the component acts in the direction opposite to that which was assumed.

## EXAMPLE 13.1



(a)

The 50-kg crate shown in Fig. 13–6a rests on a horizontal surface for which the coefficient of kinetic friction is  $\mu_k = 0.3$ . If the crate is subjected to a 400-N towing force as shown, determine the velocity of the crate in 3 s starting from rest.

## SOLUTION

Using the equations of motion, we can relate the crate's acceleration to the force causing the motion. The crate's velocity can then be determined using kinematics.

**Free-Body Diagram.** The weight of the crate is  $W = mg = 50 \text{ kg} (9.81 \text{ m/s}^2) = 490.5 \text{ N}$ . As shown in Fig. 13–6b, the frictional force has a magnitude  $F = \mu_k N_C$  and acts to the left, since it opposes the motion of the crate. The acceleration  $\mathbf{a}$  is assumed to act horizontally, in the positive  $x$  direction. There are two unknowns, namely  $N_C$  and  $a$ .

**Equations of Motion.** Using the data shown on the free-body diagram, we have

$$\pm \Sigma F_x = ma_x; \quad 400 \cos 30^\circ - 0.3N_C = 50a \quad (1)$$

$$+ \uparrow \Sigma F_y = ma_y; \quad N_C - 490.5 + 400 \sin 30^\circ = 0 \quad (2)$$

Solving Eq. 2 for  $N_C$ , substituting the result into Eq. 1, and solving for  $a$  yields

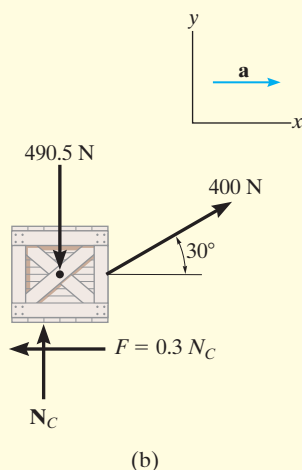
$$N_C = 290.5 \text{ N}$$

$$a = 5.185 \text{ m/s}^2$$

**Kinematics.** Notice that the acceleration is *constant*, since the applied force  $\mathbf{P}$  is constant. Since the initial velocity is zero, the velocity of the crate in 3 s is

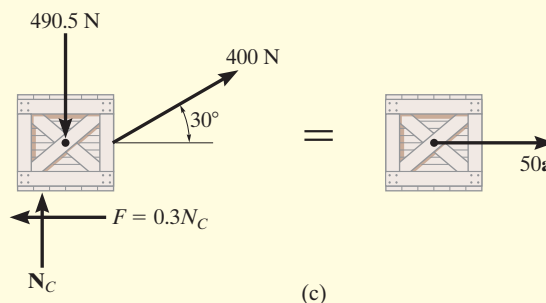
$$\begin{aligned} (\pm) \quad v &= v_0 + a_c t = 0 + 5.185(3) \\ &= 15.6 \text{ m/s} \rightarrow \end{aligned}$$

*Ans.*



(b)

Fig. 13–6



(c)

**NOTE:** We can also use the alternative procedure of drawing the crate's free-body *and* kinetic diagrams, Fig. 13–6c, prior to applying the equations of motion.

**EXAMPLE 13.2**

A 10-kg projectile is fired vertically upward from the ground, with an initial velocity of 50 m/s, Fig. 13-7a. Determine the maximum height to which it will travel if (a) atmospheric resistance is neglected; and (b) atmospheric resistance is measured as  $F_D = (0.01v^2)$  N, where  $v$  is the speed of the projectile at any instant, measured in m/s.

**SOLUTION**

In both cases the known force on the projectile can be related to its acceleration using the equation of motion. Kinematics can then be used to relate the projectile's acceleration to its position.

**Part (a) Free-Body Diagram.** As shown in Fig. 13-7b, the projectile's weight is  $W = mg = 10(9.81) = 98.1$  N. We will assume the unknown acceleration  $\mathbf{a}$  acts upward in the *positive*  $z$  direction.

**Equation of Motion.**

$$+\uparrow \Sigma F_z = ma_z; \quad -98.1 = 10a, \quad a = -9.81 \text{ m/s}^2$$

The result indicates that the projectile, like every object having free-flight motion near the earth's surface, is subjected to a *constant* downward acceleration of  $9.81 \text{ m/s}^2$ .

**Kinematics.** Initially,  $z_0 = 0$  and  $v_0 = 50 \text{ m/s}$ , and at the maximum height  $z = h$ ,  $v = 0$ . Since the acceleration is *constant*, then

$$\begin{aligned} (+\uparrow) \quad v^2 &= v_0^2 + 2a_c(z - z_0) \\ 0 &= (50)^2 + 2(-9.81)(h - 0) \\ h &= 127 \text{ m} \end{aligned}$$

*Ans.*

**Part (b) Free-Body Diagram.** Since the force  $F_D = (0.01v^2)$  N tends to retard the upward motion of the projectile, it acts downward as shown on the free-body diagram, Fig. 13-7c.

**Equation of Motion.**

$$+\uparrow \Sigma F_z = ma_z; \quad -0.01v^2 - 98.1 = 10a, \quad a = -(0.001v^2 + 9.81)$$

**Kinematics.** Here the acceleration is *not constant* since  $F_D$  depends on the velocity. Since  $a = f(v)$ , we can relate  $a$  to position using

$$(+\uparrow) a \, dz = v \, dv; \quad -(0.001v^2 + 9.81) \, dz = v \, dv$$

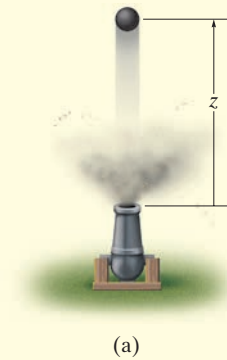
Separating the variables and integrating, realizing that initially  $z_0 = 0$ ,  $v_0 = 50 \text{ m/s}$  (positive upward), and at  $z = h$ ,  $v = 0$ , we have

$$\int_0^h dz = -\int_{50 \text{ m/s}}^0 \frac{v \, dv}{0.001v^2 + 9.81} = -500 \ln(v^2 + 9810) \Big|_{50 \text{ m/s}}^0$$

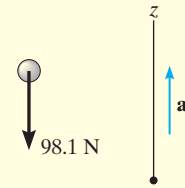
$$h = 114 \text{ m}$$

*Ans.*

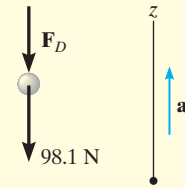
**NOTE:** The answer indicates a lower elevation than that obtained in part (a) due to atmospheric resistance or drag.



(a)



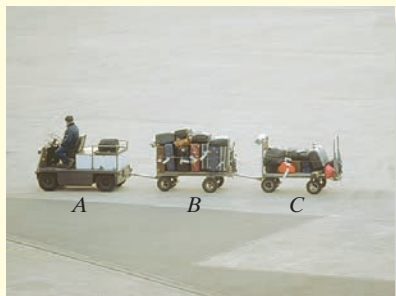
(b)



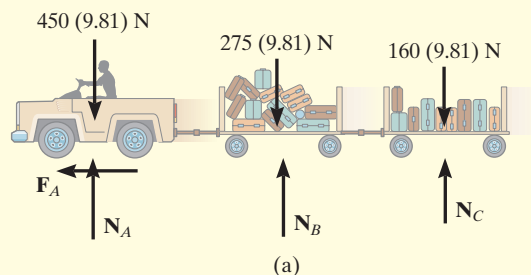
(c)

**Fig. 13-7**

## EXAMPLE 13.3



The baggage truck *A* shown in the photo has a mass of 450 kg and tows a 275-kg cart *B* and a 160-kg cart *C*. For a short time the driving frictional force developed at the wheels of the truck is  $F_A = (200t)$  N, where  $t$  is in seconds. If the truck starts from rest, determine its speed in 2 seconds. Also, what is the horizontal force acting on the coupling between the truck and cart *B* at this instant? Neglect the size of the truck and carts.



## SOLUTION

**Free-Body Diagram.** As shown in Fig. 13–8*a*, it is the frictional driving force that gives both the truck and carts an acceleration. Here we have considered all three vehicles as a single system.

**Equation of Motion.** Only motion in the horizontal direction has to be considered.

$$\begin{aligned} \pm \Sigma F_x &= ma_x; & 200t &= (450 + 275 + 160)a \\ & & a &= 0.2260t \end{aligned}$$

**Kinematics.** Since the acceleration is a function of time, the velocity of the truck is obtained using  $a = dv/dt$  with the initial condition that  $v_0 = 0$  at  $t = 0$ . We have

$$\int_0^v dv = \int_0^{2s} 0.2260t \, dt; \quad v = 0.1130t^2 \Big|_0^{2s} = 0.452 \, \text{m/s} \quad \text{Ans.}$$

**Free-Body Diagram.** In order to determine the force between the truck and cart *B*, we will consider a free-body diagram of the truck so that we can “expose” the coupling force **T** as external to the free-body diagram, Fig. 13–8*b*.

**Equation of Motion.** When  $t = 2$  s, then

$$\begin{aligned} \pm \Sigma F_x &= ma_x; & 200(2) - T &= (450)[0.2260(2)] \\ & & T &= 197 \, \text{N} \end{aligned} \quad \text{Ans.}$$

**NOTE:** Try and obtain this same result by considering a free-body diagram of carts *B* and *C* as a single system.

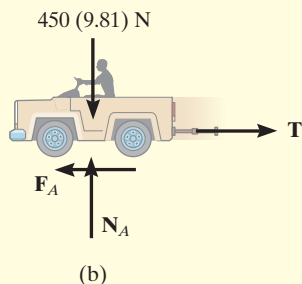


Fig. 13–8



**EXAMPLE 13.4**

A smooth 2-kg collar, shown in Fig. 13–9a, is attached to a spring having a stiffness  $k = 3 \text{ N/m}$  and an unstretched length of  $0.75 \text{ m}$ . If the collar is released from rest at  $A$ , determine its acceleration and the normal force of the rod on the collar at the instant  $y = 1 \text{ m}$ .

**SOLUTION**

**Free-Body Diagram.** The free-body diagram of the collar when it is located at the arbitrary position  $y$  is shown in Fig. 13–9b. Furthermore, the collar is *assumed* to be accelerating so that “ $a$ ” acts downward in the *positive*  $y$  direction. There are four unknowns, namely,  $N_C$ ,  $F_s$ ,  $a$ , and  $\theta$ .

**Equations of Motion.**

$$\rightarrow \Sigma F_x = ma_x; \quad -N_C + F_s \cos \theta = 0 \quad (1)$$

$$+\downarrow \Sigma F_y = ma_y; \quad 19.62 - F_s \sin \theta = 2a \quad (2)$$

From Eq. 2 it is seen that the acceleration depends on the magnitude and direction of the spring force. Solution for  $N_C$  and  $a$  is possible once  $F_s$  and  $\theta$  are known.

The magnitude of the spring force is a function of the stretch  $s$  of the spring; i.e.,  $F_s = ks$ . Here the unstretched length is  $AB = 0.75 \text{ m}$ , Fig. 13–9a; therefore,  $s = CB - AB = \sqrt{y^2 + (0.75)^2} - 0.75$ . Since  $k = 3 \text{ N/m}$ , then

$$F_s = ks = 3\left(\sqrt{y^2 + (0.75)^2} - 0.75\right) \quad (3)$$

From Fig. 13–9a, the angle  $\theta$  is related to  $y$  by trigonometry.

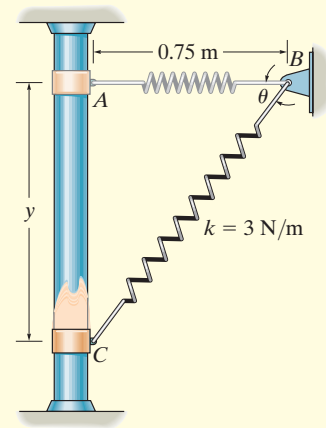
$$\tan \theta = \frac{y}{0.75}$$

Substituting  $y = 1 \text{ m}$  into Eqs. 3 and 4 yields  $F_s = 1.50 \text{ N}$  and  $\theta = 53.1^\circ$ . Substituting these results into Eqs. 1 and 2, we obtain

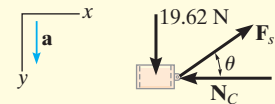
$$N_C = 0.900 \text{ N} \quad \text{Ans.}$$

$$a = 9.21 \text{ m/s}^2 \downarrow \quad \text{Ans.}$$

**NOTE:** This is not a case of constant acceleration, since the spring force changes both its magnitude and direction as the collar moves downward.



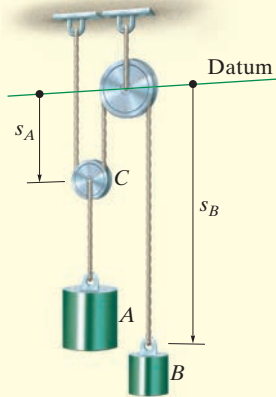
(a)



(b)

**Fig. 13–9**

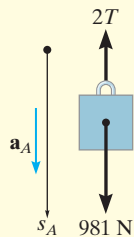
## EXAMPLE 13.5



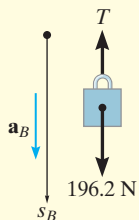
(a)



(b)



(c)



(d)

Fig. 13-10

The 100-kg block  $A$  shown in Fig. 13-10a is released from rest. If the masses of the pulleys and the cord are neglected, determine the velocity of the 20-kg block  $B$  in 2 s.

## SOLUTION

**Free-Body Diagrams.** Since the mass of the pulleys is *neglected*, then for pulley  $C$ ,  $ma = 0$  and we can apply  $\Sigma F_y = 0$ , as shown in Fig. 13-10b. The free-body diagrams for blocks  $A$  and  $B$  are shown in Fig. 13-10c and d, respectively. Notice that for  $A$  to remain stationary  $T = 490.5$  N, whereas for  $B$  to remain static  $T = 196.2$  N. Hence  $A$  will move down while  $B$  moves up. Although this is the case, we will assume both blocks accelerate downward, in the direction of  $+s_A$  and  $+s_B$ . The three unknowns are  $T$ ,  $a_A$ , and  $a_B$ .

**Equations of Motion.** Block  $A$ ,

$$+\downarrow \Sigma F_y = ma_y; \quad 981 - 2T = 100a_A \quad (1)$$

Block  $B$ ,

$$+\downarrow \Sigma F_y = ma_y; \quad 196.2 - T = 20a_B \quad (2)$$

**Kinematics.** The necessary third equation is obtained by relating  $a_A$  to  $a_B$  using a dependent motion analysis, discussed in Sec. 12.9. The coordinates  $s_A$  and  $s_B$  in Fig. 13-10a measure the positions of  $A$  and  $B$  from the fixed datum. It is seen that

$$2s_A + s_B = l$$

where  $l$  is constant and represents the total vertical length of cord. Differentiating this expression twice with respect to time yields

$$2a_A = -a_B \quad (3)$$

Notice that when writing Eqs. 1 to 3, the *positive direction was always assumed downward*. It is very important to be *consistent* in this assumption since we are seeking a simultaneous solution of equations. The results are

$$T = 327.0 \text{ N}$$

$$a_A = 3.27 \text{ m/s}^2$$

$$a_B = -6.54 \text{ m/s}^2$$

Hence when block  $A$  accelerates *downward*, block  $B$  accelerates *upward* as expected. Since  $a_B$  is constant, the velocity of block  $B$  in 2 s is thus

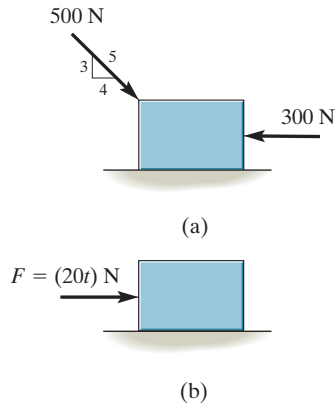
$$\begin{aligned} (+\downarrow) \quad v &= v_0 + a_B t \\ &= 0 + (-6.54)(2) \\ &= -13.1 \text{ m/s} \end{aligned}$$

*Ans.*

The negative sign indicates that block  $B$  is moving upward.

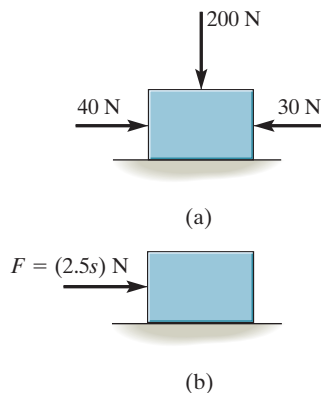
## PRELIMINARY PROBLEMS

**P13-1.** The 10-kg block is subjected to the forces shown. In each case, determine its velocity when  $t = 2$  s if  $v = 0$  when  $t = 0$ .



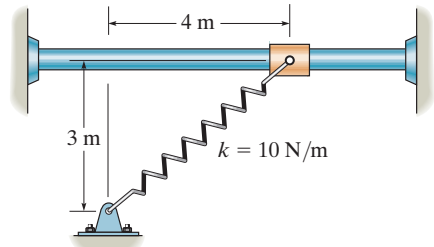
**Prob. P13-1**

**P13-2.** The 10-kg block is subjected to the forces shown. In each case, determine its velocity at  $s = 8$  m if  $v = 3$  m/s at  $s = 0$ . Motion occurs to the right.



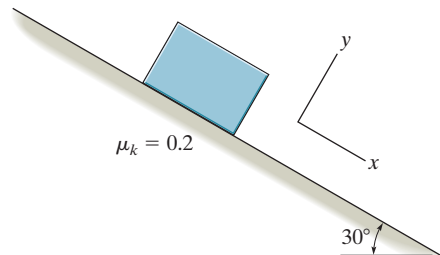
**Prob. P13-2**

**P13-3.** Determine the initial acceleration of the 10-kg smooth collar. The spring has an unstretched length of 1 m.



**Prob. P13-3**

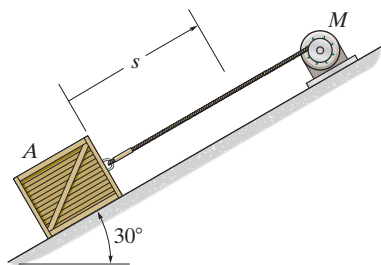
**P13-4.** Write the equations of motion in the  $x$  and  $y$  directions for the 10-kg block.



**Prob. P13-4**

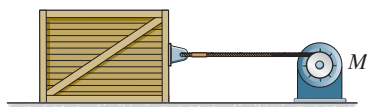
## FUNDAMENTAL PROBLEMS

**F13-1.** The motor winds in the cable with a constant acceleration, such that the 20-kg crate moves a distance  $s = 6$  m in 3 s, starting from rest. Determine the tension developed in the cable. The coefficient of kinetic friction between the crate and the plane is  $\mu_k = 0.3$ .



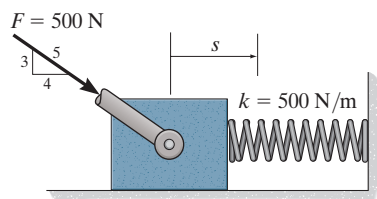
Prob. F13-1

**F13-2.** If motor  $M$  exerts a force of  $F = (10t^2 + 100)$  N on the cable, where  $t$  is in seconds, determine the velocity of the 25-kg crate when  $t = 4$  s. The coefficients of static and kinetic friction between the crate and the plane are  $\mu_s = 0.3$  and  $\mu_k = 0.25$ , respectively. The crate is initially at rest.



Prob. F13-2

**F13-3.** A spring of stiffness  $k = 500$  N/m is mounted against the 10-kg block. If the block is subjected to the force of  $F = 500$  N, determine its velocity at  $s = 0.5$  m. When  $s = 0$ , the block is at rest and the spring is uncompressed. The contact surface is smooth.



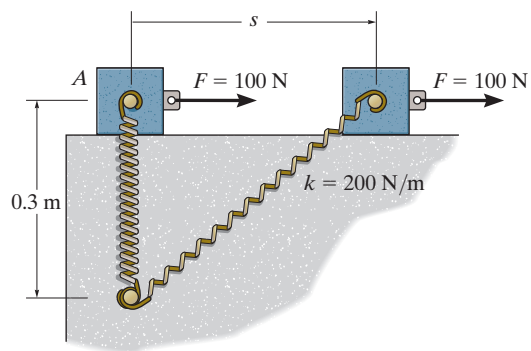
Prob. F13-3

**F13-4.** The 2-Mg car is being towed by a winch. If the winch exerts a force of  $T = 100(s + 1)$  N on the cable, where  $s$  is the displacement of the car in meters, determine the speed of the car when  $s = 10$  m, starting from rest. Neglect rolling resistance of the car.



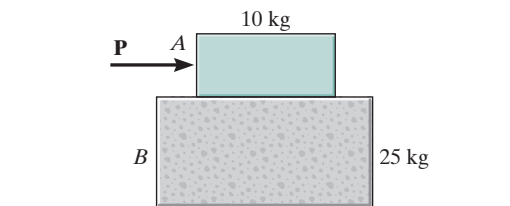
Prob. F13-4

**F13-5.** The spring has a stiffness  $k = 200$  N/m and is unstretched when the 25-kg block is at  $A$ . Determine the acceleration of the block when  $s = 0.4$  m. The contact surface between the block and the plane is smooth.



Prob. F13-5

**F13-6.** Block  $B$  rests upon a smooth surface. If the coefficients of static and kinetic friction between  $A$  and  $B$  are  $\mu_s = 0.4$  and  $\mu_k = 0.3$ , respectively, determine the acceleration of each block if  $P = 30$  N.

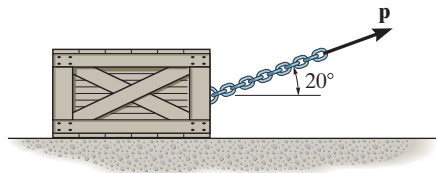


Prob. F13-6

## PROBLEMS

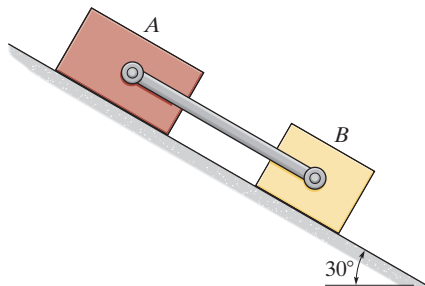
**13-1.** The crate has a mass of 80 kg and is being towed by a chain which is always directed at  $20^\circ$  from the horizontal as shown. If the magnitude of  $\mathbf{P}$  is increased until the crate begins to slide, determine the crate's initial acceleration if the coefficient of static friction is  $\mu_s = 0.5$  and the coefficient of kinetic friction is  $\mu_k = 0.3$ .

**13-2.** The crate has a mass of 80 kg and is being towed by a chain which is always directed at  $20^\circ$  from the horizontal as shown. Determine the crate's acceleration in  $t = 2$  s if the coefficient of static friction is  $\mu_s = 0.4$ , the coefficient of kinetic friction is  $\mu_k = 0.3$ , and the towing force is  $P = (90t^2)$  N, where  $t$  is in seconds.



Probs. 13-1/2

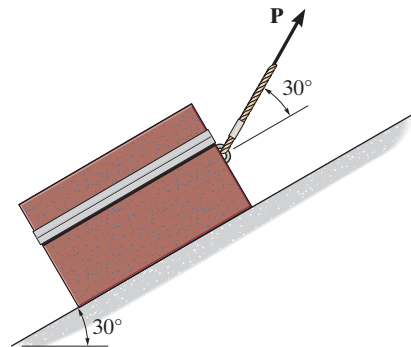
**13-3.** If blocks  $A$  and  $B$  of mass 10 kg and 6 kg respectively, are placed on the inclined plane and released, determine the force developed in the link. The coefficients of kinetic friction between the blocks and the inclined plane are  $\mu_A = 0.1$  and  $\mu_B = 0.3$ . Neglect the mass of the link.



Prob. 13-3

**\*13-4.** If  $P = 400$  N and the coefficient of kinetic friction between the 50-kg crate and the inclined plane is  $\mu_k = 0.25$ , determine the velocity of the crate after it travels 6 m up the plane. The crate starts from rest.

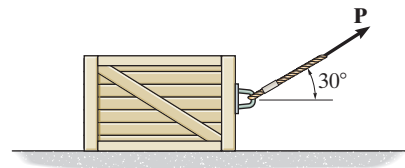
**13-5.** If the 50-kg crate starts from rest and travels a distance of 6 m up the plane in 4 s, determine the magnitude of force  $\mathbf{P}$  acting on the crate. The coefficient of kinetic friction between the crate and the ground is  $\mu_k = 0.25$ .



Probs. 13-4/5

**13-6.** If the coefficient of kinetic friction between the 50-kg crate and the ground is  $\mu_k = 0.3$ , determine the distance the crate travels and its velocity when  $t = 3$  s. The crate starts from rest, and  $P = 200$  N.

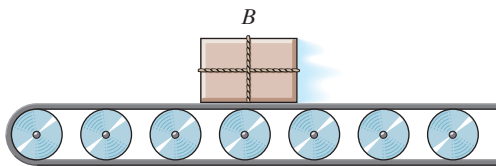
**13-7.** If the 50-kg crate starts from rest and achieves a velocity of  $v = 4$  m/s when it travels a distance of 5 m to the right, determine the magnitude of force  $\mathbf{P}$  acting on the crate. The coefficient of kinetic friction between the crate and the ground is  $\mu_k = 0.3$ .



Probs. 13-6/7

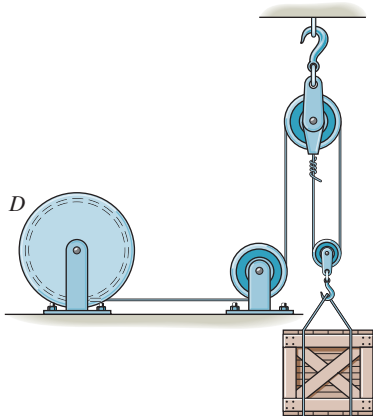
**\*13-8.** The conveyor belt is moving at 4 m/s. If the coefficient of static friction between the conveyor and the 10-kg package  $B$  is  $\mu_s = 0.2$ , determine the shortest time the belt can stop so that the package does not slide on the belt.

**13-9.** The conveyor belt is designed to transport packages of various weights. Each 10-kg package has a coefficient of kinetic friction  $\mu_k = 0.15$ . If the speed of the conveyor is 5 m/s, and then it suddenly stops, determine the distance the package will slide on the belt before coming to rest.



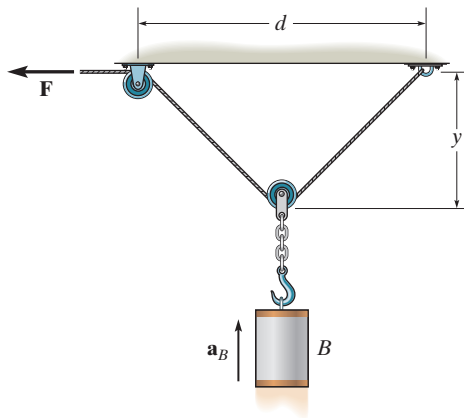
Probs. 13-8/9

**13-10.** The winding drum  $D$  is drawing in the cable at an accelerated rate of  $5 \text{ m/s}^2$ . Determine the cable tension if the suspended crate has a mass of 800 kg.



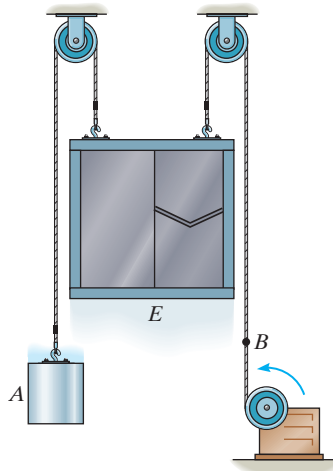
Prob. 13-10

**13-11.** Cylinder  $B$  has a mass  $m$  and is hoisted using the cord and pulley system shown. Determine the magnitude of force  $\mathbf{F}$  as a function of the block's vertical position  $y$  so that when  $\mathbf{F}$  is applied, the block rises with a constant acceleration  $\mathbf{a}_B$ . Neglect the mass of the cord and pulleys.



Prob. 13-11

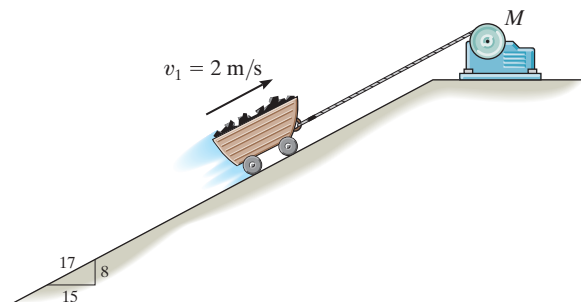
**\*13-12.** The elevator  $E$  has a mass of 500 kg and the counterweight at  $A$  has a mass of 150 kg. If the elevator attains a speed of 10 m/s after it rises 40 m, determine the constant force developed in the cable at  $B$ . Neglect the mass of the pulleys and cable.



Prob. 13-12

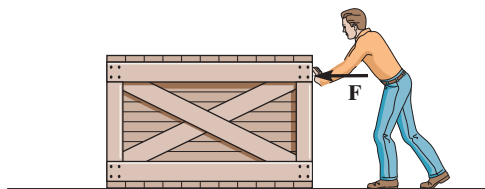
**13-13.** The 400-kg mine car is hoisted up the incline using the cable and motor  $M$ . For a short time, the force in the cable is  $F = (3200t^2)$  N, where  $t$  is in seconds. If the car has an initial velocity  $v_1 = 2$  m/s when  $t = 0$ , determine its velocity when  $t = 2$  s.

**13-14.** The 400-kg mine car is hoisted up the incline using the cable and motor  $M$ . For a short time, the force in the cable is  $F = (3200t^2)$  N where  $t$  is in seconds. If the car has an initial velocity  $v_1 = 2$  m/s at  $s = 0$  and  $t = 0$ , determine the distance it moves up the plane when  $t = 2$  s.



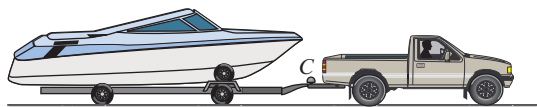
**Probs. 13-13/14**

**13-15.** The 75-kg man pushes on the 150-kg crate with a horizontal force  $\mathbf{F}$ . If the coefficients of static and kinetic friction between the crate and the surface are  $\mu_s = 0.3$  and  $\mu_k = 0.2$ , and the coefficient of static friction between the man's shoes and the surface is  $\mu_s = 0.8$ , show that the man is able to move the crate. What is the greatest acceleration the man can give the crate?



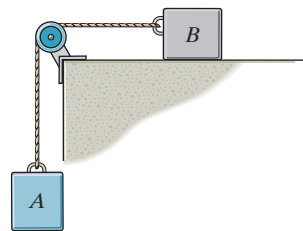
**Prob. 13-15**

**\*13-16.** The 2-Mg truck is traveling at 15 m/s when the brakes on all its wheels are applied, causing it to skid for a distance of 10 m before coming to rest. Determine the constant horizontal force developed in the coupling  $C$ , and the frictional force developed between the tires of the truck and the road during this time. The total mass of the boat and trailer is 1 Mg.



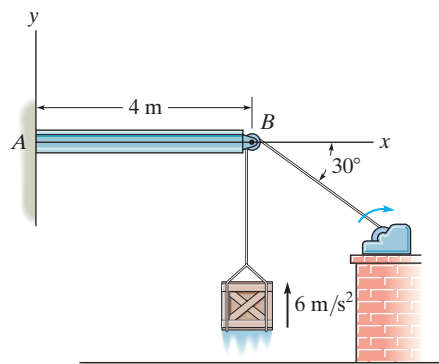
**Prob. 13-16**

**13-17.** Determine the acceleration of the blocks when the system is released. The coefficient of kinetic friction is  $\mu_k$ , and the mass of each block is  $m$ . Neglect the mass of the pulleys and cord.



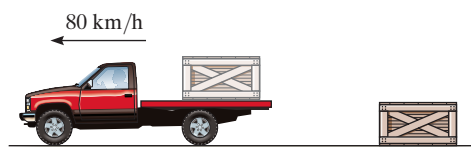
**Prob. 13-17**

**13-18.** The motor lifts the 50-kg crate with an acceleration of  $6 \text{ m/s}^2$ . Determine the components of force reaction and the couple moment at the fixed support  $A$ .



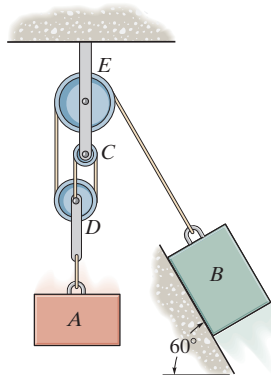
**Prob. 13-18**

**13-19.** A crate having a mass of 60 kg falls horizontally off the back of a truck which is traveling at 80 km/h. Determine the coefficient of kinetic friction between the road and the crate if the crate slides 45 m on the ground with no tumbling along the road before coming to rest. Assume that the initial speed of the crate along the road is 80 km/h.



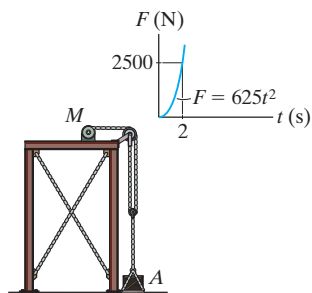
**Prob. 13-19**

**\*13–20.** Determine the required mass of block  $A$  so that when it is released from rest it moves the 5-kg block  $B$  0.75 m up along the smooth inclined plane in  $t = 2$  s. Neglect the mass of the pulleys and cords.



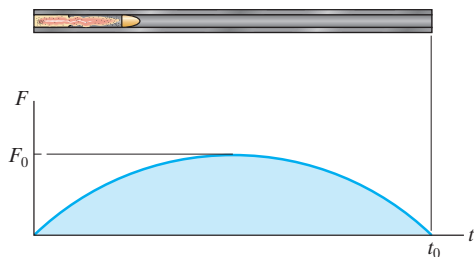
**Prob. 13–20**

**13–21.** The force of the motor  $M$  on the cable is shown in the graph. Determine the velocity of the 400-kg crate  $A$  when  $t = 2$  s.



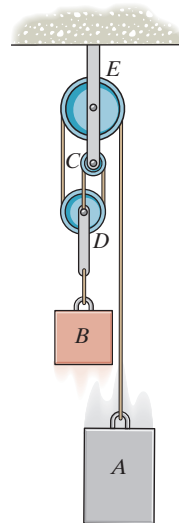
**Prob. 13–21**

**13–22.** The bullet of mass  $m$  is given a velocity due to gas pressure caused by the burning of powder within the chamber of the gun. Assuming this pressure creates a force of  $F = F_0 \sin(\pi t/t_0)$  on the bullet, determine the velocity of the bullet at any instant it is in the barrel. What is the bullet's maximum velocity? Also, determine the position of the bullet in the barrel as a function of time.



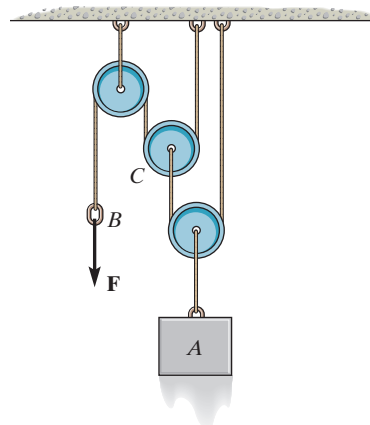
**Prob. 13–22**

**13–23.** The 50-kg block  $A$  is released from rest. Determine the velocity of the 15-kg block  $B$  in 2 s.



**Prob. 13–23**

**\*13–24.** If the supplied force  $F = 150$  N, determine the velocity of the 50-kg block  $A$  when it has risen 3 m, starting from rest.

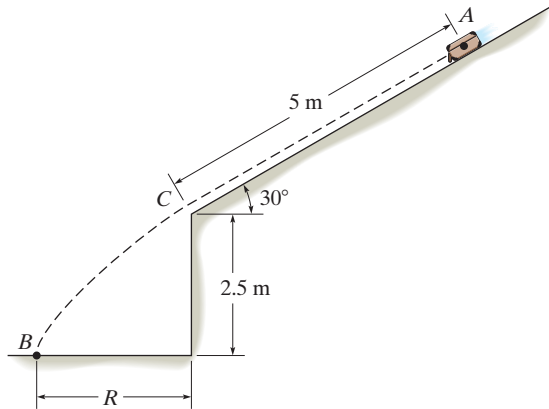


**Prob. 13–24**



**13-25.** A 60-kg suitcase slides from rest 5 m down the smooth ramp. Determine the distance  $R$  where it strikes the ground at  $B$ . How long does it take to go from  $A$  to  $B$ ?

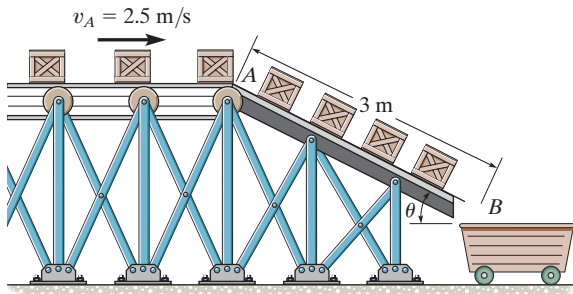
**13-26.** Solve Prob. 13-25 if the suitcase has an initial velocity down the ramp of  $v_A = 2$  m/s, and the coefficient of kinetic friction along  $AC$  is  $\mu_k = 0.2$ .



**Probs. 13-25/26**

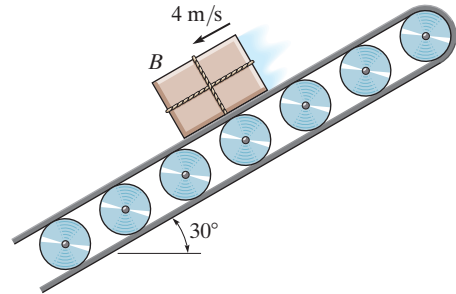
**13-27.** The conveyor belt delivers each 12-kg crate to the ramp at  $A$  such that the crate's speed is  $v_A = 2.5$  m/s, directed down *along* the ramp. If the coefficient of kinetic friction between each crate and the ramp is  $\mu_k = 0.3$ , determine the speed at which each crate slides off the ramp at  $B$ . Assume that no tipping occurs. Take  $\theta = 30^\circ$ .

**\*13-28.** The conveyor belt delivers each 12-kg crate to the ramp at  $A$  such that the crate's speed is  $v_A = 2.5$  m/s, directed down *along* the ramp. If the coefficient of kinetic friction between each crate and the ramp is  $\mu_k = 0.3$ , determine the smallest incline  $\theta$  of the ramp so that the crates will slide off and fall into the cart.



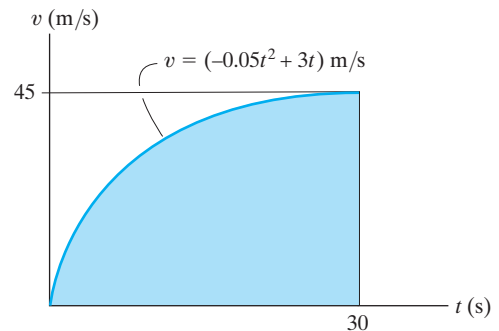
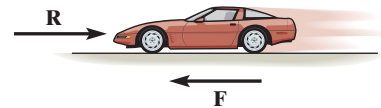
**Probs. 13-27/28**

**13-29.** The conveyor belt is moving downward at 4 m/s. If the coefficient of static friction between the conveyor and the 15-kg package  $B$  is  $\mu_s = 0.8$ , determine the shortest time the belt can stop so that the package does not slide on the belt.



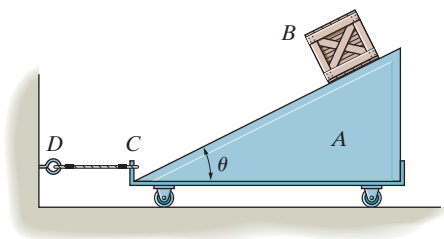
**Prob. 13-29**

**13-30.** The 1.5 Mg sports car has a tractive force of  $F = 4.5$  kN. If it produces the velocity described by  $v$ - $t$  graph shown, plot the air resistance  $R$  versus  $t$  for this time period.



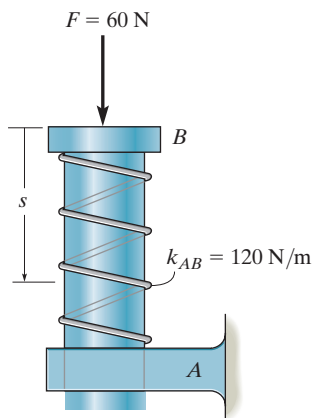
**Prob. 13-30**

**13-31.** Crate  $B$  has a mass  $m$  and is released from rest when it is on top of cart  $A$ , which has a mass  $3m$ . Determine the tension in cord  $CD$  needed to hold the cart from moving while  $B$  is sliding down  $A$ . Neglect friction.



**Prob. 13-31**

**\*13-32.** The 4-kg smooth cylinder is supported by the spring having a stiffness of  $k_{AB} = 120 \text{ N/m}$ . Determine the velocity of the cylinder when it moves downward  $s = 0.2 \text{ m}$  from its equilibrium position, which is caused by the application of the force  $F = 60 \text{ N}$ .



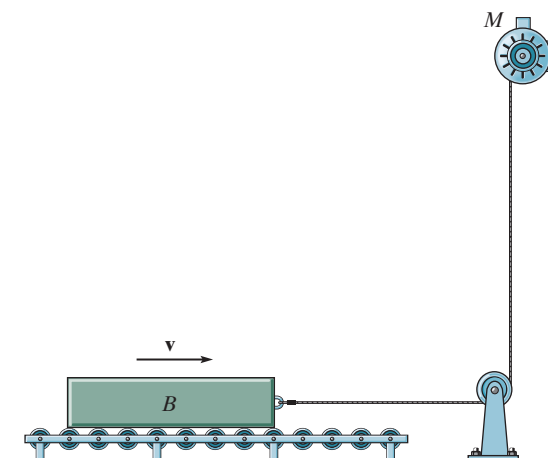
**Prob. 13-32**

**13-33.** The coefficient of static friction between the 200-kg crate and the flat bed of the truck is  $\mu_s = 0.3$ . Determine the shortest time for the truck to reach a speed of 60 km/h, starting from rest with constant acceleration, so that the crate does not slip.



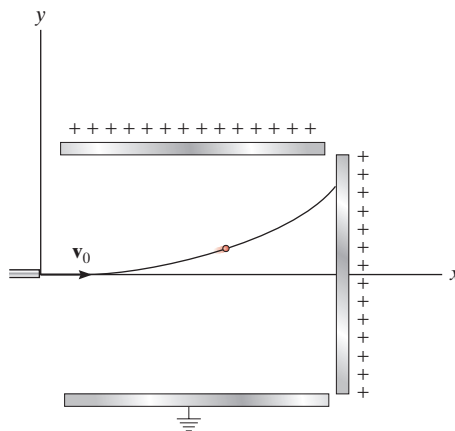
**Prob. 13-33**

**13-34.** The 300-kg bar  $B$ , originally at rest, is being towed over a series of small rollers. Determine the force in the cable when  $t = 5 \text{ s}$ , if the motor  $M$  is drawing in the cable for a short time at a rate of  $v = (0.4t^2) \text{ m/s}$ , where  $t$  is in seconds ( $0 \leq t \leq 6 \text{ s}$ ). How far does the bar move in 5 s? Neglect the mass of the cable, pulley, and the rollers.



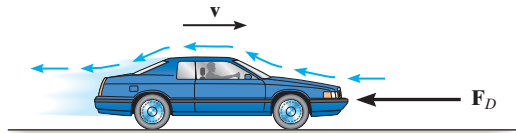
**Prob. 13-34**

**13-35.** An electron of mass  $m$  is discharged with an initial horizontal velocity of  $\mathbf{v}_0$ . If it is subjected to two fields of force for which  $F_x = F_0$  and  $F_y = 0.3F_0$ , where  $F_0$  is constant, determine the equation of the path, and the speed of the electron at any time  $t$ .



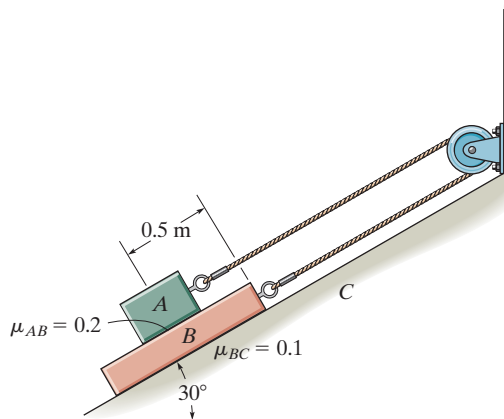
**Prob. 13-35**

**\*13–36.** A car of mass  $m$  is traveling at a slow velocity  $v_0$ . If it is subjected to the drag resistance of the wind, which is proportional to its velocity, i.e.,  $F_D = kv$ , determine the distance and the time the car will travel before its velocity becomes  $0.5v_0$ . Assume no other frictional forces act on the car.



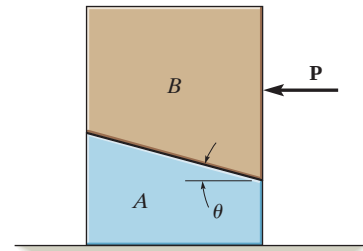
Prob. 13–36

**13–37.** The 10-kg block  $A$  rests on the 50-kg plate  $B$  in the position shown. Neglect the mass of the rope and pulley, and using the coefficients of kinetic friction indicated, determine the time needed for block  $A$  to slide 0.5 m on the plate when the system is released from rest.



Prob. 13–37

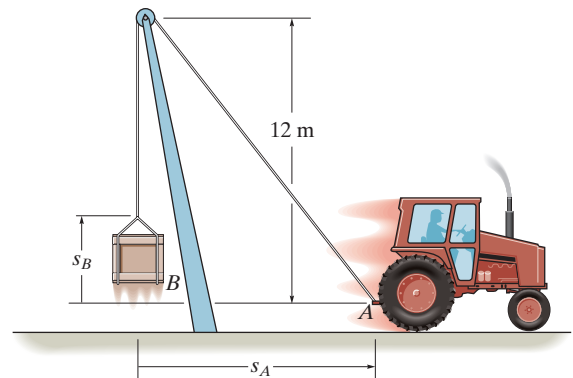
**13–38.** Blocks  $A$  and  $B$  each has a mass  $m$ . Determine the largest horizontal force  $\mathbf{P}$  which can be applied to  $B$  so that it will not slide on  $A$ . Also, what is the corresponding acceleration? The coefficient of static friction between  $A$  and  $B$  is  $\mu_s$ . Neglect any friction between  $A$  and the horizontal surface.



Prob. 13–38

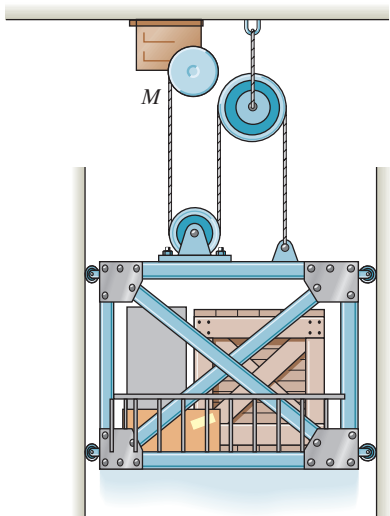
**13–39.** The tractor is used to lift the 150-kg load  $B$  with the 24-m-long rope, boom, and pulley system. If the tractor travels to the right at a constant speed of 4 m/s, determine the tension in the rope when  $s_A = 5$  m. When  $s_A = 0$ ,  $s_B = 0$ .

**\*13–40.** The tractor is used to lift the 150-kg load  $B$  with the 24-m-long rope, boom, and pulley system. If the tractor travels to the right with an acceleration of  $3 \text{ m/s}^2$  and has a velocity of 4 m/s at the instant  $s_A = 5$  m, determine the tension in the rope at this instant. When  $s_A = 0$ ,  $s_B = 0$ .



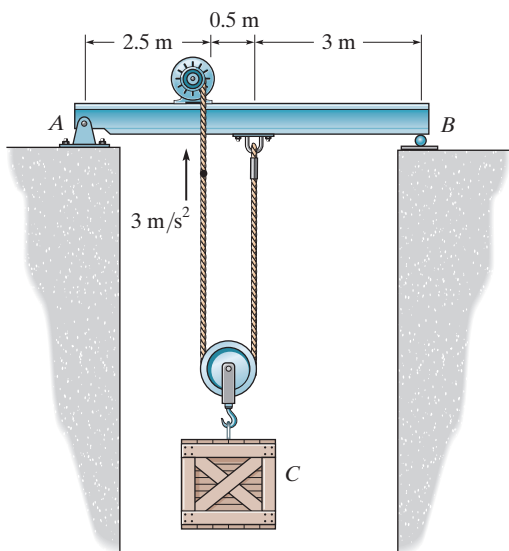
Probs. 13–39/40

**13-41.** A freight elevator, including its load, has a mass of 1 Mg. It is prevented from rotating due to the track and wheels mounted along its sides. If the motor  $M$  develops a constant tension  $T = 4$  kN in its attached cable, determine the velocity of the elevator when it has moved upward 6 m starting from rest. Neglect the mass of the pulleys and cables.



**Prob. 13-41**

**13-42.** If the motor draws in the cable with an acceleration of  $3 \text{ m/s}^2$ , determine the reactions at the supports  $A$  and  $B$ . The beam has a uniform mass of  $30 \text{ kg/m}$ , and the crate has a mass of  $200 \text{ kg}$ . Neglect the mass of the motor and pulleys.



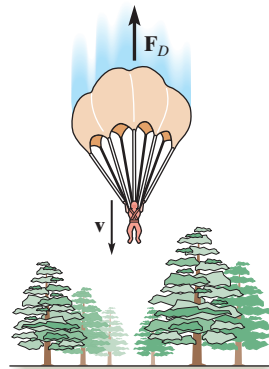
**Prob. 13-42**

**13-43.** If the force exerted on cable  $AB$  by the motor is  $F = (100t^{3/2}) \text{ N}$ , where  $t$  is in seconds, determine the 50-kg crate's velocity when  $t = 5 \text{ s}$ . The coefficients of static and kinetic friction between the crate and the ground are  $\mu_s = 0.4$  and  $\mu_k = 0.3$ , respectively. Initially the crate is at rest.



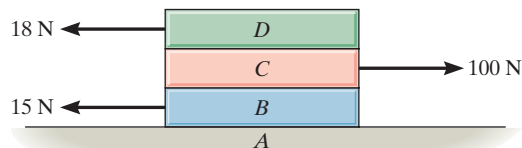
**Prob. 13-43**

**\*13-44.** A parachutist having a mass  $m$  opens his parachute from an at-rest position at a very high altitude. If the atmospheric drag resistance is  $F_D = kv^2$ , where  $k$  is a constant, determine his velocity when he has fallen for a time  $t$ . What is his velocity when he lands on the ground? This velocity is referred to as the *terminal velocity*, which is found by letting the time of fall  $t \rightarrow \infty$ .



**Prob. 13-44**

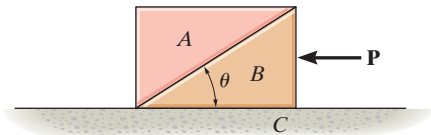
**13-45.** Each of the three plates has a mass of  $10 \text{ kg}$ . If the coefficients of static and kinetic friction at each surface of contact are  $\mu_s = 0.3$  and  $\mu_k = 0.2$ , respectively, determine the acceleration of each plate when the three horizontal forces are applied.



**Prob. 13-45**

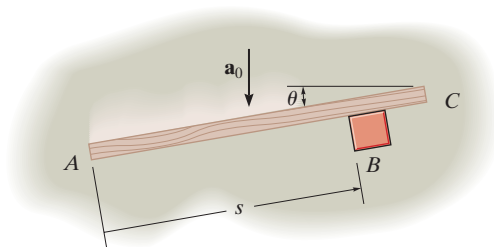
**13–46.** Blocks  $A$  and  $B$  each have a mass  $m$ . Determine the largest horizontal force  $\mathbf{P}$  which can be applied to  $B$  so that  $A$  will not move relative to  $B$ . All surfaces are smooth.

**13–47.** Blocks  $A$  and  $B$  each have a mass  $m$ . Determine the largest horizontal force  $\mathbf{P}$  which can be applied to  $B$  so that  $A$  will not slip on  $B$ . The coefficient of static friction between  $A$  and  $B$  is  $\mu_s$ . Neglect any friction between  $B$  and  $C$ .



Probs. 13–46/47

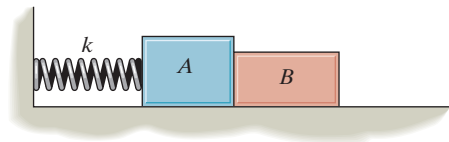
**\*13–48.** The smooth block  $B$  of negligible size has a mass  $m$  and rests on the horizontal plane. If the board  $AC$  pushes on the block at an angle  $\theta$  with a constant acceleration  $\mathbf{a}_0$ , determine the velocity of the block along the board and the distance  $s$  the block moves along the board as a function of time  $t$ . The block starts from rest when  $s = 0, t = 0$ .



Prob. 13–48

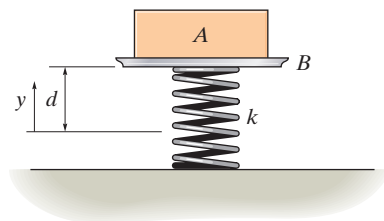
**13–49.** Block  $A$  has a mass  $m_A$  and is attached to a spring having a stiffness  $k$  and unstretched length  $l_0$ . If another block  $B$ , having a mass  $m_B$ , is pressed against  $A$  so that the spring deforms a distance  $d$ , determine the distance both blocks slide on the smooth surface before they begin to separate. What is their velocity at this instant?

**13–50.** Block  $A$  has a mass  $m_A$  and is attached to a spring having a stiffness  $k$  and unstretched length  $l_0$ . If another block  $B$ , having a mass  $m_B$ , is pressed against  $A$  so that the spring deforms a distance  $d$ , show that for separation to occur it is necessary that  $d > 2\mu_k g(m_A + m_B)/k$ , where  $\mu_k$  is the coefficient of kinetic friction between the blocks and the ground. Also, what is the distance the blocks slide on the surface before they separate?



Probs. 13–49/50

**13–51.** The block  $A$  has a mass  $m_A$  and rests on the pan  $B$ , which has a mass  $m_B$ . Both are supported by a spring having a stiffness  $k$  that is attached to the bottom of the pan and to the ground. Determine the distance  $d$  the pan should be pushed down from the equilibrium position and then released from rest so that separation of the block will take place from the surface of the pan at the instant the spring becomes unstretched.



Prob. 13–51

## 13.5 Equations of Motion: Normal and Tangential Coordinates

When a particle moves along a curved path which is known, the equation of motion for the particle may be written in the tangential, normal, and binormal directions, Fig. 13–11. Note that there is no motion of the particle in the binormal direction, since the particle is constrained to move along the path. We have

$$\Sigma \mathbf{F} = m\mathbf{a}$$

$$\Sigma F_t \mathbf{u}_t + \Sigma F_n \mathbf{u}_n + \Sigma F_b \mathbf{u}_b = m\mathbf{a}_t + m\mathbf{a}_n$$

This equation is satisfied provided

$$\begin{aligned} \Sigma F_t &= ma_t \\ \Sigma F_n &= ma_n \\ \Sigma F_b &= 0 \end{aligned} \quad (13-7)$$

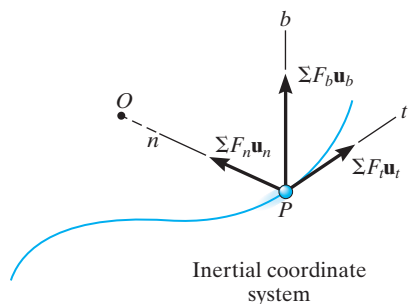


Fig. 13–11

Recall that  $a_t (= dv/dt)$  represents the time rate of change in the magnitude of velocity. So if  $\Sigma F_t$  acts in the direction of motion, the particle's speed will increase, whereas if it acts in the opposite direction, the particle will slow down. Likewise,  $a_n (= v^2/\rho)$  represents the time rate of change in the velocity's direction. It is caused by  $\Sigma F_n$ , which *always* acts in the positive  $n$  direction, i.e., toward the path's center of curvature. For this reason it is often referred to as the *centripetal force*.



As a roller coaster falls downward along the track, the cars have both a normal and a tangential component of acceleration.

## Procedure for Analysis

When a problem involves the motion of a particle along a *known curved path*, normal and tangential coordinates should be considered for the analysis since the acceleration components can be readily formulated. The method for applying the equations of motion, which relate the forces to the acceleration, has been outlined in the procedure given in Sec. 13.4. Specifically, for  $t, n, b$  coordinates it may be stated as follows:

### Free-Body Diagram.

- Establish the inertial  $t, n, b$  coordinate system at the particle and draw the particle's free-body diagram.
- The particle's normal acceleration  $\mathbf{a}_n$  *always* acts in the positive  $n$  direction.
- If the tangential acceleration  $\mathbf{a}_t$  is unknown, assume it acts in the positive  $t$  direction.
- There is no acceleration in the  $b$  direction.
- Identify the unknowns in the problem.

### Equations of Motion.

- Apply the equations of motion, Eq. 13–7.

### Kinematics.

- Formulate the tangential and normal components of acceleration; i.e.,  $a_t = dv/dt$  or  $a_t = v dv/ds$  and  $a_n = v^2/\rho$ .
- If the path is defined as  $y = f(x)$ , the radius of curvature at the point where the particle is located can be obtained from  $\rho = [1 + (dy/dx)^2]^{3/2}/|d^2y/dx^2|$ .



The unbalanced force of the rope on the skier gives him a normal component of acceleration.

**EXAMPLE 13.6**

Determine the banking angle  $\theta$  for the race track so that the wheels of the racing cars shown in Fig. 13–12a will not have to depend upon friction to prevent any car from sliding up or down the track. Assume the cars have negligible size, a mass  $m$ , and travel around the curve of radius  $\rho$  with a constant speed  $v$ .



(a)

**SOLUTION**

Before looking at the following solution, give some thought as to why it should be solved using  $t, n, b$  coordinates.

**Free-Body Diagram.** As shown in Fig. 13–12b, and as stated in the problem, no frictional force acts on the car. Here  $N_C$  represents the resultant of the ground on all four wheels. Since  $a_n$  can be calculated, the unknowns are  $N_C$  and  $\theta$ .

**Equations of Motion.** Using the  $n, b$  axes shown,

$$\rightarrow \Sigma F_n = ma_n; \quad N_C \sin \theta = m \frac{v^2}{\rho} \quad (1)$$

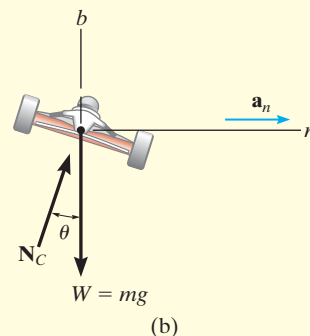
$$+\uparrow \Sigma F_b = 0; \quad N_C \cos \theta - mg = 0 \quad (2)$$

Eliminating  $N_C$  and  $m$  from these equations by dividing Eq. 1 by Eq. 2, we obtain

$$\tan \theta = \frac{v^2}{g\rho}$$

$$\theta = \tan^{-1} \left( \frac{v^2}{g\rho} \right) \quad \text{Ans.}$$

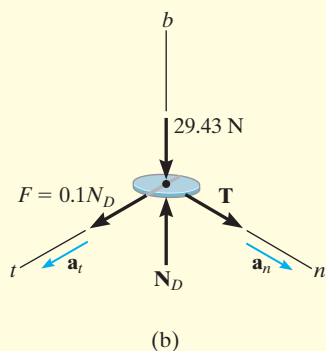
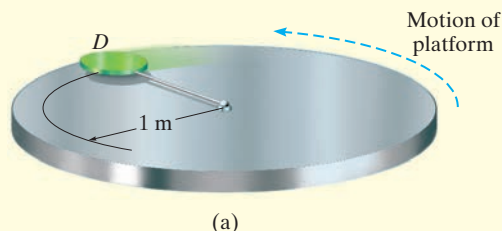
**NOTE:** The result is independent of the mass of the car. Also, a force summation in the tangential direction is of no consequence to the solution. If it were considered, then  $a_t = dv/dt = 0$ , since the car moves with *constant speed*. A further analysis of this problem is discussed in Prob. 21–52.

**Fig. 13–12**



**EXAMPLE 13.7**

The 3-kg disk  $D$  is attached to the end of a cord as shown in Fig. 13–13a. The other end of the cord is attached to a ball-and-socket joint located at the center of a platform. If the platform rotates rapidly, and the disk is placed on it and released from rest as shown, determine the time it takes for the disk to reach a speed great enough to break the cord. The maximum tension the cord can sustain is 100 N, and the coefficient of kinetic friction between the disk and the platform is  $\mu_k = 0.1$ .

**Fig. 13–13****SOLUTION**

**Free-Body Diagram.** The frictional force has a magnitude  $F = \mu_k N_D = 0.1N_D$  and a sense of direction that opposes the *relative motion* of the disk with respect to the platform. It is this force that gives the disk a tangential component of acceleration causing  $v$  to increase, thereby causing  $T$  to increase until it reaches 100 N. The weight of the disk is  $W = 3(9.81) \text{ N} = 29.43 \text{ N}$ . Since  $a_n$  can be related to  $v$ , the unknowns are  $N_D$ ,  $a_t$ , and  $v$ .

**Equations of Motion.**

$$\Sigma F_n = ma_n; \quad T = 3\left(\frac{v^2}{1}\right) \quad (1)$$

$$\Sigma F_t = ma_t; \quad 0.1N_D = 3a_t \quad (2)$$

$$\Sigma F_b = 0; \quad N_D - 29.43 = 0 \quad (3)$$

Setting  $T = 100 \text{ N}$ , Eq. 1 can be solved for the critical speed  $v_{\text{cr}}$  of the disk needed to break the cord. Solving all the equations, we obtain

$$N_D = 29.43 \text{ N}$$

$$a_t = 0.981 \text{ m/s}^2$$

$$v_{\text{cr}} = 5.77 \text{ m/s}$$

**Kinematics.** Since  $a_t$  is *constant*, the time needed to break the cord is

$$v_{\text{cr}} = v_0 + a_t t$$

$$5.77 = 0 + (0.981)t$$

$$t = 5.89 \text{ s}$$

*Ans.*

**EXAMPLE 13.8**

Design of the ski jump shown in the photo requires knowing the type of forces that will be exerted on the skier and her approximate trajectory. If in this case the jump can be approximated by the parabola shown in Fig. 13–14a, determine the normal force on the 70-kg skier the instant she arrives at the end of the jump, point A, where her velocity is 20 m/s. Also, what is her acceleration at this point?

**SOLUTION**

Why consider using  $n, t$  coordinates to solve this problem?

**Free-Body Diagram.** Since  $dy/dx = x/30|_{x=0} = 0$ , the slope at A is horizontal. The free-body diagram of the skier when she is at A is shown in Fig. 13–14b. Since the path is *curved*, there are two components of acceleration,  $\mathbf{a}_n$  and  $\mathbf{a}_t$ . Since  $a_n$  can be calculated, the unknowns are  $a_t$  and  $N_A$ .

**Equations of Motion.**

$$+\uparrow \Sigma F_n = ma_n; \quad N_A - 70(9.81) = 70 \left[ \frac{(20)^2}{\rho} \right] \quad (1)$$

$$\leftarrow \Sigma F_t = ma_t; \quad 0 = 70a_t \quad (2)$$

The radius of curvature  $\rho$  for the path must be determined at point A(0, -60 m). Here  $y = \frac{1}{60}x^2 - 60$ ,  $dy/dx = \frac{1}{30}x$ ,  $d^2y/dx^2 = \frac{1}{30}$ , so that at  $x = 0$ ,

$$\rho = \frac{[1 + (dy/dx)^2]^{3/2}}{|d^2y/dx^2|} \bigg|_{x=0} = \frac{[1 + (0)^2]^{3/2}}{|\frac{1}{30}|} = 30 \text{ m}$$

Substituting this into Eq. 1 and solving for  $N_A$ , we obtain

$$N_A = 1620 \text{ N} \quad \text{Ans.}$$

**Kinematics.** From Eq. 2,

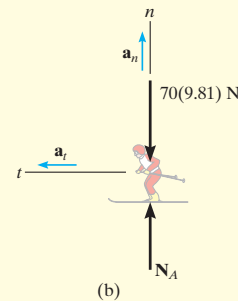
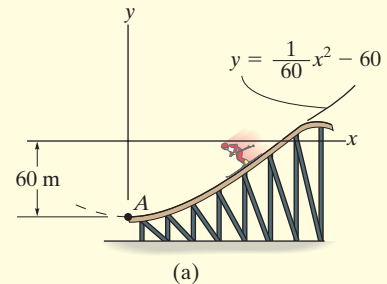
$$a_t = 0$$

Thus,

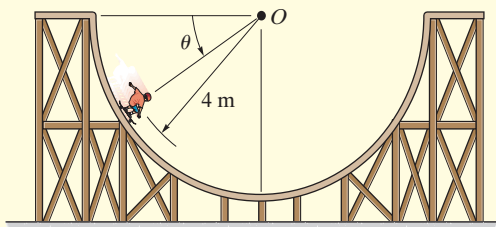
$$a_n = \frac{v^2}{\rho} = \frac{(20)^2}{30} = 13.33 \text{ m/s}^2$$

$$a_A = a_n = 13.3 \text{ m/s}^2 \uparrow \quad \text{Ans.}$$

**NOTE:** Apply the equation of motion in the  $y$  direction and show that when the skier is in midair, her downward acceleration is  $9.81 \text{ m/s}^2$ .



**Fig. 13–14**

**EXAMPLE 13.9**

The 60-kg skateboarder in Fig. 13–15a coasts down the circular track. If he starts from rest when  $\theta = 0^\circ$ , determine the magnitude of the normal reaction the track exerts on him when  $\theta = 60^\circ$ . Neglect his size for the calculation.

**SOLUTION**

**Free-Body Diagram.** The free-body diagram of the skateboarder when he is at an *arbitrary position*  $\theta$  is shown in Fig. 13–15b. At  $\theta = 60^\circ$  there are three unknowns,  $N_s$ ,  $a_t$ , and  $a_n$  (or  $v$ ).

**Equations of Motion.**

$$+\nearrow \Sigma F_n = ma_n; \quad N_s - [60(9.81)\text{N}] \sin \theta = (60 \text{ kg}) \left( \frac{v^2}{4 \text{ m}} \right) \quad (1)$$

$$+\searrow \Sigma F_t = ma_t; \quad [60(9.81)\text{N}] \cos \theta = (60 \text{ kg}) a_t$$

$$a_t = 9.81 \cos \theta$$

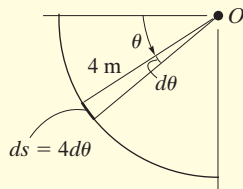
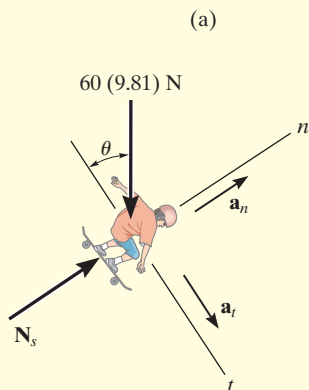
**Kinematics.** Since  $a_t$  is expressed in terms of  $\theta$ , the equation  $v dv = a_t ds$  must be used to determine the speed of the skateboarder when  $\theta = 60^\circ$ . Using the geometric relation  $s = r\theta$ , where  $ds = r d\theta = (4 \text{ m}) d\theta$ , Fig. 13–15c, and the initial condition  $v = 0$  at  $\theta = 0^\circ$ , we have,

$$\begin{aligned} v dv &= a_t ds \\ \int_0^v v dv &= \int_0^{60^\circ} 9.81 \cos \theta (4 d\theta) \\ \frac{v^2}{2} \Big|_0^v &= 39.24 \sin \theta \Big|_0^{60^\circ} \\ \frac{v^2}{2} - 0 &= 39.24(\sin 60^\circ - 0) \\ v^2 &= 67.97 \text{ m}^2/\text{s}^2 \end{aligned}$$

Substituting this result and  $\theta = 60^\circ$  into Eq. (1), yields

$$N_s = 1529.23 \text{ N} = 1.53 \text{ kN}$$

*Ans.*



(c)

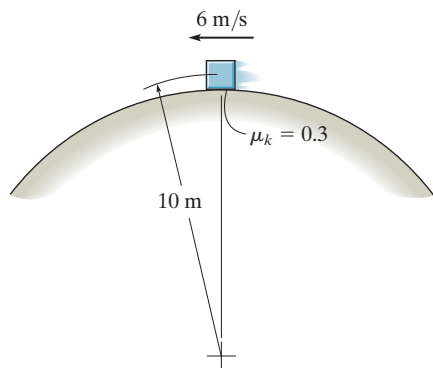
**Fig. 13–15**

## PRELIMINARY PROBLEMS

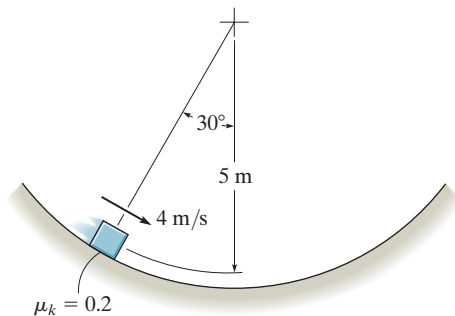
**P13-5.** Set up the  $n, t$  axes and write the equations of motion for the 10-kg block along each of these axes.

**P13-6.** Set up the  $n, b, t$  axes and write the equations of motion for the 10-kg block along each of these axes.

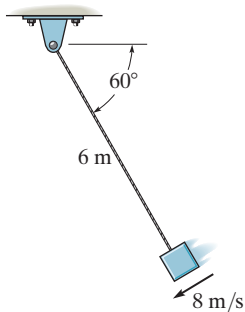
13



(a)

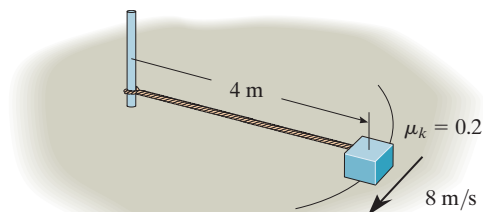


(b)

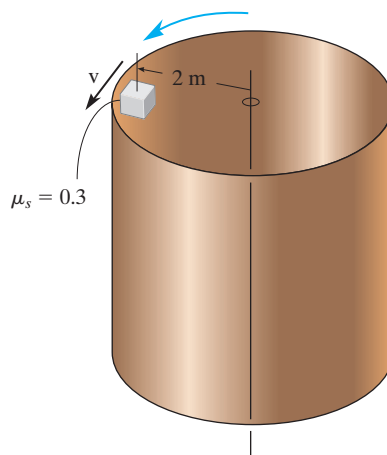


(c)

Prob. P13-5



(a)



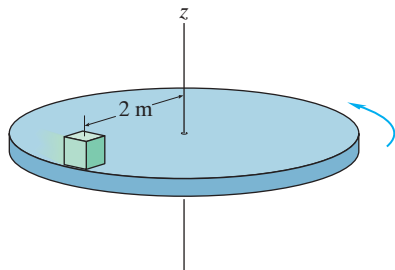
Constant rotation  
Block has impending motion

(b)

Prob. P13-6

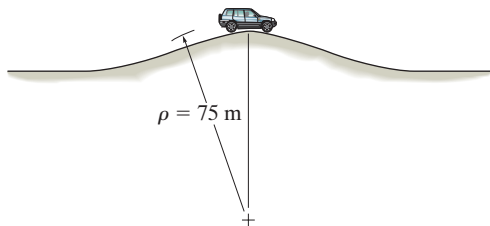
## FUNDAMENTAL PROBLEMS

**F13-7.** The block rests at a distance of 2 m from the center of the platform. If the coefficient of static friction between the block and the platform is  $\mu_s = 0.3$ , determine the maximum speed which the block can attain before it begins to slip. Assume the angular motion of the disk is slowly increasing.



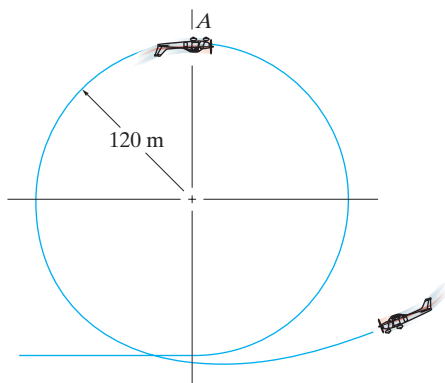
Prob. F13-7

**F13-8.** Determine the maximum speed that the jeep can travel over the crest of the hill and not lose contact with the road.



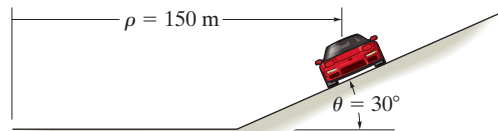
Prob. F13-8

**F13-9.** A pilot weighs 70 kg and is traveling at a constant speed of 36 m/s. Determine the normal force he exerts on the seat of the plane when he is upside down at A. The loop has a radius of curvature of 120 m.



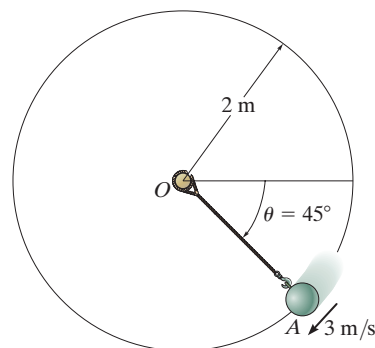
Prob. F13-9

**F13-10.** The sports car is traveling along a  $30^\circ$  banked road having a radius of curvature of  $\rho = 150$  m. If the coefficient of static friction between the tires and the road is  $\mu_s = 0.2$ , determine the maximum safe speed so no slipping occurs. Neglect the size of the car.



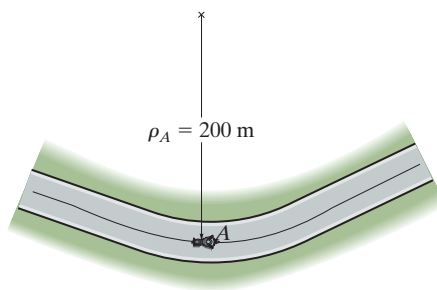
Prob. F13-10

**F13-11.** If the 10-kg ball has a velocity of 3 m/s when it is at the position A, along the vertical path, determine the tension in the cord and the increase in the speed of the ball at this position.



Prob. F13-11

**F13-12.** The motorcycle has a mass of 0.5 Mg and a negligible size. It passes point A traveling with a speed of 15 m/s, which is increasing at a constant rate of  $1.5 \text{ m/s}^2$ . Determine the resultant frictional force exerted by the road on the tires at this instant.

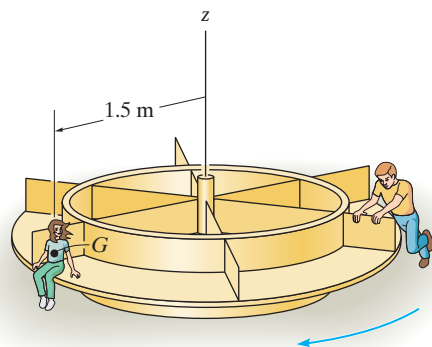


Prob. F13-12

## PROBLEMS

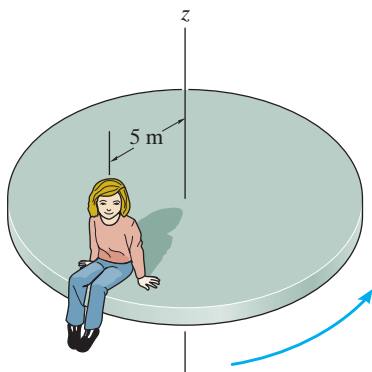
13

**\*13-52.** A girl having a mass of 25 kg sits at the edge of the merry-go-round so her center of mass  $G$  is at a distance of 1.5 m from the axis of rotation. If the angular motion of the platform is *slowly* increased so that the girl's tangential component of acceleration can be neglected, determine the maximum speed which she can have before she begins to slip off the merry-go-round. The coefficient of static friction between the girl and the merry-go-round is  $\mu_s = 0.3$ .



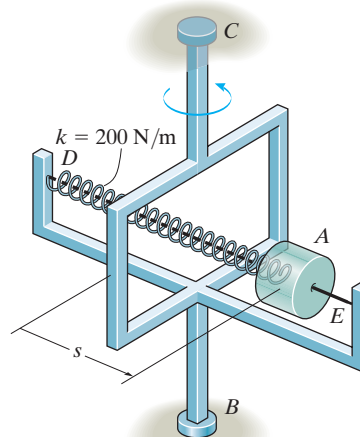
Prob. 13-52

**13-53.** A girl, having a mass of 15 kg, sits motionless relative to the surface of a horizontal platform at a distance of  $r = 5$  m from the platform's center. If the angular motion of the platform is *slowly* increased so that the girl's tangential component of acceleration can be neglected, determine the maximum speed which the girl will have before she begins to slip off the platform. The coefficient of static friction between the girl and the platform is  $\mu = 0.2$ .



Prob. 13-53

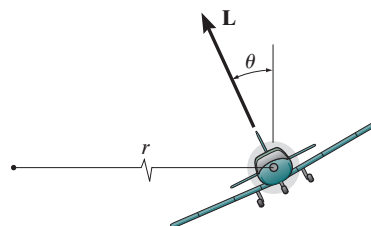
**13-54.** The collar  $A$ , having a mass of 0.75 kg, is attached to a spring having a stiffness of  $k = 200$  N/m. When rod  $BC$  rotates about the vertical axis, the collar slides outward along the smooth rod  $DE$ . If the spring is unstretched when  $s = 0$ , determine the constant speed of the collar in order that  $s = 100$  mm. Also, what is the normal force of the rod on the collar? Neglect the size of the collar.



Prob. 13-54

**13-55.** A 5-Mg airplane is flying at a constant speed of 350 km/h along a horizontal circular path of radius  $r = 3000$  m. Determine the uplift force  $\mathbf{L}$  acting on the airplane and the banking angle  $\theta$ . Neglect the size of the airplane.

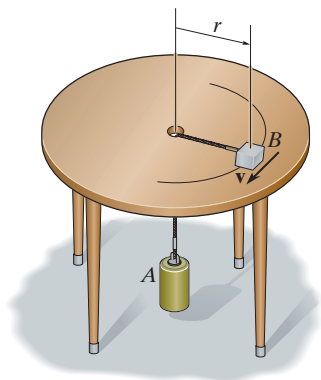
**\*13-56.** A 5-Mg airplane is flying at a constant speed of 350 km/h along a horizontal circular path. If the banking angle  $\theta = 15^\circ$ , determine the uplift force  $\mathbf{L}$  acting on the airplane and the radius  $r$  of the circular path. Neglect the size of the airplane.



Probs. 13-55/56

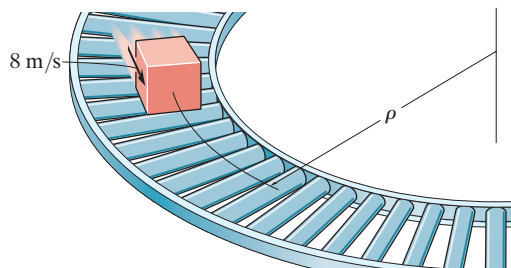
**13-57.** The 2-kg block  $B$  and 15-kg cylinder  $A$  are connected to a light cord that passes through a hole in the center of the smooth table. If the block is given a speed of  $v = 10$  m/s, determine the radius  $r$  of the circular path along which it travels.

**13-58.** The 2-kg block  $B$  and 15-kg cylinder  $A$  are connected to a light cord that passes through a hole in the center of the smooth table. If the block travels along a circular path of radius  $r = 1.5$  m, determine the speed of the block.



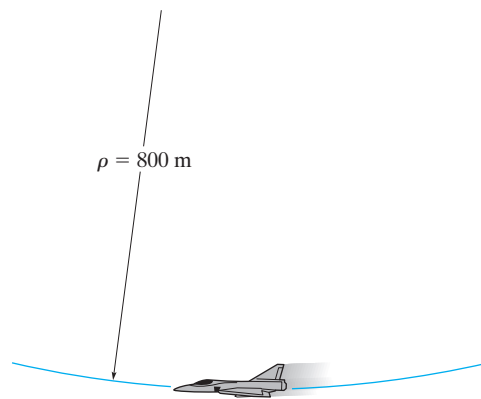
**Probs. 13-57/58**

**13-59.** Cartons having a mass of 5 kg are required to move along the assembly line at a constant speed of 8 m/s. Determine the smallest radius of curvature,  $\rho$ , for the conveyor so the cartons do not slip. The coefficients of static and kinetic friction between a carton and the conveyor are  $\mu_s = 0.7$  and  $\mu_k = 0.5$ , respectively.



**Prob. 13-59**

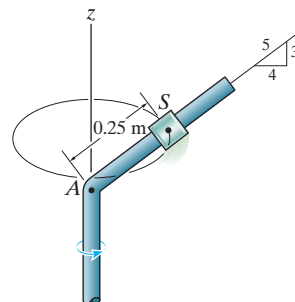
**\*13-60.** Determine the maximum constant speed at which the pilot can travel around the vertical curve having a radius of curvature  $\rho = 800$  m, so that he experiences a maximum acceleration  $a_n = 8g = 78.5$  m/s<sup>2</sup>. If he has a mass of 70 kg, determine the normal force he exerts on the seat of the airplane when the plane is traveling at this speed and is at its lowest point.



**Prob. 13-60**

**13-61.** The 2-kg spool  $S$  fits loosely on the inclined rod for which the coefficient of static friction is  $\mu_s = 0.2$ . If the spool is located 0.25 m from  $A$ , determine the minimum constant speed the spool can have so that it does not slip down the rod.

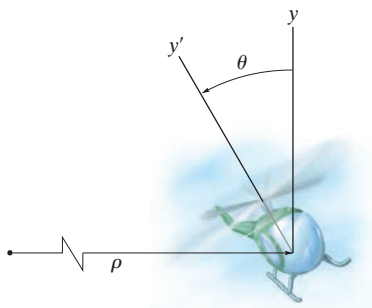
**13-62.** The 2-kg spool  $S$  fits loosely on the inclined rod for which the coefficient of static friction is  $\mu_s = 0.2$ . If the spool is located 0.25 m from  $A$ , determine the maximum constant speed the spool can have so that it does not slip up the rod.



**Probs. 13-61/62**

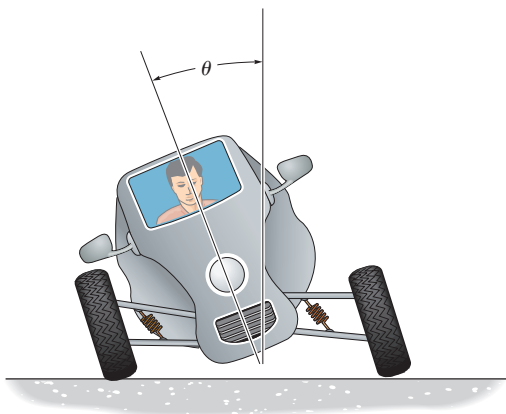
**13–63.** The 1.40-Mg helicopter is traveling at a constant speed of 40 m/s along the horizontal curved path while banking at  $\theta = 30^\circ$ . Determine the force acting normal to the blade, i.e., in the  $y'$  direction, and the radius of curvature of the path.

**13** **\*13–64.** The 1.40-Mg helicopter is traveling at a constant speed of 33 m/s along the horizontal curved path having a radius of curvature of  $\rho = 300$  m. Determine the force the blade exerts on the frame and the bank angle  $\theta$ .



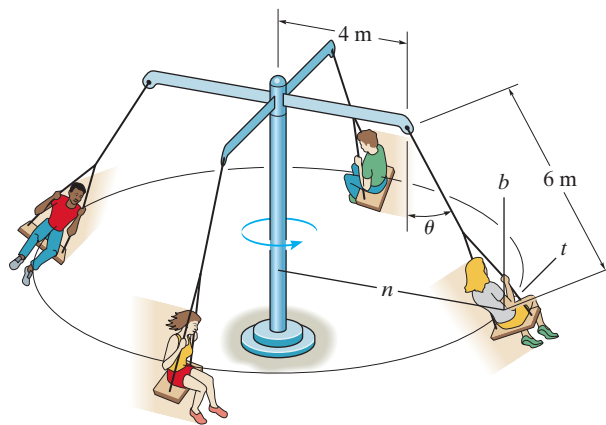
Probs. 13–63/64

**13–65.** The vehicle is designed to combine the feel of a motorcycle with the comfort and safety of an automobile. If the vehicle is traveling at a constant speed of 80 km/h along a circular curved road of radius 100 m, determine the tilt angle  $\theta$  of the vehicle so that only a normal force from the seat acts on the driver. Neglect the size of the driver.



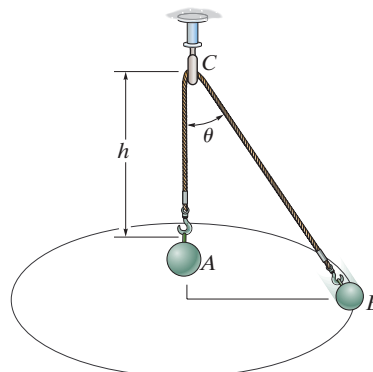
Prob. 13–65

**13–66.** Determine the constant speed of the passengers on the amusement-park ride if it is observed that the supporting cables are directed at  $\theta = 30^\circ$  from the vertical. Each chair including its passenger has a mass of 80 kg. Also, what are the components of force in the  $n$ ,  $t$ , and  $b$  directions which the chair exerts on a 50-kg passenger during the motion?



Prob. 13–66

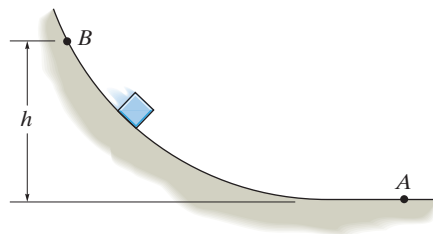
**13–67.** Bobs  $A$  and  $B$  of mass  $m_A$  and  $m_B$  ( $m_A > m_B$ ) are connected to an inextensible light string of length  $l$  that passes through the smooth ring at  $C$ . If bob  $B$  moves as a conical pendulum such that  $A$  is suspended a distance of  $h$  from  $C$ , determine the angle  $\theta$  and the speed of bob  $B$ . Neglect the size of both bobs.



Prob. 13–67

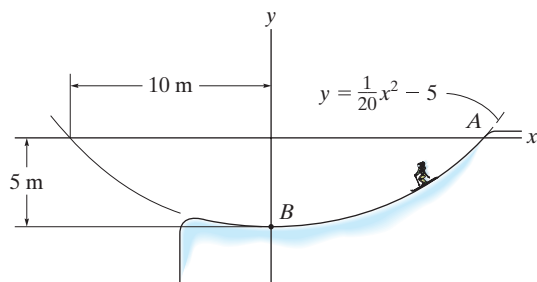


**\*13–68.** Prove that if the block is released from rest at point  $B$  of a smooth path of *arbitrary shape*, the speed it attains when it reaches point  $A$  is equal to the speed it attains when it falls freely through a distance  $h$ ; i.e.,  $v = \sqrt{2gh}$ .



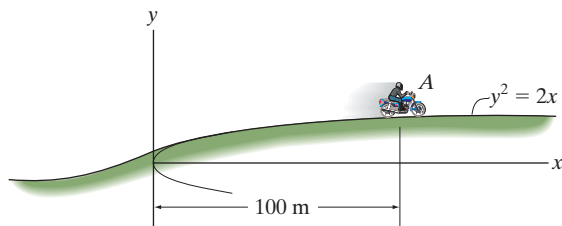
**Prob. 13–68**

**13–69.** The skier starts from rest at  $A(10 \text{ m}, 0)$  and descends the smooth slope, which may be approximated by a parabola. If she has a mass of 52 kg, determine the normal force the ground exerts on the skier at the instant she arrives at point  $B$ . Neglect the size of the skier.



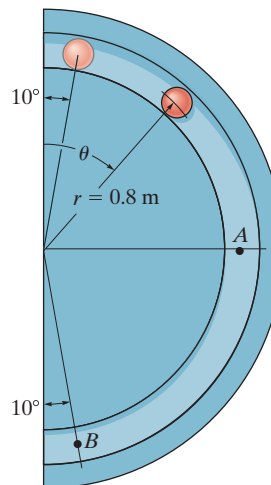
**Prob. 13–69**

**13–70.** The 800-kg motorbike travels with a constant speed of 80 km/h up the hill. Determine the normal force the surface exerts on its wheels when it reaches point  $A$ . Neglect its size.



**Prob. 13–70**

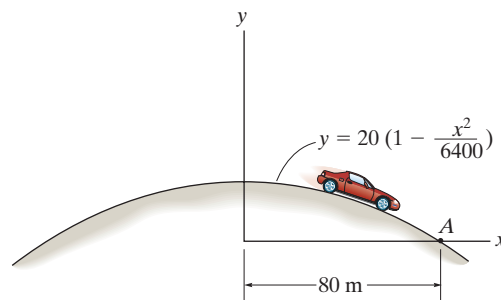
**13–71.** A ball having a mass 2 kg and negligible size moves within a smooth vertical circular slot. If it is released from rest when  $\theta = 10^\circ$ , determine the force of the slot on the ball when the ball arrives at points  $A$  and  $B$ .



**Prob. 13–71**

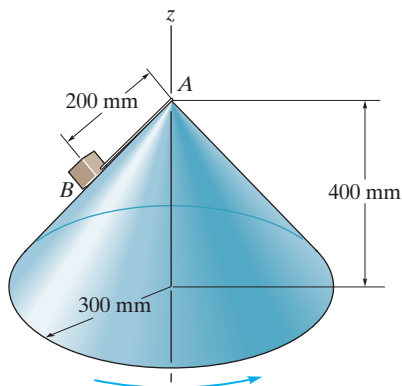
**\*13–72.** The 0.8-Mg car travels over the hill having the shape of a parabola. If the driver maintains a constant speed of 9 m/s, determine both the resultant normal force and the resultant frictional force that all the wheels of the car exert on the road at the instant it reaches point  $A$ . Neglect the size of the car.

**13–73.** The 0.8-Mg car travels over the hill having the shape of a parabola. When the car is at point  $A$ , it is traveling at 9 m/s and increasing its speed at  $3 \text{ m/s}^2$ . Determine both the resultant normal force and the resultant frictional force that all the wheels of the car exert on the road at this instant. Neglect the size of the car.



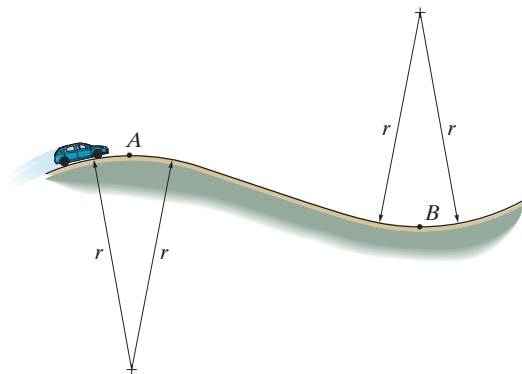
**Probs. 13–72/73**

**13-74.** The block  $B$ , having a mass of 0.2 kg, is attached to the vertex  $A$  of the right circular cone using a light cord. The cone is rotating at a constant angular rate about the  $z$  axis such that the block attains a speed of 0.5 m/s. At this speed, determine the tension in the cord and the reaction which the cone exerts on the block. Neglect the size of the block and the effect of friction.



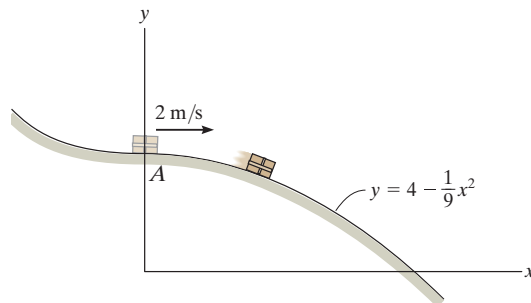
**Prob. 13-74**

**13-75.** Determine the maximum speed at which the car with mass  $m$  can pass over the top point  $A$  of the vertical curved road and still maintain contact with the road. If the car maintains this speed, what is the normal reaction the road exerts on the car when it passes the lowest point  $B$  on the road?



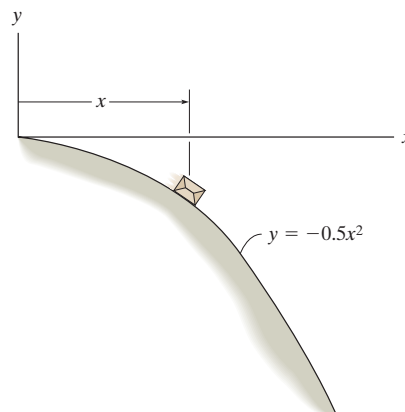
**Prob. 13-75**

**\*13-76.** The 35-kg box has a speed 2 m/s when it is at  $A$  on the smooth ramp. If the surface is in the shape of a parabola, determine the normal force on the box at the instant  $x = 3$  m. Also, what is the rate of increase in its speed at this instant?



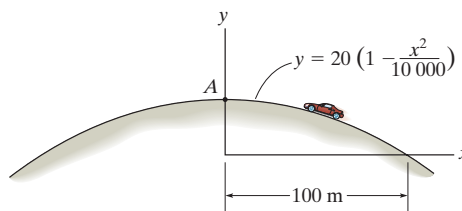
**Prob. 13-76**

**13-77.** The box has a mass  $m$  and slides down the smooth chute having the shape of a parabola. If it has an initial velocity of  $v_0$  at the origin, determine its velocity as a function of  $x$ . Also, what is the normal force on the box, and the tangential acceleration as a function of  $x$ ?



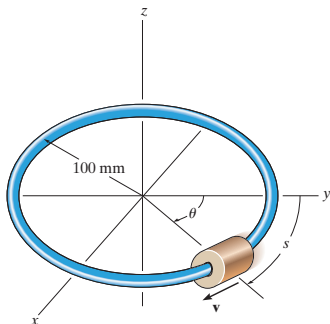
**Prob. 13-77**

**13-78.** Determine the maximum constant speed at which the 2-Mg car can travel over the crest of the hill at  $A$  without leaving the surface of the road. Neglect the size of the car in the calculation.



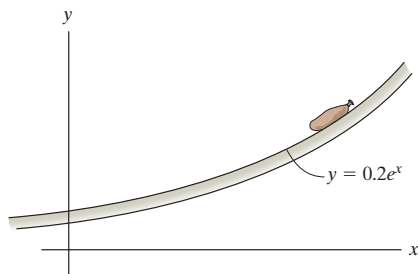
**Prob. 13-78**

**13-79.** A collar having a mass  $0.75\text{ kg}$  and negligible size slides over the surface of a horizontal circular rod for which the coefficient of kinetic friction is  $\mu_k = 0.3$ . If the collar is given a speed of  $4\text{ m/s}$  and then released at  $\theta = 0^\circ$ , determine how far,  $s$ , it slides on the rod before coming to rest.



**Prob. 13-79**

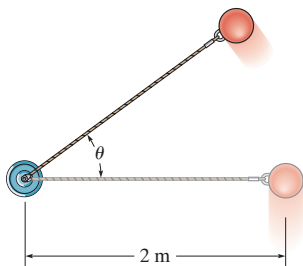
**\*13-80.** The  $8\text{-kg}$  sack slides down the smooth ramp. If it has a speed of  $1.5\text{ m/s}$  when  $y = 0.2\text{ m}$ , determine the normal reaction the ramp exerts on the sack and the rate of increase in the speed of sack at this instant.



**Prob. 13-80**

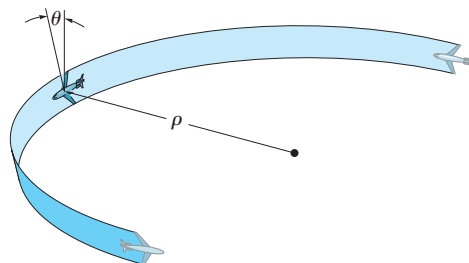
**13-81.** The  $2\text{-kg}$  pendulum bob moves in the vertical plane with a velocity of  $8\text{ m/s}$  when  $\theta = 0^\circ$ . Determine the initial tension in the cord and also at the instant the bob reaches  $\theta = 30^\circ$ . Neglect the size of the bob.

**13-82.** The  $2\text{-kg}$  pendulum bob moves in the vertical plane with a velocity of  $6\text{ m/s}$  when  $\theta = 0^\circ$ . Determine the angle  $\theta$  where the tension in the cord becomes zero.



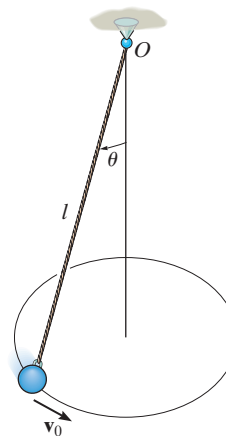
**Probs. 13-81/82**

**13-83.** The airplane, traveling at a constant speed of  $50\text{ m/s}$ , is executing a horizontal turn. If the plane is banked at  $\theta = 15^\circ$ , when the pilot experiences only a normal force on the seat of the plane, determine the radius of curvature  $\rho$  of the turn. Also, what is the normal force of the seat on the pilot if he has a mass of  $70\text{ kg}$ .



**Prob. 13-83**

**\*13-84.** The ball has a mass  $m$  and is attached to the cord of length  $l$ . The cord is tied at the top to a swivel and the ball is given a velocity  $\mathbf{v}_0$ . Show that the angle  $\theta$  which the cord makes with the vertical as the ball travels around the circular path must satisfy the equation  $\tan \theta \sin \theta = v_0^2 / gl$ . Neglect air resistance and the size of the ball.



**Prob. 13-84**

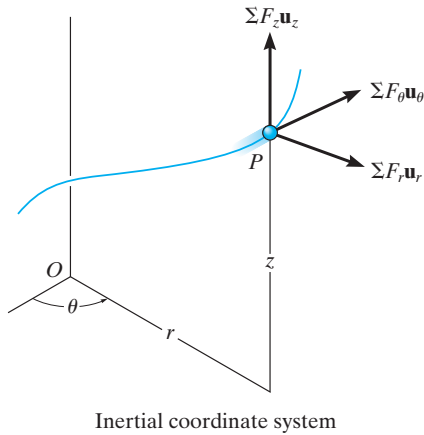


Fig. 13-16

## 13.6 Equations of Motion: Cylindrical Coordinates

When all the forces acting on a particle are resolved into cylindrical components, i.e., along the unit-vector directions  $\mathbf{u}_r$ ,  $\mathbf{u}_\theta$ ,  $\mathbf{u}_z$ , Fig. 13-16, the equation of motion can be expressed as

$$\Sigma \mathbf{F} = m\mathbf{a}$$

$$\Sigma F_r \mathbf{u}_r + \Sigma F_\theta \mathbf{u}_\theta + \Sigma F_z \mathbf{u}_z = m a_r \mathbf{u}_r + m a_\theta \mathbf{u}_\theta + m a_z \mathbf{u}_z$$

To satisfy this equation, we require

$$\begin{aligned} \Sigma F_r &= m a_r \\ \Sigma F_\theta &= m a_\theta \\ \Sigma F_z &= m a_z \end{aligned} \quad (13-8)$$

If the particle is constrained to move only in the  $r$ - $\theta$  plane, then only the first two of Eq. 13-8 are used to specify the motion.

**Tangential and Normal Forces.** The most straightforward type of problem involving cylindrical coordinates requires the determination of the resultant force components  $\Sigma F_r$ ,  $\Sigma F_\theta$ ,  $\Sigma F_z$  which cause a particle to move with a *known* acceleration. If, however, the particle's accelerated motion is not completely specified at the given instant, then some information regarding the directions or magnitudes of the forces acting on the particle must be known or calculated in order to solve Eq. 13-8. For example, the force  $\mathbf{P}$  causes the particle in Fig. 13-17a to move along a path  $r = f(\theta)$ . The *normal force*  $\mathbf{N}$  which the path exerts on the particle is always *perpendicular to the tangent of the path*, whereas the frictional force  $\mathbf{F}$  always acts along the tangent in the opposite direction of motion. The *directions* of  $\mathbf{N}$  and  $\mathbf{F}$  can be specified relative to the radial coordinate by using the angle  $\psi$ , Fig. 13-17b, which is defined between the *extended* radial line and the tangent to the curve.



Motion of the roller coaster along this spiral can be studied using cylindrical coordinates.

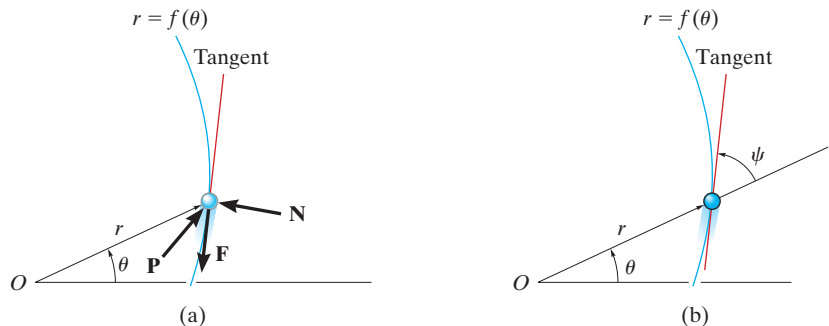
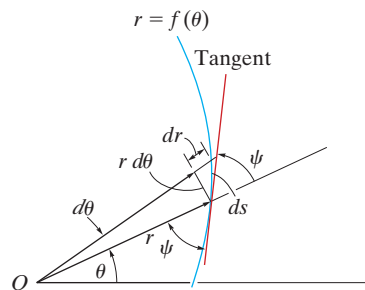


Fig. 13-17

This angle can be obtained by noting that when the particle is displaced a distance  $ds$  along the path, Fig. 13–17c, the component of displacement in the radial direction is  $dr$  and the component of displacement in the transverse direction is  $r d\theta$ . Since these two components are mutually perpendicular, the angle  $\psi$  can be determined from  $\tan \psi = r d\theta/dr$ , or

$$\tan \psi = \frac{r}{dr/d\theta} \quad (13-9)$$

If  $\psi$  is calculated as a positive quantity, it is measured from the *extended radial line* to the tangent in a counterclockwise sense or in the positive direction of  $\theta$ . If it is negative, it is measured in the opposite direction to positive  $\theta$ . For example, consider the cardioid  $r = a(1 + \cos \theta)$ , shown in Fig. 13–18. Because  $dr/d\theta = -a \sin \theta$ , then when  $\theta = 30^\circ$ ,  $\tan \psi = a(1 + \cos 30^\circ)/(-a \sin 30^\circ) = -3.732$ , or  $\psi = -75^\circ$ , measured clockwise, opposite to  $+\theta$  as shown in the figure.



(c)

Fig. 13–17

## Procedure for Analysis

Cylindrical or polar coordinates are a suitable choice for the analysis of a problem for which data regarding the angular motion of the radial line  $r$  are given, or in cases where the path can be conveniently expressed in terms of these coordinates. Once these coordinates have been established, the equations of motion can then be applied in order to relate the forces acting on the particle to its acceleration components. The method for doing this has been outlined in the procedure for analysis given in Sec. 13.4. The following is a summary of this procedure.

### Free-Body Diagram.

- Establish the  $r, \theta, z$  inertial coordinate system and draw the particle's free-body diagram.
- Assume that  $\mathbf{a}_r, \mathbf{a}_\theta, \mathbf{a}_z$  act in the positive directions of  $r, \theta, z$  if they are unknown.
- Identify all the unknowns in the problem.

### Equations of Motion.

- Apply the equations of motion, Eq. 13–8.

### Kinematics.

- Use the methods of Sec. 12.8 to determine  $r$  and the time derivatives  $\dot{r}, \ddot{r}, \dot{\theta}, \ddot{\theta}, \dot{z}$ , and then evaluate the acceleration components  $a_r = \ddot{r} - r\dot{\theta}^2, a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta}, a_z = \ddot{z}$ .
- If any of the acceleration components is computed as a negative quantity, it indicates that it acts in its negative coordinate direction.
- When taking the time derivatives of  $r = f(\theta)$ , it is very important to use the chain rule of calculus, which is discussed in Appendix C.

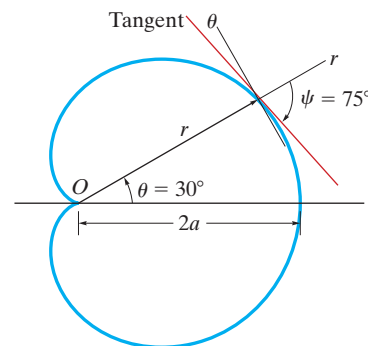
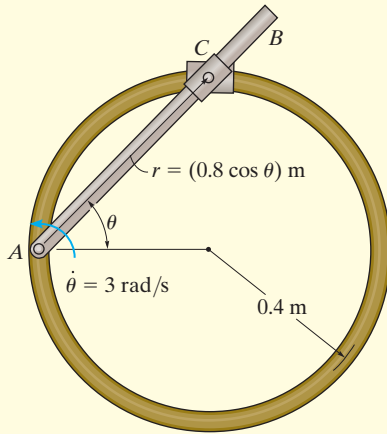
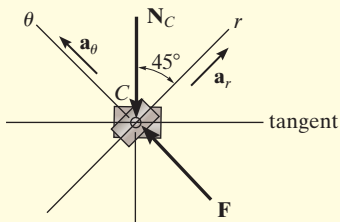


Fig. 13–18

## EXAMPLE 13.10



(a)



(b)

Fig. 13–19

The smooth 0.5-kg double-collar in Fig. 13–19a can freely slide on arm  $AB$  and the circular guide rod. If the arm rotates with a constant angular velocity of  $\dot{\theta} = 3 \text{ rad/s}$ , determine the force the arm exerts on the collar at the instant  $\theta = 45^\circ$ . Motion is in the horizontal plane.

## SOLUTION

**Free-Body Diagram.** The normal reaction  $N_C$  of the circular guide rod and the force  $\mathbf{F}$  of arm  $AB$  act on the collar in the plane of motion, Fig. 13–19b. Note that  $\mathbf{F}$  acts perpendicular to the axis of arm  $AB$ , that is, in the direction of the  $\theta$  axis, while  $N_C$  acts perpendicular to the tangent of the circular path at  $\theta = 45^\circ$ . The four unknowns are  $N_C$ ,  $F$ ,  $a_r$ ,  $a_\theta$ .

## Equations of Motion.

$$+\nearrow \Sigma F_r = ma_r: \quad -N_C \cos 45^\circ = (0.5 \text{ kg}) a_r \quad (1)$$

$$+\searrow \Sigma F_\theta = ma_\theta: \quad F - N_C \sin 45^\circ = (0.5 \text{ kg}) a_\theta \quad (2)$$

**Kinematics.** Using the chain rule (see Appendix C), the first and second time derivatives of  $r$  when  $\theta = 45^\circ$ ,  $\dot{\theta} = 3 \text{ rad/s}$ ,  $\ddot{\theta} = 0$ , are

$$r = 0.8 \cos \theta = 0.8 \cos 45^\circ = 0.5657 \text{ m}$$

$$\dot{r} = -0.8 \sin \theta \dot{\theta} = -0.8 \sin 45^\circ (3) = -1.6971 \text{ m/s}$$

$$\begin{aligned} \ddot{r} &= -0.8 [\sin \theta \ddot{\theta} + \cos \theta \dot{\theta}^2] \\ &= -0.8 [\sin 45^\circ (0) + \cos 45^\circ (3^2)] = -5.091 \text{ m/s}^2 \end{aligned}$$

We have

$$a_r = \ddot{r} - r\dot{\theta}^2 = -5.091 \text{ m/s}^2 - (0.5657 \text{ m})(3 \text{ rad/s})^2 = -10.18 \text{ m/s}^2$$

$$\begin{aligned} a_\theta &= r\ddot{\theta} + 2\dot{r}\dot{\theta} = (0.5657 \text{ m})(0) + 2(-1.6971 \text{ m/s})(3 \text{ rad/s}) \\ &= -10.18 \text{ m/s}^2 \end{aligned}$$

Substituting these results into Eqs. (1) and (2) and solving, we get

$$N_C = 7.20 \text{ N}$$

$$F = 0$$

*Ans.*

**EXAMPLE 13.11**

The smooth 2-kg cylinder  $C$  in Fig. 13–20a has a pin  $P$  through its center which passes through the slot in arm  $OA$ . If the arm is forced to rotate in the *vertical plane* at a constant rate  $\dot{\theta} = 0.5$  rad/s, determine the force that the arm exerts on the peg at the instant  $\theta = 60^\circ$ .

**SOLUTION**

Why is it a good idea to use polar coordinates to solve this problem?

**Free-Body Diagram.** The free-body diagram for the cylinder is shown in Fig. 13–20b. The force on the peg,  $\mathbf{F}_P$ , acts perpendicular to the slot in the arm. As usual,  $\mathbf{a}_r$  and  $\mathbf{a}_\theta$  are assumed to act in the directions of *positive*  $r$  and  $\theta$ , respectively. Identify the four unknowns.

**Equations of Motion.** Using the data in Fig. 13–20b, we have

$$+\swarrow \Sigma F_r = ma_r; \quad 19.62 \sin \theta - N_C \sin \theta = 2a_r \quad (1)$$

$$+\searrow \Sigma F_\theta = ma_\theta; \quad 19.62 \cos \theta + F_P - N_C \cos \theta = 2a_\theta \quad (2)$$

**Kinematics.** From Fig. 13–20a,  $r$  can be related to  $\theta$  by the equation

$$r = \frac{0.4}{\sin \theta} = 0.4 \csc \theta$$

Since  $d(\csc \theta) = -(\csc \theta \cot \theta) d\theta$  and  $d(\cot \theta) = -(\csc^2 \theta) d\theta$ , then  $r$  and the necessary time derivatives become

$$\dot{\theta} = 0.5 \quad r = 0.4 \csc \theta$$

$$\ddot{\theta} = 0 \quad \dot{r} = -0.4(\csc \theta \cot \theta)\dot{\theta}$$

$$= -0.2 \csc \theta \cot \theta$$

$$\ddot{r} = -0.2(-\csc \theta \cot \theta)(\dot{\theta}) \cot \theta - 0.2 \csc \theta(-\csc^2 \theta)\dot{\theta}$$

$$= 0.1 \csc \theta(\cot^2 \theta + \csc^2 \theta)$$

Evaluating these formulas at  $\theta = 60^\circ$ , we get

$$\dot{\theta} = 0.5 \quad r = 0.462$$

$$\ddot{\theta} = 0 \quad \dot{r} = -0.133$$

$$\ddot{r} = 0.192$$

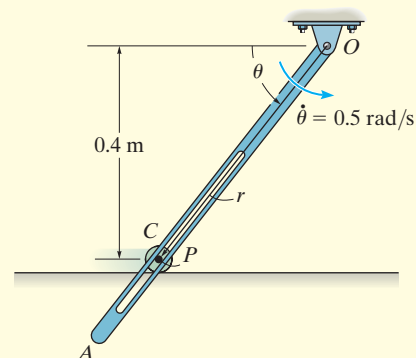
$$a_r = \ddot{r} - r\dot{\theta}^2 = 0.192 - 0.462(0.5)^2 = 0.0770$$

$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = 0 + 2(-0.133)(0.5) = -0.133$$

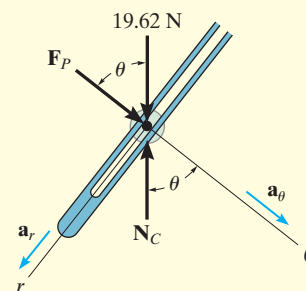
Substituting these results into Eqs. 1 and 2 with  $\theta = 60^\circ$  and solving yields

$$N_C = 19.4 \text{ N} \quad F_P = -0.356 \text{ N} \quad \text{Ans.}$$

The negative sign indicates that  $\mathbf{F}_P$  acts opposite to the direction shown in Fig. 13–20b.



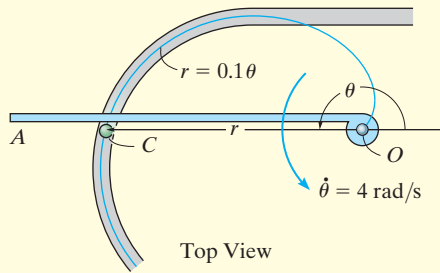
(a)



(b)

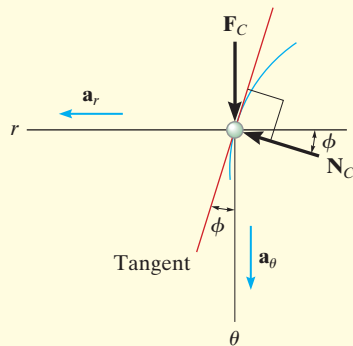
**Fig. 13–20**

## EXAMPLE 13.12



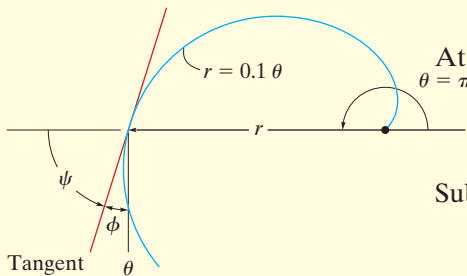
Top View

(a)



Tangent

(b)



Tangent

(c)

Fig. 13–21

A can  $C$ , having a mass of  $0.5 \text{ kg}$ , moves along a grooved horizontal slot shown in Fig. 13–21a. The slot is in the form of a spiral, which is defined by the equation  $r = (0.1\theta) \text{ m}$ , where  $\theta$  is in radians. If the arm  $OA$  rotates with a constant rate  $\dot{\theta} = 4 \text{ rad/s}$  in the horizontal plane, determine the force it exerts on the can at the instant  $\theta = \pi \text{ rad}$ . Neglect friction and the size of the can.

## SOLUTION

**Free-Body Diagram.** The driving force  $\mathbf{F}_C$  acts perpendicular to the arm  $OA$ , whereas the normal force of the wall of the slot on the can,  $\mathbf{N}_C$ , acts perpendicular to the tangent to the curve at  $\theta = \pi \text{ rad}$ , Fig. 13–21b. As usual,  $\mathbf{a}_r$  and  $\mathbf{a}_\theta$  are assumed to act in the *positive directions* of  $r$  and  $\theta$ , respectively. Since the path is specified, the angle  $\psi$  which the extended radial line  $r$  makes with the tangent, Fig. 13–21c, can be determined from Eq. 13–9. We have  $r = 0.1\theta$ , so that  $dr/d\theta = 0.1$ , and therefore

$$\tan \psi = \frac{r}{dr/d\theta} = \frac{0.1\theta}{0.1} = \theta$$

When  $\theta = \pi$ ,  $\psi = \tan^{-1}\pi = 72.3^\circ$ , so that  $\phi = 90^\circ - \psi = 17.7^\circ$ , as shown in Fig. 13–21c. Identify the four unknowns in Fig. 13–21b.

**Equations of Motion.** Using  $\phi = 17.7^\circ$  and the data shown in Fig. 13–21b, we have

$$+\uparrow \Sigma F_r = ma_r; \quad N_C \cos 17.7^\circ = 0.5a_r \quad (1)$$

$$+\downarrow \Sigma F_\theta = ma_\theta; \quad F_C - N_C \sin 17.7^\circ = 0.5a_\theta \quad (2)$$

**Kinematics.** The time derivatives of  $r$  and  $\theta$  are

$$\dot{\theta} = 4 \text{ rad/s}$$

$$r = 0.1\theta$$

$$\ddot{\theta} = 0$$

$$\dot{r} = 0.1\dot{\theta} = 0.1(4) = 0.4 \text{ m/s}$$

$$\ddot{r} = 0.1\ddot{\theta} = 0$$

At the instant  $\theta = \pi \text{ rad}$ ,

$$a_r = \ddot{r} - r\dot{\theta}^2 = 0 - 0.1(\pi)(4)^2 = -5.03 \text{ m/s}^2$$

$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = 0 + 2(0.4)(4) = 3.20 \text{ m/s}^2$$

Substituting these results into Eqs. 1 and 2 and solving yields

$$N_C = -2.64 \text{ N}$$

$$F_C = 0.800 \text{ N}$$

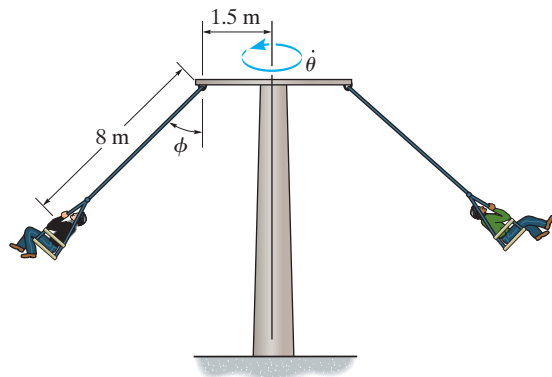
*Ans.*

What does the negative sign for  $N_C$  indicate?



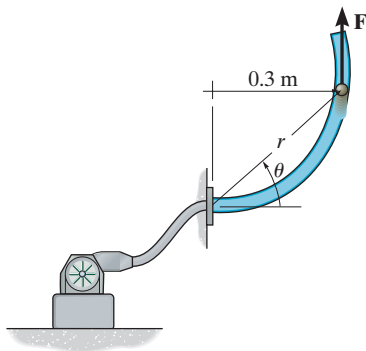
## FUNDAMENTAL PROBLEMS

**F13-13.** Determine the constant angular velocity  $\dot{\theta}$  of the vertical shaft of the amusement ride if  $\phi = 45^\circ$ . Neglect the mass of the cables and the size of the passengers.



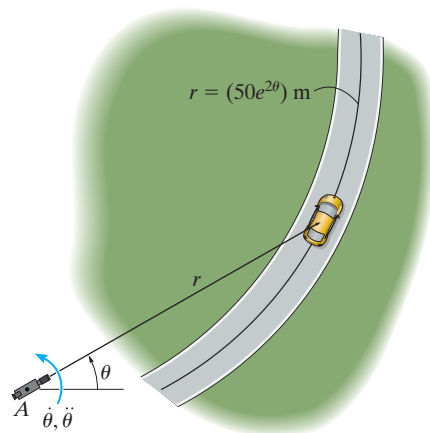
**Prob. F13-13**

**F13-14.** The 0.2-kg ball is blown through the smooth vertical circular tube whose shape is defined by  $r = (0.6 \sin \theta)$  m, where  $\theta$  is in radians. If  $\theta = (\pi t^2)$  rad, where  $t$  is in seconds, determine the magnitude of force  $\mathbf{F}$  exerted by the blower on the ball when  $t = 0.5$  s.



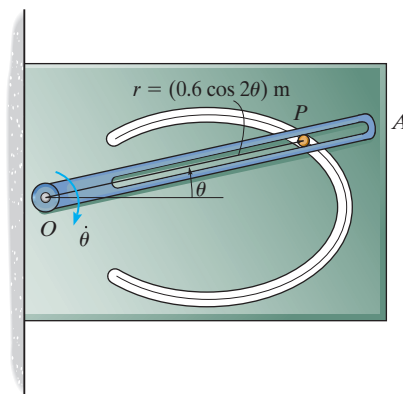
**Prob. F13-14**

**F13-15.** The 2-Mg car is traveling along the curved road described by  $r = (50e^{2\theta})$  m, where  $\theta$  is in radians. If a camera is located at  $A$  and it rotates with an angular velocity of  $\dot{\theta} = 0.05$  rad/s and an angular acceleration of  $\ddot{\theta} = 0.01$  rad/s<sup>2</sup> at the instant  $\theta = \frac{\pi}{6}$  rad, determine the resultant friction force developed between the tires and the road at this instant.



**Prob. F13-15**

**F13-16.** The 0.2-kg pin  $P$  is constrained to move in the smooth curved slot, which is defined by the lemniscate  $r = (0.6 \cos 2\theta)$  m. Its motion is controlled by the rotation of the slotted arm  $OA$ , which has a constant clockwise angular velocity of  $\dot{\theta} = -3$  rad/s. Determine the force arm  $OA$  exerts on the pin  $P$  when  $\theta = 0^\circ$ . Motion is in the vertical plane.



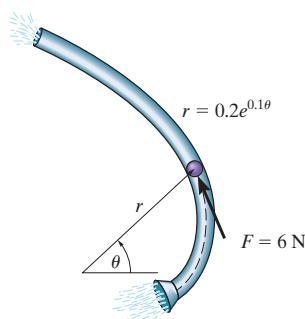
**Prob. F13-16**

## PROBLEMS

13

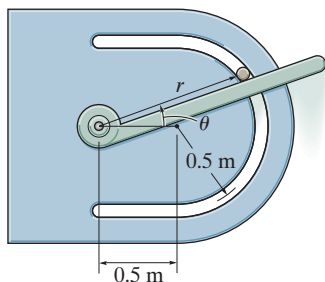
**13–85.** Using air pressure, the 0.5-kg ball is forced to move through the tube lying in the *horizontal plane* and having the shape of a logarithmic spiral. If the tangential force exerted on the ball due to the air is 6 N, determine the rate of increase in the ball's speed at the instant  $\theta = \pi/2$ . What direction does it act in?

**13–86.** Solve Prob. 13–85 if the tube lies in a *vertical plane*.



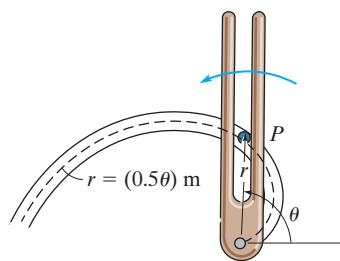
Probs. 13–85/86

**13–87.** The 0.75-kg smooth can is guided along the circular path using the arm guide. If the arm has an angular velocity  $\dot{\theta} = 2 \text{ rad/s}$  and an angular acceleration  $\ddot{\theta} = 0.4 \text{ rad/s}^2$  at the instant  $\theta = 30^\circ$ , determine the force of the guide on the can. Motion occurs in the *horizontal plane*.



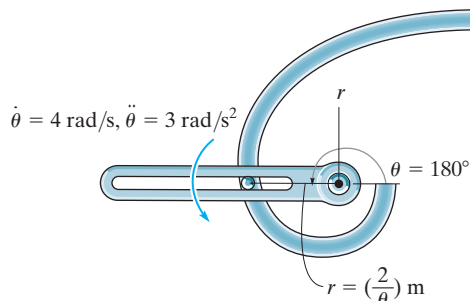
Prob. 13–87

**\*13–88.** Using a forked rod, a 0.5-kg smooth peg  $P$  is forced to move along the *vertical slotted path*  $r = (0.5\theta) \text{ m}$ , where  $\theta$  is in radians. If the angular position of the arm is  $\theta = (\frac{\pi}{8}t^2) \text{ rad}$ , where  $t$  is in seconds, determine the force of the rod on the peg and the normal force of the slot on the peg at the instant  $t = 2 \text{ s}$ . The peg is in contact with only *one edge* of the rod and slot at any instant.



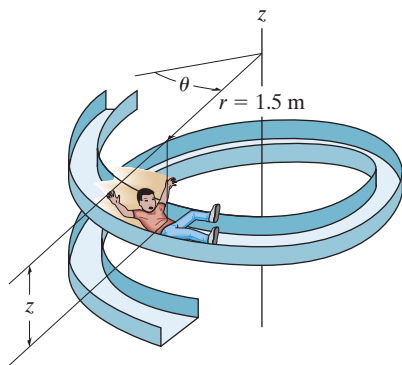
Prob. 13–88

**13–89.** The arm is rotating at a rate of  $\dot{\theta} = 4 \text{ rad/s}$  when  $\ddot{\theta} = 3 \text{ rad/s}^2$  and  $\theta = 180^\circ$ . Determine the force it must exert on the 0.5-kg smooth cylinder if it is confined to move along the slotted path. Motion occurs in the *horizontal plane*.



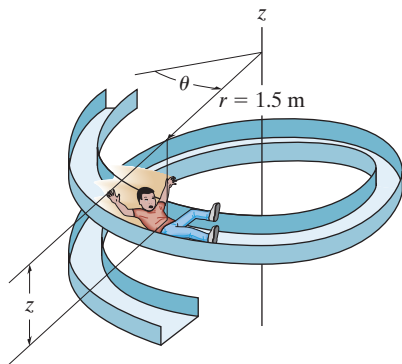
Prob. 13–89

**13–90.** The boy of mass 40 kg is sliding down the spiral slide at a constant speed such that his position, measured from the top of the chute, has components  $r = 1.5$  m,  $\theta = (0.7t)$  rad, and  $z = (-0.5t)$  m, where  $t$  is in seconds. Determine the components of force  $\mathbf{F}_r$ ,  $\mathbf{F}_\theta$ , and  $\mathbf{F}_z$  which the slide exerts on him at the instant  $t = 2$  s. Neglect the size of the boy.



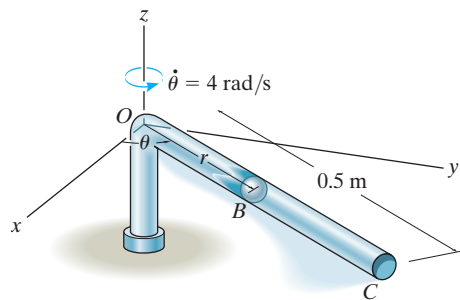
**Prob. 13–90**

**13–91.** The 40-kg boy is sliding down the smooth spiral slide such that  $\dot{z} = -2$  m/s and his speed is 2 m/s. Determine the  $r$ ,  $\theta$ ,  $z$  components of force the slide exerts on him at this instant. Neglect the size of the boy.



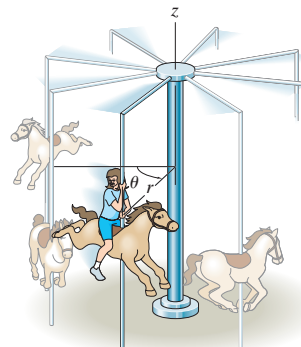
**Prob. 13–91**

**\*13–92.** The tube rotates in the horizontal plane at a constant rate of  $\dot{\theta} = 4$  rad/s. If a 0.2-kg ball  $B$  starts at the origin  $O$  with an initial radial velocity  $\dot{r} = 1.5$  m/s and moves outward through the tube, determine the radial and transverse components of the ball's velocity at the instant it leaves the outer end at  $C$ ,  $r = 0.5$  m. *Hint:* Show that the equation of motion in the  $r$  direction is  $\ddot{r} - 16r = 0$ . The solution is of the form  $r = Ae^{-4t} + Be^{4t}$ . Evaluate the integration constants  $A$  and  $B$ , and determine the time  $t$  when  $r = 0.5$  m. Proceed to obtain  $v_r$  and  $v_\theta$ .



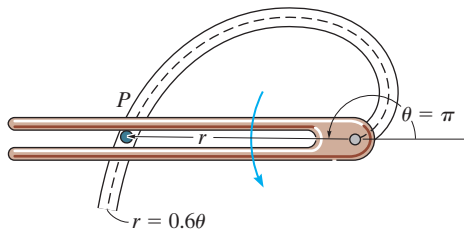
**Prob. 13–92**

**13–93.** The girl has a mass of 50 kg. She is seated on the horse of the merry-go-round which undergoes constant rotational motion  $\dot{\theta} = 1.5$  rad/s. If the path of the horse is defined by  $r = 4$  m,  $z = (0.5 \sin \theta)$  m, determine the maximum and minimum force  $F_z$  the horse exerts on her during the motion.



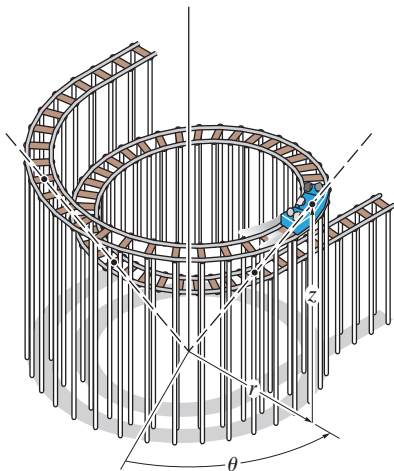
**Prob. 13–93**

**13-94.** Using a forked rod, a smooth cylinder  $P$ , having a mass of  $0.4 \text{ kg}$ , is forced to move along the *vertical slotted* path  $r = (0.6\theta) \text{ m}$ , where  $\theta$  is in radians. If the cylinder has a constant speed of  $v_C = 2 \text{ m/s}$ , determine the force of the rod and the normal force of the slot on the cylinder at the instant  $\theta = \pi \text{ rad}$ . Assume the cylinder is in contact with only *one* edge of the rod and slot at any instant. *Hint:* To obtain the time derivatives necessary to compute the cylinder's acceleration components  $a_r$  and  $a_\theta$ , take the first and second time derivatives of  $r = 0.6\theta$ . Then, for further information, use Eq. 12-26 to determine  $\dot{\theta}$ . Also, take the time derivative of Eq. 12-26, noting that  $\dot{v} = 0$  to determine  $\ddot{\theta}$ .



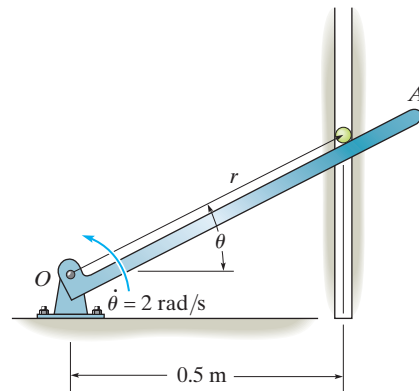
**Prob. 13-94**

**13-95.** A car of a roller coaster travels along a track which for a short distance is defined by a conical spiral,  $r = \frac{3}{4}z$ ,  $\theta = -1.5z$ , where  $r$  and  $z$  are in meters and  $\theta$  in radians. If the angular motion  $\dot{\theta} = 1 \text{ rad/s}$  is always maintained, determine the  $r$ ,  $\theta$ ,  $z$  components of reaction exerted on the car by the track at the instant  $z = 6 \text{ m}$ . The car and passengers have a total mass of  $200 \text{ kg}$ .



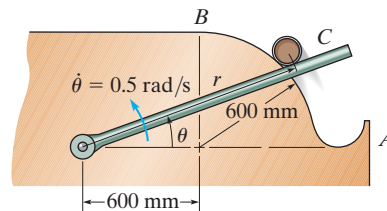
**Prob. 13-95**

**\*13-96.** The particle has a mass of  $0.5 \text{ kg}$  and is confined to move along the smooth vertical slot due to the rotation of the arm  $OA$ . Determine the force of the rod on the particle and the normal force of the slot on the particle when  $\theta = 30^\circ$ . The rod is rotating with a constant angular velocity  $\dot{\theta} = 2 \text{ rad/s}$ . Assume the particle contacts only one side of the slot at any instant.



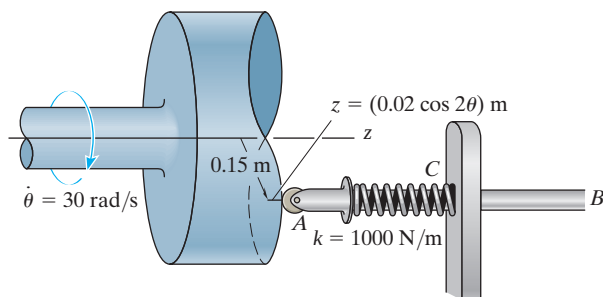
**Prob. 13-96**

**13-97.** A smooth can  $C$ , having a mass of  $3 \text{ kg}$ , is lifted from a feed at  $A$  to a ramp at  $B$  by a rotating rod. If the rod maintains a constant angular velocity of  $\dot{\theta} = 0.5 \text{ rad/s}$ , determine the force which the rod exerts on the can at the instant  $\theta = 30^\circ$ . Neglect the effects of friction in the calculation and the size of the can so that  $r = (1.2 \cos \theta) \text{ m}$ . The ramp from  $A$  to  $B$  is circular, having a radius of  $600 \text{ mm}$ .



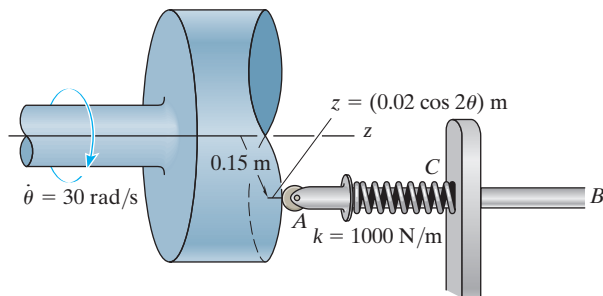
**Prob. 13-97**

**13–98.** The spring-held follower  $AB$  has a mass of  $0.5 \text{ kg}$  and moves back and forth as its end rolls on the contoured surface of the cam, where  $r = 0.15 \text{ m}$  and  $z = (0.02 \cos 2\theta) \text{ m}$ . If the cam is rotating at a constant rate of  $30 \text{ rad/s}$ , determine the force component  $F_z$  at the end  $A$  of the follower when  $\theta = 30^\circ$ . The spring is uncompressed when  $\theta = 90^\circ$ . Neglect friction at the bearing  $C$ .



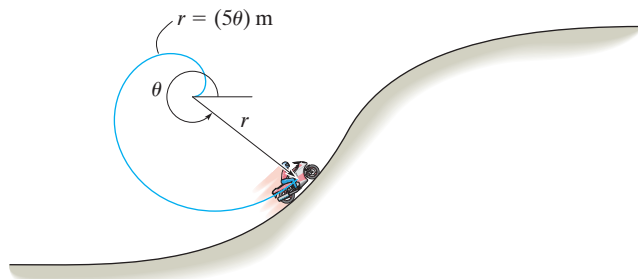
**Prob. 13–98**

**13–99.** The spring-held follower  $AB$  has a mass of  $0.5 \text{ kg}$  and moves back and forth as its end rolls on the contoured surface of the cam, where  $r = 0.15 \text{ m}$  and  $z = (0.02 \cos 2\theta) \text{ m}$ . If the cam is rotating at a constant rate of  $30 \text{ rad/s}$ , determine the maximum and minimum force components  $F_z$  the follower exerts on the cam if the spring is uncompressed when  $\theta = 90^\circ$ .



**Prob. 13–99**

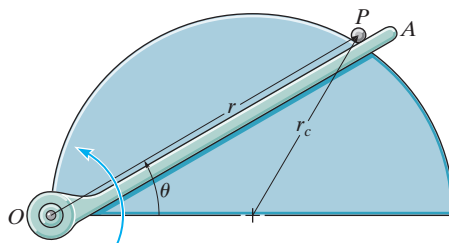
**\*13–100.** Determine the normal and frictional driving forces that the partial spiral track exerts on the  $200\text{-kg}$  motorcycle at the instant  $\theta = \frac{5}{3}\pi \text{ rad}$ ,  $\dot{\theta} = 0.4 \text{ rad/s}$ ,  $\ddot{\theta} = 0.8 \text{ rad/s}^2$ . Neglect the size of the motorcycle.



**Prob. 13–100**

**13–101.** The  $0.5\text{-kg}$  ball is guided along the vertical circular path  $r = 2r_c \cos \theta$  using the arm  $OA$ . If the arm has an angular velocity  $\dot{\theta} = 0.4 \text{ rad/s}$  and an angular acceleration  $\ddot{\theta} = 0.8 \text{ rad/s}^2$  at the instant  $\theta = 30^\circ$ , determine the force of the arm on the ball. Neglect friction and the size of the ball. Set  $r_c = 0.4 \text{ m}$ .

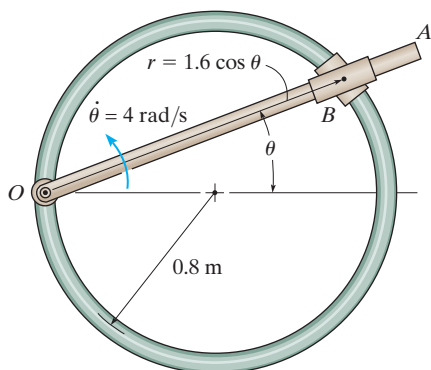
**13–102.** The ball of mass  $m$  is guided along the vertical circular path  $r = 2r_c \cos \theta$  using the arm  $OA$ . If the arm has a constant angular velocity  $\dot{\theta}_0$ , determine the angle  $\theta \leq 45^\circ$  at which the ball starts to leave the surface of the semicylinder. Neglect friction and the size of the ball.



**Probs. 13–101/102**

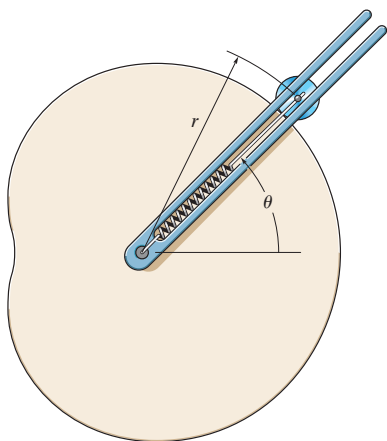
**13–103.** Rod  $OA$  rotates counterclockwise at a constant angular rate  $\dot{\theta} = 4 \text{ rad/s}$ . The double collar  $B$  is pin-connected together such that one collar slides over the rotating rod and the other collar slides over the circular rod described by the equation  $r = (1.6 \cos \theta) \text{ m}$ . If *both* collars have a mass of  $0.5 \text{ kg}$ , determine the force which the circular rod exerts on one of the collars and the force that  $OA$  exerts on the other collar at the instant  $\theta = 45^\circ$ . Motion is in the horizontal plane.

**\*13–104.** Solve Prob. 13–103 if motion is in the vertical plane.



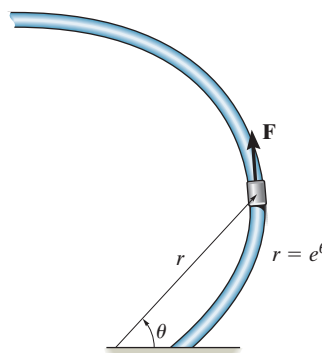
**Probs. 13–103/104**

**13–105.** The smooth surface of the vertical cam is defined in part by the curve  $r = (0.2 \cos \theta + 0.3) \text{ m}$ . The forked rod is rotating with an angular acceleration of  $\ddot{\theta} = 2 \text{ rad/s}^2$ , and when  $\theta = 45^\circ$  the angular velocity is  $\dot{\theta} = 6 \text{ rad/s}$ . Determine the force the cam and the rod exert on the  $2\text{-kg}$  roller at this instant. The attached spring has a stiffness  $k = 100 \text{ N/m}$  and an unstretched length of  $0.1 \text{ m}$ .



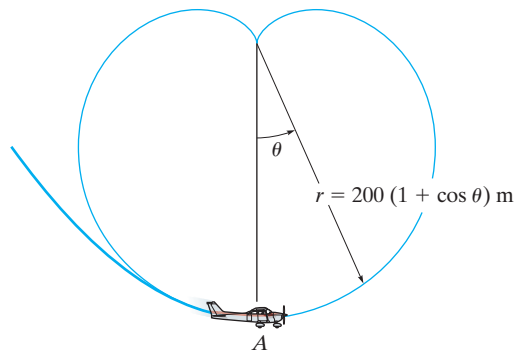
**Prob. 13–105**

**13–106.** The collar has a mass of  $2 \text{ kg}$  and travels along the smooth horizontal rod defined by the equiangular spiral  $r = (e^\theta) \text{ m}$ , where  $\theta$  is in radians. Determine the tangential force  $F$  and the normal force  $N$  acting on the collar when  $\theta = 45^\circ$ , if the force  $F$  maintains a constant angular motion  $\dot{\theta} = 2 \text{ rad/s}$ .



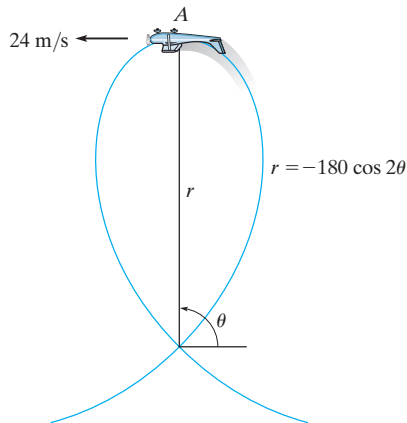
**Prob. 13–106**

**13–107.** The pilot of the airplane executes a vertical loop which in part follows the path of a cardioid,  $r = 200(1 + \cos \theta) \text{ m}$ , where  $\theta$  is in radians. If his speed at  $A$  is a constant  $v_p = 85 \text{ m/s}$ , determine the vertical reaction the seat of the plane exerts on the pilot when the plane is at  $A$ . He has a mass of  $80 \text{ kg}$ . *Hint:* To determine the time derivatives necessary to calculate the acceleration components  $a_r$  and  $a_\theta$ , take the first and second time derivatives of  $r = 200(1 + \cos \theta)$ . Then, for further information, use Eq. 12–26 to determine  $\dot{\theta}$ .



**Prob. 13–107**

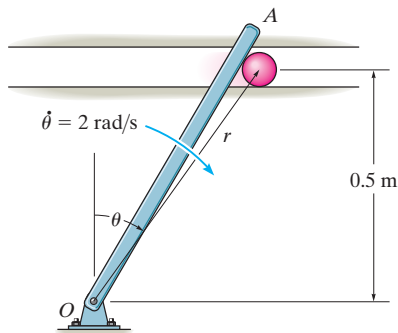
**\*13–108.** The pilot of an airplane executes a vertical loop which in part follows the path of a “four-leaved rose,”  $r = (-180 \cos 2\theta)$  m, where  $\theta$  is in radians. If his speed at  $A$  is a constant  $v_p = 24$  m/s, determine the vertical reaction the seat of the plane exerts on the pilot when the plane is at  $A$ . He weighs 650 N. *Hint:* To determine the time derivatives necessary to compute the acceleration components  $a_r$  and  $a_\theta$ , take the first and second time derivatives of  $r = -180(1 + \cos \theta)$ . Then, for further information, use Eq. 12–26 to determine  $\dot{\theta}$ . Also, take the time derivative of Eq. 12–26, noting that  $\dot{v}_C = 0$  to determine  $\ddot{\theta}$ .



**Prob. 13–108**

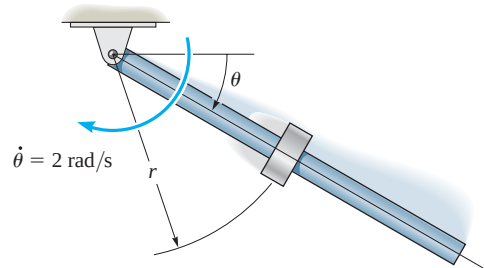
**13–109.** The particle has a mass of 0.5 kg and is confined to move along the smooth horizontal slot due to the rotation of the arm  $OA$ . Determine the force of the rod on the particle and the normal force of the slot on the particle when  $\theta = 30^\circ$ . The rod is rotating with a constant angular velocity  $\dot{\theta} = 2$  rad/s. Assume the particle contacts only one side of the slot at any instant.

**13–110.** Solve Prob. 13–109 if the arm has an angular acceleration of  $\ddot{\theta} = 3$  rad/s<sup>2</sup> when  $\dot{\theta} = 2$  rad/s at  $\theta = 30^\circ$ .



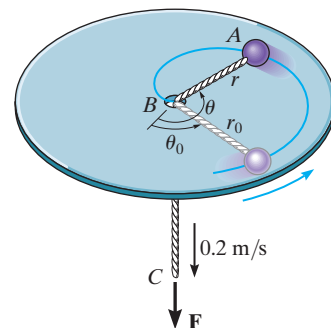
**Probs. 13–109/110**

**13–111.** A 0.2-kg spool slides down along a smooth rod. If the rod has a constant angular rate of rotation  $\dot{\theta} = 2$  rad/s in the vertical plane, show that the equations of motion for the spool are  $\ddot{r} - 4r - 9.81 \sin \theta = 0$  and  $0.8\ddot{r} + N_s - 1.962 \cos \theta = 0$ , where  $N_s$  is the magnitude of the normal force of the rod on the spool. Using the methods of differential equations, it can be shown that the solution of the first of these equations is  $r = C_1 e^{-2t} + C_2 e^{2t} - (9.81/8) \sin 2t$ . If  $r$ ,  $\dot{r}$ , and  $\theta$  are zero when  $t = 0$ , evaluate the constants  $C_1$  and  $C_2$  to determine  $r$  at the instant  $\theta = \pi/4$  rad.



**Prob. 13–111**

**\*13–112.** The ball has a mass of 2 kg and a negligible size. It is originally traveling around the horizontal circular path of radius  $r_0 = 0.5$  m such that the angular rate of rotation is  $\dot{\theta}_0 = 1$  rad/s. If the attached cord  $ABC$  is drawn down through the hole at a constant speed of 0.2 m/s, determine the tension the cord exerts on the ball at the instant  $r = 0.25$  m. Also, compute the angular velocity of the ball at this instant. Neglect the effects of friction between the ball and horizontal plane. *Hint:* First show that the equation of motion in the  $\theta$  direction yields  $a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = (1/r)(d(r^2\dot{\theta})/dt) = 0$ . When integrated,  $r^2\dot{\theta} = c$ , where the constant  $c$  is determined from the problem data.



**Prob. 13–112**

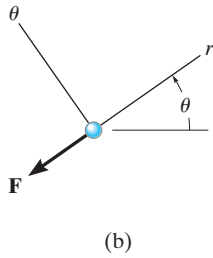
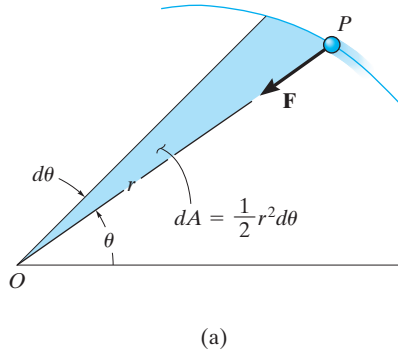


Fig. 13-22

## \*13.7 Central-Force Motion and Space Mechanics

If a particle is moving only under the influence of a force having a line of action which is always directed toward a fixed point, the motion is called *central-force motion*. This type of motion is commonly caused by electrostatic and gravitational forces.

In order to analyze the motion, we will consider the particle  $P$  shown in Fig. 13-22a, which has a mass  $m$  and is acted upon only by the central force  $\mathbf{F}$ . The free-body diagram for the particle is shown in Fig. 13-22b. Using polar coordinates  $(r, \theta)$ , the equations of motion, Eq. 13-8, become

$$\begin{aligned}\Sigma F_r &= ma_r; \\ -F &= m \left[ \frac{d^2 r}{dt^2} - r \left( \frac{d\theta}{dt} \right)^2 \right]\end{aligned}\quad (13-10)$$

$$\Sigma F_\theta = ma_\theta; \quad 0 = m \left( r \frac{d^2 \theta}{dt^2} + 2 \frac{dr}{dt} \frac{d\theta}{dt} \right)$$

The second of these equations may be written in the form

$$\frac{1}{r} \left[ \frac{d}{dt} \left( r^2 \frac{d\theta}{dt} \right) \right] = 0$$

so that integrating yields

$$r^2 \frac{d\theta}{dt} = h \quad (13-11)$$

Here  $h$  is the constant of integration.

From Fig. 13-22a notice that the shaded area described by the radius  $r$ , as  $r$  moves through an angle  $d\theta$ , is  $dA = \frac{1}{2} r^2 d\theta$ . If the *areal velocity* is defined as

$$\frac{dA}{dt} = \frac{1}{2} r^2 \frac{d\theta}{dt} = \frac{h}{2} \quad (13-12)$$

then it is seen that the areal velocity for a particle subjected to central-force motion is *constant*. In other words, the particle will sweep out equal segments of area per unit of time as it travels along the path. To obtain the *path of motion*,  $r = f(\theta)$ , the independent variable  $t$  must be eliminated from Eq. 13-10. Using the chain rule of calculus and Eq. 13-11, the time derivatives of Eq. 13-10 may be replaced by

$$\frac{dr}{dt} = \frac{dr}{d\theta} \frac{d\theta}{dt} = \frac{h}{r^2} \frac{dr}{d\theta}$$

$$\frac{d^2 r}{dt^2} = \frac{d}{dt} \left( \frac{h}{r^2} \frac{dr}{d\theta} \right) = \frac{d}{d\theta} \left( \frac{h}{r^2} \frac{dr}{d\theta} \right) \frac{d\theta}{dt} = \left[ \frac{d}{d\theta} \left( \frac{h}{r^2} \frac{dr}{d\theta} \right) \right] \frac{h}{r^2}$$



Substituting a new dependent variable (xi)  $\xi = 1/r$  into the second equation, we have

$$\frac{d^2 r}{dt^2} = -h^2 \xi^2 \frac{d^2 \xi}{d\theta^2}$$

Also, the square of Eq. 13-11 becomes

$$\left(\frac{d\theta}{dt}\right)^2 = h^2 \xi^4$$

Substituting these two equations into the first of Eq. 13-10 yields

$$-h^2 \xi^2 \frac{d^2 \xi}{d\theta^2} - h^2 \xi^3 = -\frac{F}{m}$$

or

$$\frac{d^2 \xi}{d\theta^2} + \xi = \frac{F}{mh^2 \xi^2} \quad (13-13)$$

This differential equation defines the path over which the particle travels when it is subjected to the central force  $\mathbf{F}$ .\*

For application, the force of gravitational attraction will be considered. Some common examples of central-force systems which depend on gravitation include the motion of the moon and artificial satellites about the earth, and the motion of the planets about the sun. As a typical problem in space mechanics, consider the trajectory of a space satellite or space vehicle launched into free-flight orbit with an initial velocity  $\mathbf{v}_0$ , Fig. 13-23. It will be assumed that this velocity is initially *parallel* to the tangent at the surface of the earth, as shown in the figure.† Just after the satellite is released into free flight, the only force acting on it is the gravitational force of the earth. (Gravitational attractions involving other bodies such as the moon or sun will be neglected, since for orbits close to the earth their effect is small in comparison with the earth's gravitation.) According to Newton's law of gravitation, force  $\mathbf{F}$  will always act between the mass centers of the earth and the satellite, Fig. 13-23. From Eq. 13-1, this force of attraction has a magnitude of

$$F = G \frac{M_e m}{r^2}$$

where  $M_e$  and  $m$  represent the mass of the earth and the satellite, respectively,  $G$  is the gravitational constant, and  $r$  is the distance between



This satellite is subjected to a central force and its orbital motion can be closely predicted using the equations developed in this section. (UniversalImagesGroup/Getty Images)

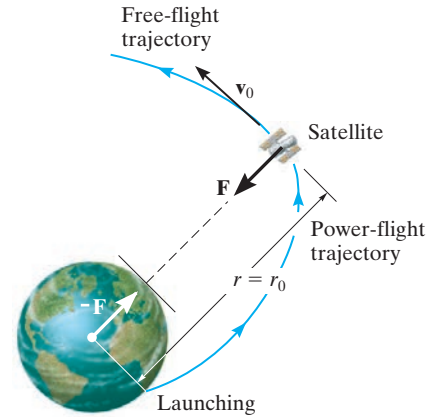


Fig. 13-23

\*In the derivation,  $\mathbf{F}$  is considered positive when it is directed toward point  $O$ . If  $\mathbf{F}$  is oppositely directed, the right side of Eq. 13-13 should be negative.

†The case where  $\mathbf{v}_0$  acts at some initial angle  $\theta$  to the tangent is best described using the conservation of angular momentum.

the mass centers. To obtain the orbital path, we set  $\xi = 1/r$  in the foregoing equation and substitute the result into Eq. 13-13. We obtain

$$\frac{d^2\xi}{d\theta^2} + \xi = \frac{GM_e}{h^2} \quad (13-14)$$

This second-order differential equation has constant coefficients and is nonhomogeneous. The solution is the sum of the complementary and particular solutions given by

$$\xi = \frac{1}{r} = C \cos(\theta - \phi) + \frac{GM_e}{h^2} \quad (13-15)$$

This equation represents the *free-flight trajectory* of the satellite. It is the equation of a conic section expressed in terms of polar coordinates.

A geometric interpretation of Eq. 13-15 requires knowledge of the equation for a conic section. As shown in Fig. 13-24, a conic section is defined as the locus of a point  $P$  that moves in such a way that the ratio of its distance to a *focus*, or fixed point  $F$ , to its perpendicular distance to a fixed line  $DD$  called the *directrix*, is constant. This constant ratio will be denoted as  $e$  and is called the *eccentricity*. By definition

$$e = \frac{FP}{PA}$$

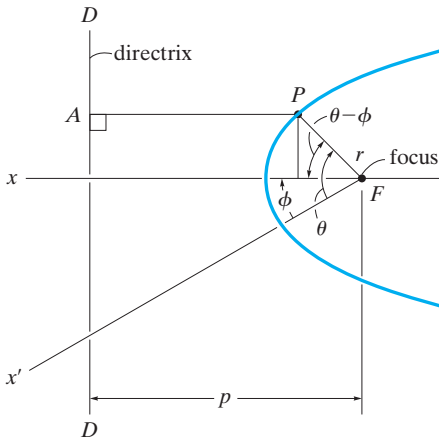


Fig. 13-24

From Fig. 13-24,

$$FP = r = e(PA) = e[p - r \cos(\theta - \phi)]$$

or

$$\frac{1}{r} = \frac{1}{p} \cos(\theta - \phi) + \frac{1}{ep}$$

Comparing this equation with Eq. 13-15, it is seen that the fixed distance from the focus to the directrix is

$$p = \frac{1}{C} \quad (13-16)$$

And the eccentricity of the conic section for the trajectory is

$$e = \frac{Ch^2}{GM_e} \quad (13-17)$$

Provided the polar angle  $\theta$  is measured from the  $x$  axis (an axis of symmetry since it is perpendicular to the directrix), the angle  $\phi$  is zero, Fig. 13-24, and therefore Eq. 13-15 reduces to

$$\frac{1}{r} = C \cos \theta + \frac{GM_e}{h^2} \quad (13-18)$$

The constants  $h$  and  $C$  are determined from the data obtained for the position and velocity of the satellite at the end of the *power-flight trajectory*. For example, if the initial height or distance to the space vehicle is  $r_0$ , measured from the center of the earth, and its initial speed is  $v_0$  at the beginning of its free flight, Fig. 13-25, then the constant  $h$  may be obtained from Eq. 13-11. When  $\theta = \phi = 0^\circ$ , the velocity  $\mathbf{v}_0$  has no radial component; therefore, from Eq. 12-25,  $v_0 = r_0(d\theta/dt)$ , so that

$$h = r_0^2 \frac{d\theta}{dt}$$

or

$$h = r_0 v_0 \quad (13-19)$$

To determine  $C$ , use Eq. 13-18 with  $\theta = 0^\circ$ ,  $r = r_0$ , and substitute Eq. 13-19 for  $h$ :

$$C = \frac{1}{r_0} \left( 1 - \frac{GM_e}{r_0 v_0^2} \right) \quad (13-20)$$

The equation for the free-flight trajectory therefore becomes

$$\frac{1}{r} = \frac{1}{r_0} \left( 1 - \frac{GM_e}{r_0 v_0^2} \right) \cos \theta + \frac{GM_e}{r_0^2 v_0^2} \quad (13-21)$$

The type of path traveled by the satellite is determined from the value of the eccentricity of the conic section as given by Eq. 13-17. If

$$\begin{array}{ll} e = 0 & \text{free-flight trajectory is a circle} \\ e = 1 & \text{free-flight trajectory is a parabola} \\ e < 1 & \text{free-flight trajectory is an ellipse} \\ e > 1 & \text{free-flight trajectory is a hyperbola} \end{array} \quad (13-22)$$

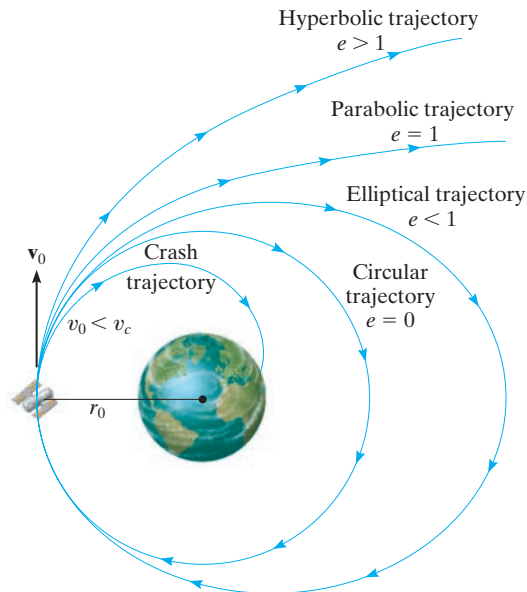
**Parabolic Path.** Each of these trajectories is shown in Fig. 13–25. From the curves it is seen that when the satellite follows a parabolic path, it is “on the border” of never returning to its initial starting point. The initial launch velocity,  $v_0$ , required for the satellite to follow a parabolic path is called the *escape velocity*. The speed,  $v_e$ , can be determined by using the second of Eq. 13–22,  $e = 1$ , with Eqs. 13–17, 13–19, and 13–20. It is left as an exercise to show that

$$v_e = \sqrt{\frac{2GM_e}{r_0}} \quad (13-23)$$

**Circular Orbit.** The speed  $v_c$  required to launch a satellite into a *circular orbit* can be found using the first of Eq. 13–22,  $e = 0$ . Since  $e$  is related to  $h$  and  $C$ , Eq. 13–17,  $C$  must be zero to satisfy this equation (from Eq. 13–19,  $h$  cannot be zero); and therefore, using Eq. 13–20, we have

$$v_c = \sqrt{\frac{GM_e}{r_0}} \quad (13-24)$$

Provided  $r_0$  represents a minimum height for launching, in which frictional resistance from the atmosphere is neglected, speeds at launch which are less than  $v_c$  will cause the satellite to reenter the earth’s atmosphere and either burn up or crash, Fig. 13–25.



**Fig. 13–25**

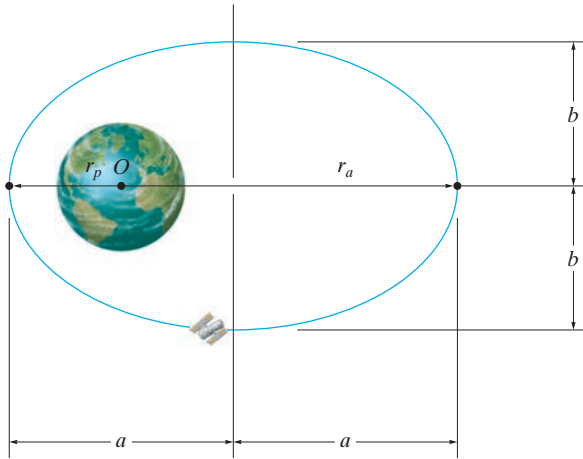


Fig. 13-26

**Elliptical Orbit.** All the trajectories attained by planets and most satellites are elliptical, Fig. 13-26. For a satellite's orbit about the earth, the *minimum distance* from the orbit to the center of the earth  $O$  (which is located at one of the foci of the ellipse) is  $r_p$  and can be found using Eq. 13-21 with  $\theta = 0^\circ$ . Therefore;

$$r_p = r_0 \quad (13-25)$$

This minimum distance is called the *perigee* of the orbit. The *apogee* or maximum distance  $r_a$  can be found using Eq. 13-21 with  $\theta = 180^\circ$ .\* Thus,

$$r_a = \frac{r_0}{(2GM_e/r_0v_0^2) - 1} \quad (13-26)$$

With reference to Fig. 13-26, the half-length of the major axis of the ellipse is

$$a = \frac{r_p + r_a}{2} \quad (13-27)$$

Using analytical geometry, it can be shown that the half-length of the minor axis is determined from the equation

$$b = \sqrt{r_p r_a} \quad (13-28)$$

\*Actually, the terminology perigee and apogee pertains only to orbits about the *earth*. If any other heavenly body is located at the focus of an elliptical orbit, the minimum and maximum distances are referred to respectively as the *periapsis* and *apoapsis* of the orbit.

Furthermore, by direct integration, the area of an ellipse is

$$A = \pi ab = \frac{\pi}{2}(r_p + r_a)\sqrt{r_p r_a} \quad (13-29)$$

The areal velocity has been defined by Eq. 13-12,  $dA/dt = h/2$ . Integrating yields  $A = hT/2$ , where  $T$  is the *period* of time required to make one orbital revolution. From Eq. 13-29, the period is

$$T = \frac{\pi}{h}(r_p + r_a)\sqrt{r_p r_a} \quad (13-30)$$

In addition to predicting the orbital trajectory of earth satellites, the theory developed in this section is valid, to a surprisingly close approximation, at predicting the actual motion of the planets traveling around the sun. In this case the mass of the sun,  $M_s$ , should be substituted for  $M_e$  when the appropriate formulas are used.

The fact that the planets do indeed follow elliptic orbits about the sun was discovered by the German astronomer Johannes Kepler in the early seventeenth century. His discovery was made *before* Newton had developed the laws of motion and the law of gravitation, and so at the time it provided important proof as to the validity of these laws. Kepler's laws, developed after 20 years of planetary observation, are summarized as follows:

1. Every planet travels in its orbit such that the line joining it to the center of the sun sweeps over equal areas in equal intervals of time, whatever the line's length.
2. The orbit of every planet is an ellipse with the sun placed at one of its foci.
3. The square of the period of any planet is directly proportional to the cube of the major axis of its orbit.

A mathematical statement of the first and second laws is given by Eqs. 13-12 and 13-21, respectively. The third law can be shown from Eq. 13-30 using Eqs. 13-18, 13-27, and 13-28. (See Prob. 13-117.)

## PROBLEMS

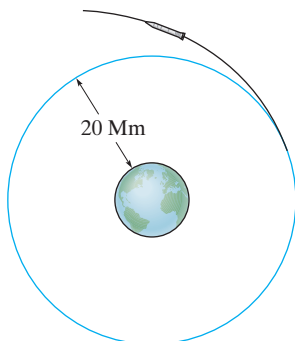
In the following problems, except where otherwise indicated, assume that the radius of the earth is 6378 km, the earth's mass is  $5.976(10^{24})$  kg, the mass of the sun is  $1.99(10^{30})$  kg, and the gravitational constant is  $G = 66.73(10^{-12}) \text{ m}^3/(\text{kg} \cdot \text{s}^2)$ .

**13-113.** The earth has an orbit with eccentricity 0.0167 around the sun. Knowing that the earth's minimum distance from the sun is  $146(10^6)$  km, find the speed at which the earth travels when it is at this distance. Determine the equation in polar coordinates which describes the earth's orbit about the sun.

**13-114.** A communications satellite is in a circular orbit above the earth such that it always remains directly over a point on the earth's surface. As a result, the period of the satellite must equal the rotation of the earth, which is approximately 24 hours. Determine the satellite's altitude  $h$  above the earth's surface and its orbital speed.

**13-115.** The speed of a satellite launched into a circular orbit about the earth is given by Eq. 13-24. Determine the speed of a satellite launched parallel to the surface of the earth so that it travels in a circular orbit 800 km from the earth's surface.

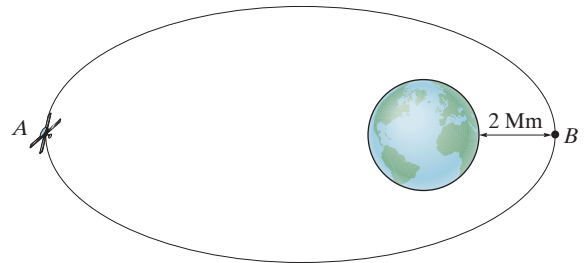
**\*13-116.** The rocket is in circular orbit about the earth at an altitude of 20 Mm. Determine the minimum increment in speed it must have in order to escape the earth's gravitational field.



**Prob. 13-116**

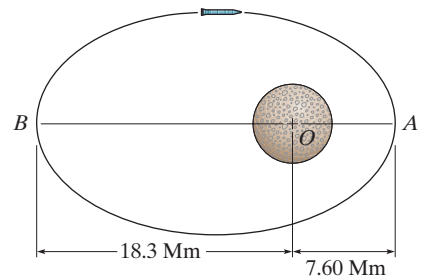
**13-117.** Prove Kepler's third law of motion. *Hint:* Use Eqs. 13-18, 13-27, 13-28, and 13-30.

**13-118.** The satellite is moving in an elliptical orbit with an eccentricity  $e = 0.25$ . Determine its speed when it is at its maximum distance  $A$  and minimum distance  $B$  from the earth.



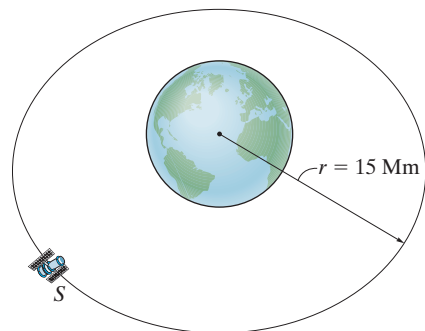
**Prob. 13-118**

**13-119.** The rocket is traveling in free flight along the elliptical orbit. The planet has no atmosphere, and its mass is 0.60 times that of the earth. If the rocket has the orbit shown, determine the rocket's speed when it is at  $A$  and at  $B$ .



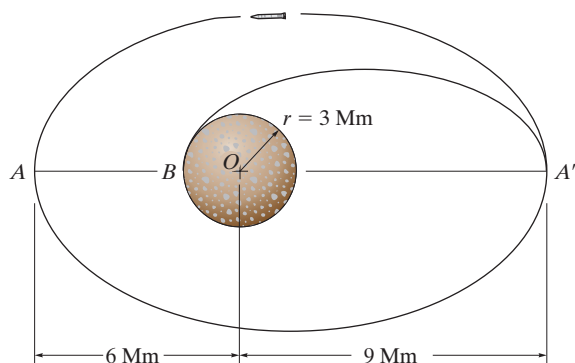
**Prob. 13-119**

**\*13-120.** Determine the constant speed of satellite  $S$  so that it circles the earth with an orbit of radius  $r = 15$  Mm. *Hint:* Use Eq. 13-1.



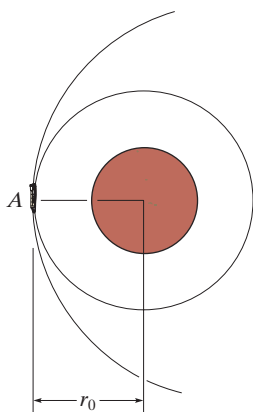
**Prob. 13-120**

**13-121.** The rocket is in free flight along an elliptical trajectory  $A'A$ . The planet has no atmosphere, and its mass is 0.70 times that of the earth. If the rocket has an apoapsis and periapsis as shown in the figure, determine the speed of the rocket when it is at point  $A$ .



**Prob. 13-121**

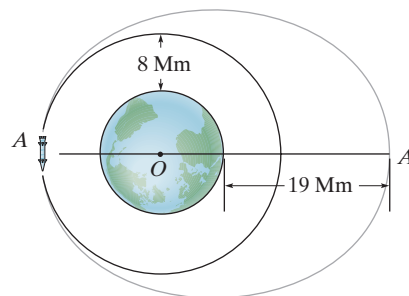
**13-122.** The Viking Explorer approaches the planet Mars on a parabolic trajectory as shown. When it reaches point  $A$  its velocity is 10 Mm/h. Determine  $r_0$  and the required change in velocity at  $A$  so that it can then maintain a circular orbit as shown. The mass of Mars is 0.1074 times the mass of the earth.



**Prob. 13-122**

**13-123.** The rocket is initially in free-flight circular orbit around the earth. Determine the speed of the rocket at  $A$ . What change in the speed at  $A$  is required so that it can move in an elliptical orbit to reach point  $A'$ ?

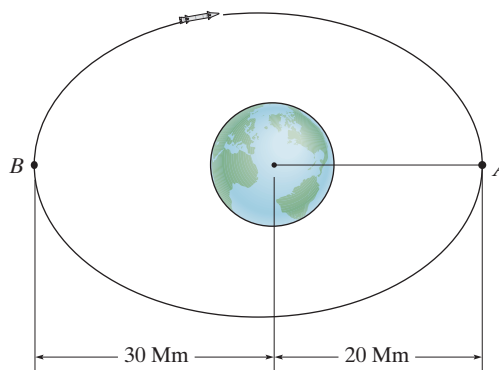
**\*13-124.** The rocket is in free-flight circular orbit around the earth. Determine the time needed for the rocket to travel from the inner orbit at  $A$  to the outer orbit at  $A'$ .



**Probs. 13-123/124**

**13-125.** A rocket is in a free-flight elliptical orbit about the earth such that the eccentricity of its orbit is  $e$  and its perigee is  $r_0$ . Determine the minimum increment of speed it should have in order to escape the earth's gravitational field when it is at this point along its orbit.

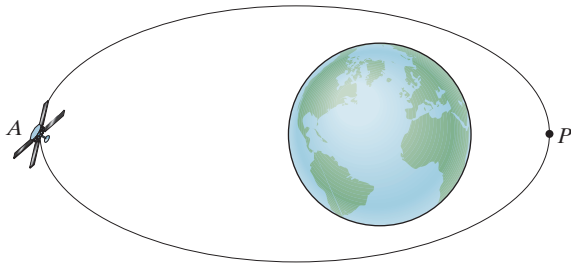
**13-126.** The rocket is traveling around the earth in free flight along the elliptical orbit. If the rocket has the orbit shown, determine the speed of the rocket when it is at  $A$  and at  $B$ .



**Prob. 13-126**

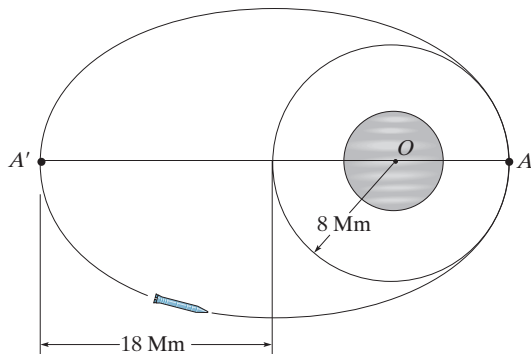


**13–127.** An elliptical path of a satellite has an eccentricity  $e = 0.130$ . If it has a speed of 15 Mm/h when it is at perigee,  $P$ , determine its speed when it arrives at apogee,  $A$ . Also, how far is it from the earth's surface when it is at  $A$ ?



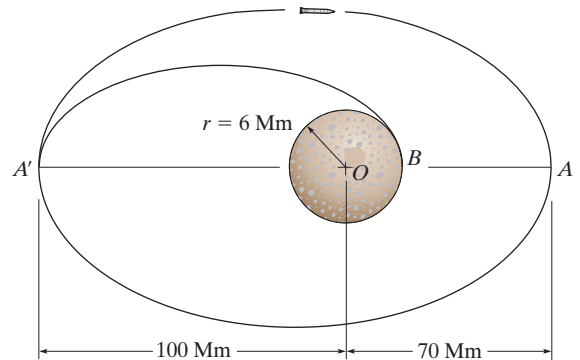
**Prob. 13–127**

**\*13–128.** A rocket is in free-flight elliptical orbit around the planet Venus. Knowing that the periapsis and apoapsis of the orbit are 8 Mm and 26 Mm, respectively, determine (a) the speed of the rocket at point  $A'$ , (b) the required speed it must attain at  $A$  just after braking so that it undergoes an 8-Mm free-flight circular orbit around Venus, and (c) the periods of both the circular and elliptical orbits. The mass of Venus is 0.816 times the mass of the earth.



**Prob. 13–128**

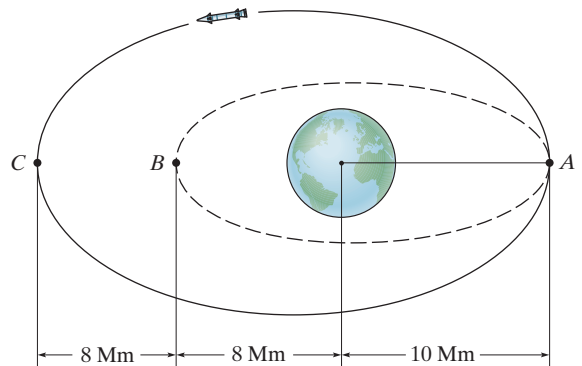
**13–129.** The rocket is traveling in a free flight along an elliptical trajectory  $A'A$ . The planet has no atmosphere, and its mass is 0.60 times that of the earth. If the rocket has the orbit shown, determine the rocket's velocity when it is at point  $A$ .



**Probs. 13–129/130**

**13–131.** The rocket is traveling around the earth in free flight along an elliptical orbit  $AC$ . If the rocket has the orbit shown, determine the rocket's velocity when it is at point  $A$ .

**\*13–132.** The rocket is traveling around the earth in free flight along the elliptical orbit  $AC$ . Determine its change in speed when it reaches  $A$  so that it travels along the elliptical orbit  $AB$ .



**Probs. 13–131/132**

## CONCEPTUAL PROBLEMS

**C13-1.** If the box is released from rest at *A*, use numerical values to show how you would estimate the time for it to arrive at *B*. Also, list the assumptions for your analysis.



**Prob. C13-1**

**C13-2.** The tugboat has a known mass and its propeller provides a known maximum thrust. When the tug is fully powered you observe the time it takes for the tug to reach a speed of known value starting from rest. Show how you could determine the mass of the barge. Neglect the drag force of the water on the tug. Use numerical values to explain your answer.



**Prob. C13-2**

**C13-3.** Determine the smallest speed of each car *A* and *B* so that the passengers do not lose contact with the seat while the arms turn at a constant rate. What is the largest normal force of the seat on each passenger? Use numerical values to explain your answer.



**Prob. C13-3**

**C13-4.** Each car is pin connected at its ends to the rim of the wheel which turns at a constant speed. Using numerical values, show how to determine the resultant force the seat exerts on the passenger located in the top car *A*. The passengers are seated toward the center of the wheel. Also, list the assumptions for your analysis.



**Prob. C13-4**

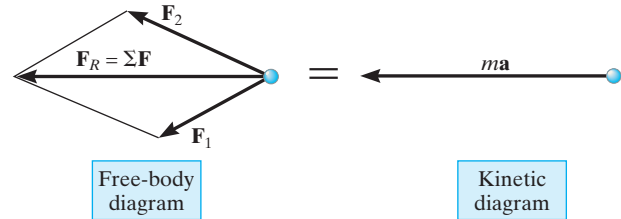
## CHAPTER REVIEW

13

### Kinetics

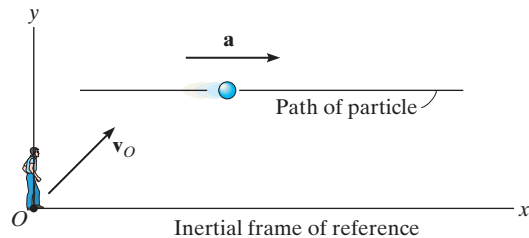
Kinetics is the study of the relation between forces and the acceleration they cause. This relation is based on Newton's second law of motion, expressed mathematically as  $\Sigma \mathbf{F} = m\mathbf{a}$ .

Before applying the equation of motion, it is important to first draw the particle's *free-body diagram* in order to account for all of the forces that act on the particle. Graphically, this diagram is equal to the *kinetic diagram*, which shows the result of the forces, that is, the  $m\mathbf{a}$  vector.



### Inertial Coordinate Systems

When applying the equation of motion, it is important to measure the acceleration from an inertial coordinate system. This system has axes that do not rotate but are either fixed or translate with a constant velocity. Various types of inertial coordinate systems can be used to apply  $\Sigma \mathbf{F} = m\mathbf{a}$  in component form.



Rectangular  $x$ ,  $y$ ,  $z$  axes are used to describe the motion along each of the axes.

$$\Sigma F_x = ma_x, \Sigma F_y = ma_y, \Sigma F_z = ma_z$$

Normal, tangential, and binormal axes  $n$ ,  $t$ ,  $b$ , are often used when the path is known. Recall that  $\mathbf{a}_n$  is always directed in the  $+n$  direction. It indicates the change in the velocity direction. Also recall that  $\mathbf{a}_t$  is tangent to the path. It indicates the change in the velocity magnitude.

$$\Sigma F_t = ma_t, \Sigma F_n = ma_n, \Sigma F_b = 0$$

$$a_t = dv/dt \quad \text{or} \quad a_t = v dv/ds$$

$$a_n = v^2/\rho \quad \text{where} \quad \rho = \frac{[1 + (dy/dx)^2]^{3/2}}{|d^2y/dx^2|}$$

Cylindrical coordinates are useful when angular motion of the radial line  $r$  is specified or when the path can conveniently be described with these coordinates.

$$\Sigma F_r = m(\ddot{r} - r\dot{\theta}^2)$$

$$\Sigma F_\theta = m(r\ddot{\theta} + 2\dot{r}\dot{\theta})$$

$$\Sigma F_z = m\ddot{z}$$

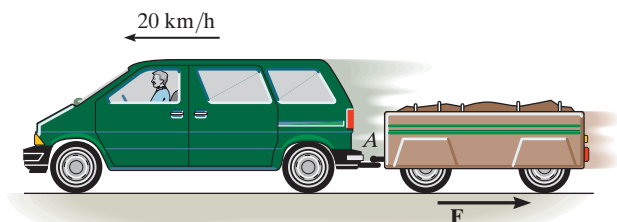
### Central-Force Motion

When a single force acts upon a particle, such as during the free-flight trajectory of a satellite in a gravitational field, then the motion is referred to as central-force motion. The orbit depends upon the eccentricity  $e$ ; and as a result, the trajectory can either be circular, parabolic, elliptical, or hyperbolic.

## REVIEW PROBLEMS

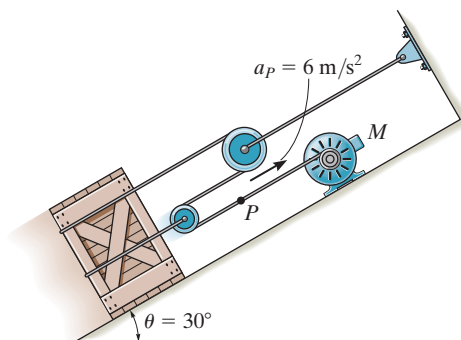
13

**R13-1.** The van is traveling at 20 km/h when the coupling of the trailer at  $A$  fails. If the trailer has a mass of 250 kg and coasts 45 m before coming to rest, determine the constant horizontal force  $F$  created by rolling friction which causes the trailer to stop.



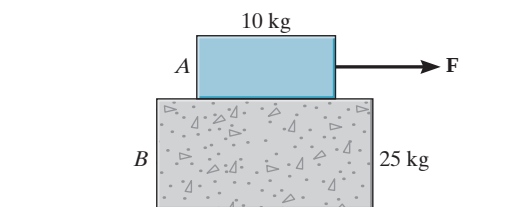
Prob. R13-1

**R13-2.** The motor  $M$  pulls in its attached rope with an acceleration  $a_p = 6 \text{ m/s}^2$ . Determine the towing force exerted by  $M$  on the rope in order to move the 50-kg crate up the inclined plane. The coefficient of kinetic friction between the crate and the plane is  $\mu_k = 0.3$ . Neglect the mass of the pulleys and rope.



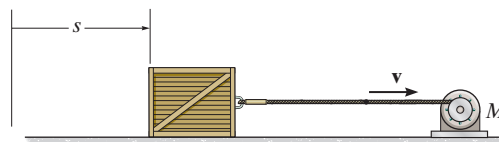
Prob. R13-2

**R13-3.** Block  $B$  rests on a smooth surface. If the coefficients of friction between  $A$  and  $B$  are  $\mu_s = 0.4$  and  $\mu_k = 0.3$ , determine the acceleration of each block if  $F = 250 \text{ N}$ .



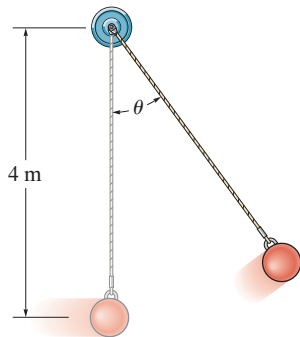
Prob. R13-3

**R13-4.** If the motor draws in the cable at a rate of  $v = (0.05s^{3/2}) \text{ m/s}$ , where  $s$  is in meters, determine the tension developed in the cable when  $s = 10 \text{ m}$ . The crate has a mass of 20 kg, and the coefficient of kinetic friction between the crate and the ground is  $\mu_k = 0.2$ .



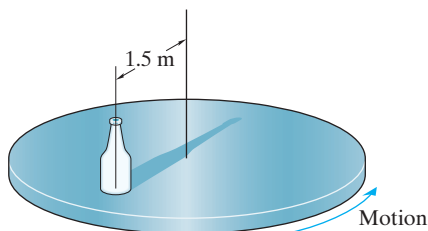
Prob. R13-4

**R13-5.** The ball has a mass of 30 kg and a speed  $v = 4$  m/s at the instant it is at its lowest point,  $\theta = 0^\circ$ . Determine the tension in the cord and the rate at which the ball's speed is decreasing at the instant  $\theta = 20^\circ$ . Neglect the size of the ball.



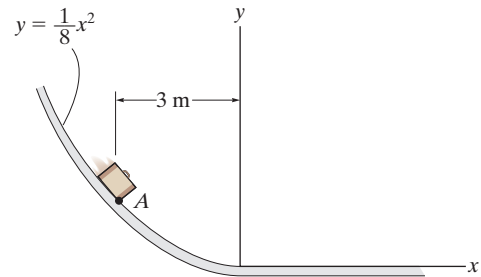
**Prob. R13-5**

**R13-6.** The bottle rests at a distance of 1.5 m from the center of the horizontal platform. If the coefficient of static friction between the bottle and the platform is  $\mu_s = 0.3$ , determine the maximum speed that the bottle can attain before slipping. Assume the angular motion of the platform is slowly increasing.



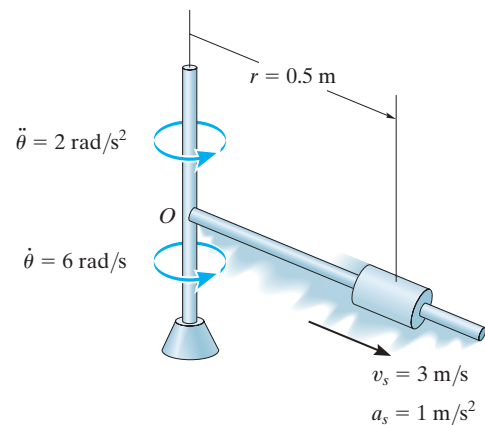
**Prob. R13-6**

**R13-7.** The 5-kg suitcase slides down the curved ramp for which the coefficient of kinetic friction is  $\mu_k = 0.2$ . If at the instant it reaches point A it has a speed of 2 m/s, determine the normal force on the suitcase and the rate of increase of its speed.



**Prob. R13-7**

**R13-8.** The spool, which has a mass of 4 kg, slides along the rotating rod. At the instant shown, the angular rate of rotation of the rod is  $\dot{\theta} = 6$  rad/s and this rotation is increasing at  $\ddot{\theta} = 2$  rad/s<sup>2</sup>. At this same instant, the spool has a velocity of 3 m/s and an acceleration of 1 m/s<sup>2</sup>, both measured relative to the rod and directed away from the center O when  $r = 0.5$  m. Determine the radial frictional force and the normal force, both exerted by the rod on the spool at this instant.



**Prob. R13-8**



# Chapter 14



(© Oliver Furrer/Ocean/Corbis)

As the woman falls, her energy will have to be absorbed by the bungee cord.  
The principles of work and energy can be used to predict the motion.

# Kinetics of a Particle: Work and Energy

## CHAPTER OBJECTIVES

- To develop the principle of work and energy and apply it to solve problems that involve force, velocity, and displacement.
- To study problems that involve power and efficiency.
- To introduce the concept of a conservative force and apply the theorem of conservation of energy to solve kinetic problems.



Video Solutions are available for selected questions in this chapter.

## 14.1 The Work of a Force

In this chapter, we will analyze motion of a particle using the concepts of work and energy. The resulting equation will be useful for solving problems that involve force, velocity, and displacement. Before we do this, however, we must first define the work of a force. Specifically, a force  $\mathbf{F}$  will do *work* on a particle only when the particle undergoes a *displacement in the direction of the force*. For example, if the force  $\mathbf{F}$  in Fig. 14–1 causes the particle to move along the path  $s$  from position  $\mathbf{r}$  to a new position  $\mathbf{r}'$ , the displacement is then  $d\mathbf{r} = \mathbf{r}' - \mathbf{r}$ . The magnitude of  $d\mathbf{r}$  is  $ds$ , the length of the differential segment along the path. If the angle between the tails of  $d\mathbf{r}$  and  $\mathbf{F}$  is  $\theta$ , Fig. 14–1, then the work done by  $\mathbf{F}$  is a *scalar quantity*, defined by

$$dU = F ds \cos \theta$$

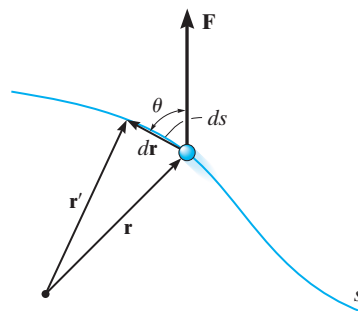


Fig. 14–1

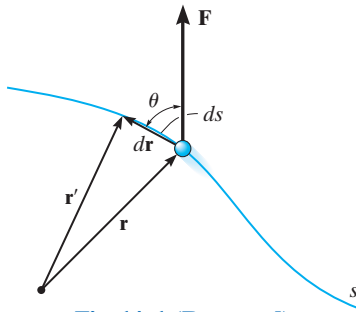


Fig. 14-1 (Repeated)

By definition of the dot product (see Eq. B-14) this equation can also be written as

$$dU = \mathbf{F} \cdot d\mathbf{r}$$

This result may be interpreted in one of two ways: either as the product of  $F$  and the component of displacement  $ds \cos \theta$  in the direction of the force, or as the product of  $ds$  and the component of force,  $F \cos \theta$ , in the direction of displacement. Note that if  $0^\circ \leq \theta < 90^\circ$ , then the force component and the displacement have the *same sense* so that the work is *positive*; whereas if  $90^\circ < \theta \leq 180^\circ$ , these vectors will have *opposite sense*, and therefore the work is *negative*. Also,  $dU = 0$  if the force is *perpendicular* to displacement, since  $\cos 90^\circ = 0$ , or if the force is applied at a *fixed point*, in which case the displacement is zero.

The unit of work in SI units is the joule (J), which is the amount of work done by a one-newton force when it moves through a distance of one meter in the direction of the force ( $1 \text{ J} = 1 \text{ N} \cdot \text{m}$ ).

**Work of a Variable Force.** If the particle acted upon by the force  $\mathbf{F}$  undergoes a finite displacement along its path from  $\mathbf{r}_1$  to  $\mathbf{r}_2$  or  $s_1$  to  $s_2$ , Fig. 14-2a, the work of force  $\mathbf{F}$  is determined by integration. Provided  $\mathbf{F}$  and  $\theta$  can be expressed as a function of position, then

$$U_{1-2} = \int_{\mathbf{r}_1}^{\mathbf{r}_2} \mathbf{F} \cdot d\mathbf{r} = \int_{s_1}^{s_2} F \cos \theta \, ds \quad (14-1)$$

Sometimes, this relation may be obtained by using experimental data to plot a graph of  $F \cos \theta$  vs.  $s$ . Then the area under this graph bounded by  $s_1$  and  $s_2$  represents the total work, Fig. 14-2b.

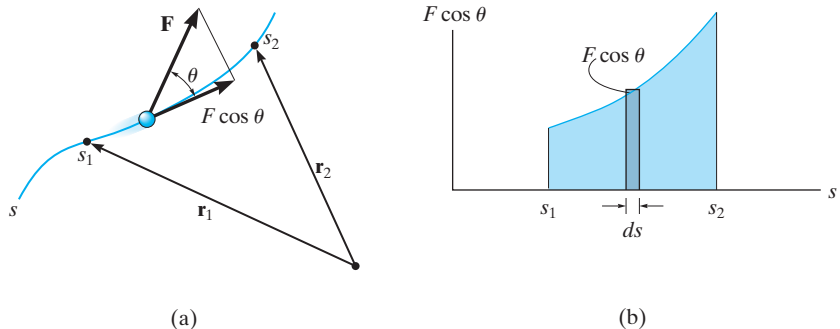


Fig. 14-2



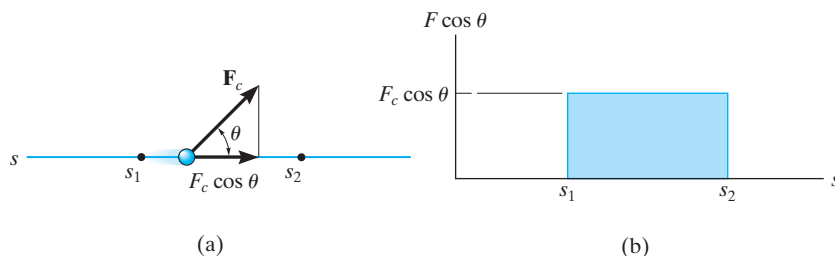


Fig. 14-3

### Work of a Constant Force Moving Along a Straight Line.

If the force  $\mathbf{F}_c$  has a constant magnitude and acts at a constant angle  $\theta$  from its straight-line path, Fig. 14-3a, then the component of  $\mathbf{F}_c$  in the direction of displacement is always  $F_c \cos \theta$ . The work done by  $\mathbf{F}_c$  when the particle is displaced from  $s_1$  to  $s_2$  is determined from Eq. 14-1, in which case

$$U_{1-2} = F_c \cos \theta \int_{s_1}^{s_2} ds$$

or

$$U_{1-2} = F_c \cos \theta (s_2 - s_1) \quad (14-2)$$

Here the work of  $\mathbf{F}_c$  represents the *area of the rectangle* in Fig. 14-3b.

**Work of a Weight.** Consider a particle of weight  $\mathbf{W}$ , which moves up along the path  $s$  shown in Fig. 14-4 from position  $s_1$  to position  $s_2$ . At an intermediate point, the displacement  $d\mathbf{r} = dx\mathbf{i} + dy\mathbf{j} + dz\mathbf{k}$ . Since  $\mathbf{W} = -W\mathbf{j}$ , applying Eq. 14-1 we have

$$\begin{aligned} U_{1-2} &= \int_{\mathbf{r}_1}^{\mathbf{r}_2} \mathbf{F} \cdot d\mathbf{r} = \int_{\mathbf{r}_1}^{\mathbf{r}_2} (-W\mathbf{j}) \cdot (dx\mathbf{i} + dy\mathbf{j} + dz\mathbf{k}) \\ &= \int_{y_1}^{y_2} -W dy = -W(y_2 - y_1) \end{aligned}$$

or

$$U_{1-2} = -W \Delta y \quad (14-3)$$

Thus, the work is independent of the path and is equal to the magnitude of the particle's weight times its vertical displacement. In the case shown in Fig. 14-4 the work is *negative*, since  $W$  is downward and  $\Delta y$  is upward. Note, however, that if the particle is displaced *downward* ( $-\Delta y$ ), the work of the weight is *positive*. Why?



The crane must do work in order to hoist the weight of the pipe.

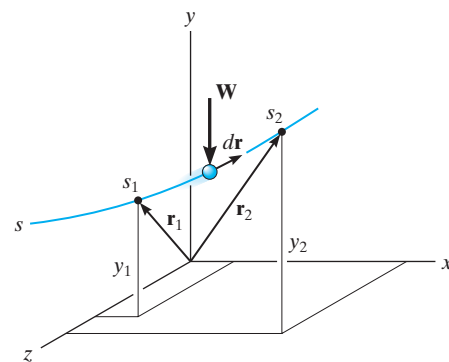


Fig. 14-4

**Work of a Spring Force.** If an elastic spring is elongated a distance  $ds$ , Fig. 14–5a, then the work done by the force that acts on the attached particle is  $dU = -F_s ds = -ks ds$ . The work is *negative* since  $\mathbf{F}_s$  acts in the opposite sense to  $ds$ . If the particle displaces from  $s_1$  to  $s_2$ , the work of  $\mathbf{F}_s$  is then

$$U_{1-2} = \int_{s_1}^{s_2} F_s ds = \int_{s_1}^{s_2} -ks ds$$

$$U_{1-2} = -\left(\frac{1}{2}ks_2^2 - \frac{1}{2}ks_1^2\right) \quad (14-4)$$

This work represents the trapezoidal area under the line  $F_s = ks$ , Fig. 14–5b.

A mistake in sign can be avoided when applying this equation if one simply notes the direction of the spring force acting on the particle and compares it with the sense of direction of displacement of the particle—if both are in the *same sense*, *positive work* results; if they are *opposite* to one another, the *work is negative*.

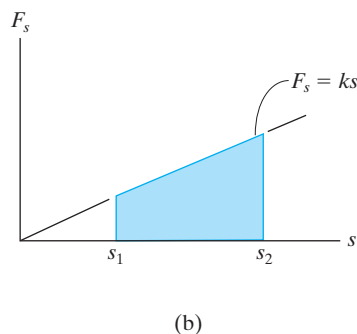
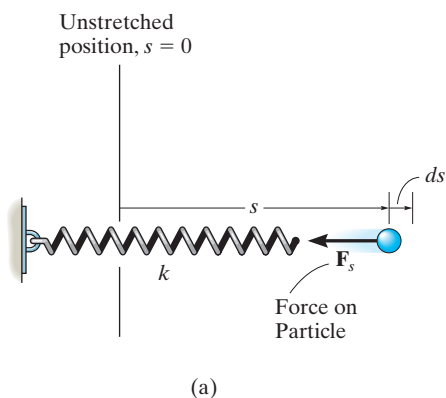
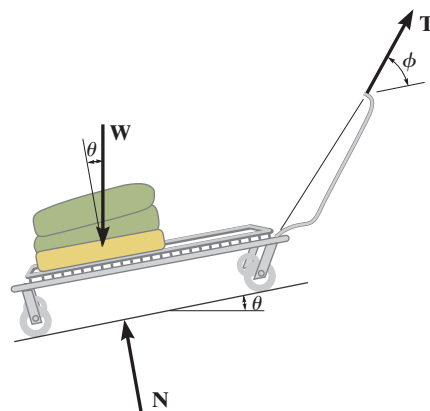


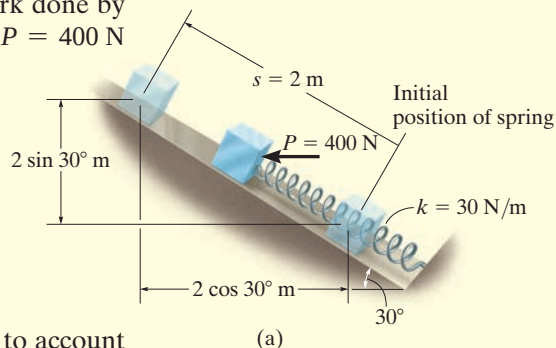
Fig. 14–5

The forces acting on the cart, as it is pulled a distance  $s$  up the incline, are shown on its free-body diagram. The constant towing force  $\mathbf{T}$  does positive work of  $U_T = (T \cos \phi)s$ , the weight does negative work of  $U_W = -(W \sin \theta)s$ , and the normal force  $\mathbf{N}$  does no work since there is no displacement of this force along its line of action.



**EXAMPLE 14.1**

The 10-kg block shown in Fig. 14–6*a* rests on the smooth incline. If the spring is originally stretched 0.5 m, determine the total work done by all the forces acting on the block when a horizontal force  $P = 400$  N pushes the block up the plane  $s = 2$  m.

**SOLUTION**

First the free-body diagram of the block is drawn in order to account for all the forces that act on the block, Fig. 14–6*b*.

**Horizontal Force  $P$ .** Since this force is *constant*, the work is determined using Eq. 14–2. The result can be calculated as the force times the component of displacement in the direction of the force; i.e.,

$$U_P = 400 \text{ N} (2 \text{ m} \cos 30^\circ) = 692.8 \text{ J}$$

or the displacement times the component of force in the direction of displacement, i.e.,

$$U_P = 400 \text{ N} \cos 30^\circ (2 \text{ m}) = 692.8 \text{ J}$$

**Spring Force  $F_s$ .** In the initial position the spring is stretched  $s_1 = 0.5$  m and in the final position it is stretched  $s_2 = 0.5 \text{ m} + 2 \text{ m} = 2.5$  m. We require the work to be negative since the force and displacement are opposite to each other. The work of  $F_s$  is thus

$$U_s = -\left[\frac{1}{2}(30 \text{ N/m})(2.5 \text{ m})^2 - \frac{1}{2}(30 \text{ N/m})(0.5 \text{ m})^2\right] = -90 \text{ J}$$

**Weight  $W$ .** Since the weight acts in the opposite sense to its vertical displacement, the work is negative; i.e.,

$$U_W = -(98.1 \text{ N}) (2 \text{ m} \sin 30^\circ) = -98.1 \text{ J}$$

Note that it is also possible to consider the component of weight in the direction of displacement; i.e.,

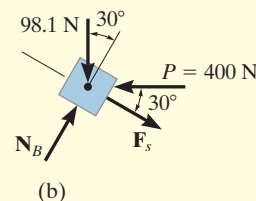
$$U_W = -(98.1 \sin 30^\circ \text{ N}) (2 \text{ m}) = -98.1 \text{ J}$$

**Normal Force  $N_B$ .** This force does *no work* since it is *always* perpendicular to the displacement.

**Total Work.** The work of all the forces when the block is displaced 2 m is therefore

$$U_T = 692.8 \text{ J} - 90 \text{ J} - 98.1 \text{ J} = 505 \text{ J}$$

*Ans.*



**Fig. 14–6**

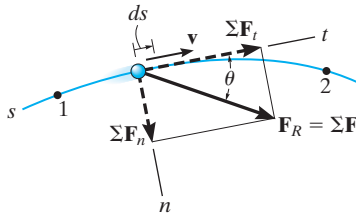


Fig. 14-7

## 14.2 Principle of Work and Energy

Consider the particle in Fig. 14-7, which is located on the path defined relative to an inertial coordinate system. If the particle has a mass  $m$  and is subjected to a system of external forces represented by the resultant  $\mathbf{F}_R = \Sigma \mathbf{F}$ , then the equation of motion for the particle in the tangential direction is  $\Sigma F_t = ma_t$ . Applying the kinematic equation  $a_t = v dv/ds$  and integrating both sides, assuming initially that the particle has a position  $s = s_1$  and a speed  $v = v_1$ , and later at  $s = s_2$ ,  $v = v_2$ , we have

$$\begin{aligned}\Sigma \int_{s_1}^{s_2} F_t ds &= \int_{v_1}^{v_2} mv dv \\ \Sigma \int_{s_1}^{s_2} F_t ds &= \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2\end{aligned}\quad (14-5)$$

From Fig. 14-7, note that  $\Sigma F_t = \Sigma F \cos \theta$ , and since work is defined from Eq. 14-1, the final result can be written as

$$\Sigma U_{1-2} = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2 \quad (14-6)$$

This equation represents the *principle of work and energy* for the particle. The term on the left is the sum of the work done by *all* the forces acting on the particle as the particle moves from point 1 to point 2. The two terms on the right side, which are of the form  $T = \frac{1}{2}mv^2$ , define the particle's final and initial *kinetic energy*, respectively. Like work, kinetic energy is a *scalar* and has units of joules (J). However, unlike work, which can be either positive or negative, the kinetic energy is *always positive*, regardless of the direction of motion of the particle.

When Eq. 14-6 is applied, it is often expressed in the form

$$T_1 + \Sigma U_{1-2} = T_2 \quad (14-7)$$

which states that the particle's initial kinetic energy plus the work done by all the forces acting on the particle as it moves from its initial to its final position is equal to the particle's final kinetic energy.

As noted from the derivation, the principle of work and energy represents an integrated form of  $\Sigma F_t = ma_t$ , obtained by using the kinematic equation  $a_t = v dv/ds$ . As a result, this principle will provide a convenient *substitution* for  $\Sigma F_t = ma_t$  when solving those types of kinetic problems which involve *force*, *velocity*, and *displacement* since these quantities are involved in Eq. 14-7. For application, it is suggested that the following procedure be used.

## Procedure for Analysis

### Work (Free-Body Diagram).

- Establish the inertial coordinate system and draw a free-body diagram of the particle in order to account for all the forces that do work on the particle as it moves along its path.

### Principle of Work and Energy.

- Apply the principle of work and energy,  $T_1 + \Sigma U_{1-2} = T_2$ .
- The kinetic energy at the initial and final points is *always positive*, since it involves the speed squared ( $T = \frac{1}{2}mv^2$ ).
- A force does work when it moves through a displacement in the direction of the force.
- Work is *positive* when the force component is in the *same sense of direction* as its displacement, otherwise it is negative.
- Forces that are functions of displacement must be integrated to obtain the work. Graphically, the work is equal to the area under the force-displacement curve.
- The work of a weight is the product of the weight magnitude and the vertical displacement,  $U_W = \pm Wy$ . It is positive when the weight moves downwards.
- The work of a spring is of the form  $U_s = \frac{1}{2}ks^2$ , where  $k$  is the spring stiffness and  $s$  is the stretch or compression of the spring.

Numerical application of this procedure is illustrated in the examples following Sec. 14.3.

If an oncoming car strikes these crash barrels, the car's kinetic energy will be transformed into work, which causes the barrels, and to some extent the car, to be deformed. By knowing the amount of energy that can be absorbed by each barrel it is possible to design a crash cushion such as this.



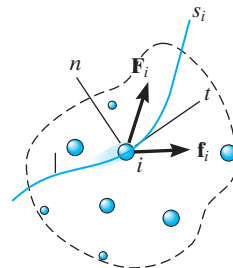
### 14.3 Principle of Work and Energy for a System of Particles

The principle of work and energy can be extended to include a system of particles isolated within an enclosed region of space as shown in Fig. 14–8. Here the arbitrary  $i$ th particle, having a mass  $m_i$ , is subjected to a resultant external force  $\mathbf{F}_i$  and a resultant internal force  $\mathbf{f}_i$  which all the other particles exert on the  $i$ th particle. If we apply the principle of work and energy to this and each of the other particles in the system, then since work and energy are scalar quantities, the equations can be summed algebraically, which gives

$$\Sigma T_1 + \Sigma U_{1-2} = \Sigma T_2 \quad (14-8)$$

In this case, the initial kinetic energy of the system plus the work done by all the external and internal forces acting on the system is equal to the final kinetic energy of the system.

If the system represents a *translating rigid body*, or a series of connected translating bodies, then all the particles in each body will undergo the *same displacement*. Therefore, the work of all the internal forces will occur in equal but opposite collinear pairs and so it will cancel out. On the other hand, if the body is assumed to be *nonrigid*, the particles of the body may be displaced along *different paths*, and some of the energy due to force interactions would be given off and lost as heat or stored in the body if permanent deformations occur. We will discuss these effects briefly at the end of this section and in Sec. 15.4. Throughout this text, however, the principle of work and energy will be applied to problems where direct accountability of such energy losses does not have to be considered.



Inertial coordinate system

Fig. 14–8

**Work of Friction Caused by Sliding.** A special class of problems will now be investigated which requires a careful application of Eq. 14–8. These problems involve cases where a body slides over the surface of another body in the presence of friction. Consider, for example, a block which is translating a distance  $s$  over a rough surface as shown in Fig. 14–9a. If the applied force  $\mathbf{P}$  just balances the *resultant* frictional force  $\mu_k N$ , Fig. 14–9b, then due to equilibrium a constant velocity  $\mathbf{v}$  is maintained, and one would expect Eq. 14–8 to be applied as follows:

$$\frac{1}{2}mv^2 + Ps - \mu_k Ns = \frac{1}{2}mv^2$$

Indeed this equation is satisfied if  $P = \mu_k N$ ; however, as one realizes from experience, the sliding motion will *generate heat*, a form of energy which seems not to be accounted for in the work-energy equation. In order to explain this paradox and thereby more closely represent the nature of friction, we should actually model the block so that the surfaces of contact are *deformable* (nonrigid).<sup>\*</sup> Recall that the rough portions at the bottom of the block act as “teeth,” and when the block slides these teeth *deform slightly* and either break off or vibrate as they pull away from “teeth” at the contacting surface, Fig. 14–9c. As a result, frictional forces that act on the block at these points are displaced slightly, due to the localized deformations, and later they are replaced by other frictional forces as other points of contact are made. At any instant, the *resultant*  $\mathbf{F}$  of all these frictional forces remains essentially constant, i.e.,  $\mu_k N$ ; however, due to the many *localized deformations*, the actual displacement  $s'$  of  $\mu_k N$  is *not* the same as the displacement  $s$  of the applied force  $\mathbf{P}$ . Instead,  $s'$  will be *less* than  $s$  ( $s' < s$ ), and therefore the *external work* done by the resultant frictional force will be  $\mu_k Ns'$  and not  $\mu_k Ns$ . The remaining amount of work,  $\mu_k N(s - s')$ , manifests itself as an increase in *internal energy*, which in fact causes the block’s temperature to rise.

In summary then, Eq. 14–8 can be applied to problems involving sliding friction; however, it should be fully realized that the work of the resultant frictional force is not represented by  $\mu_k Ns$ ; instead, this term represents *both* the external work of friction ( $\mu_k Ns'$ ) *and* internal work [ $\mu_k N(s - s')$ ] which is converted into various forms of internal energy, such as heat.<sup>†</sup>

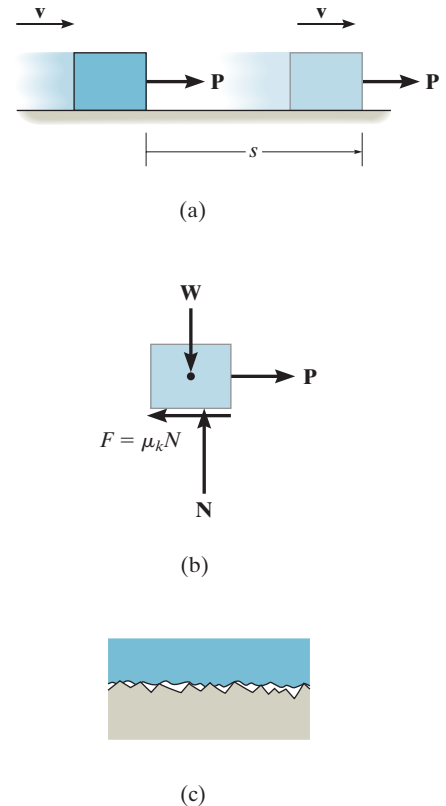


Fig. 14–9

<sup>\*</sup>See Chapter 8 of *Engineering Mechanics: Statics*.

<sup>†</sup>See B. A. Sherwood and W. H. Bernard, “Work and Heat Transfer in the Presence of Sliding Friction,” *Am. J. Phys.* 52, 1001 (1984).

## EXAMPLE 14.2

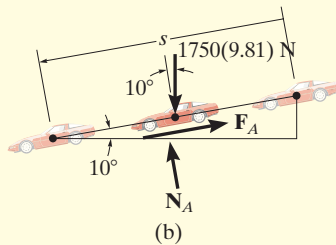
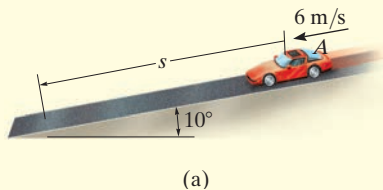


Fig. 14–10

The 1750-kg automobile shown in Fig. 14–10*a* travels down the  $10^\circ$  inclined road at a speed of 6 m/s. If the driver jams on the brakes, causing his wheels to lock, determine how far  $s$  the tires skid on the road. The coefficient of kinetic friction between the wheels and the road is  $\mu_k = 0.5$ .

## SOLUTION

This problem can be solved using the principle of work and energy, since it involves force, velocity, and displacement.

**Work (Free-Body Diagram).** As shown in Fig. 14–10*b*, the normal force  $N_A$  does no work since it never undergoes displacement along its line of action. The weight,  $1750(9.81)$  N, is displaced  $s \sin 10^\circ$  and does positive work. Why? The frictional force  $F_A$  does both external and internal work when it undergoes a displacement  $s$ . This work is negative since it is in the opposite sense of direction to the displacement. Applying the equation of equilibrium normal to the road, we have

$$+\curvearrowright \Sigma F_n = 0; \quad N_A - 1750(9.81) \cos 10^\circ \text{ N} = 0 \quad N_A = 16\,906.7 \text{ N}$$

Thus,

$$F_A = \mu_k N_A = 0.5 (16\,906.7 \text{ N}) = 8453.35 \text{ N}$$

## Principle of Work and Energy.

$$T_1 + \Sigma U_{1-2} = T_2$$

$$\frac{1}{2} (1750 \text{ kg}) (6 \text{ m/s})^2 + 1750(9.81) \text{ N} (s \sin 10^\circ) - (8453.35 \text{ N}) s = 0$$

Solving for  $s$  yields

$$s = 5.76 \text{ m} \quad \text{Ans.}$$

**NOTE:** If this problem is solved by using the equation of motion, *two steps* are involved. First, from the free-body diagram, Fig. 14–10*b*, the equation of motion is applied along the incline. This yields

$$+\searrow \Sigma F_s = ma_s; \quad 1750(9.81) \sin 10^\circ \text{ N} - 8453.35 \text{ N} = (1750 \text{ kg}) a$$

$$a = -3.127 \text{ m/s}^2$$

Then, since  $a$  is constant, we have

$$(+\searrow) \quad v^2 = v_0^2 + 2a_c(s - s_0);$$

$$(0)^2 = (6 \text{ m/s})^2 + 2(-3.127 \text{ m/s}^2)(s - 0)$$

$$s = 5.76 \text{ m} \quad \text{Ans.}$$



**EXAMPLE 14.3**

For a short time the crane in Fig. 14–11*a* lifts the 2.50-Mg beam with a force of  $F = (28 + 3s^2)$  kN. Determine the speed of the beam when it has risen  $s = 3$  m. Also, how much time does it take to attain this height starting from rest?

**SOLUTION**

We can solve part of this problem using the principle of work and energy since it involves force, velocity, and displacement. Kinematics must be used to determine the time. Note that at  $s = 0$ ,  $F = 28(10^3)$  N  $>$   $W = 2.50(10^3)(9.81)$  N, so motion will occur.

**Work (Free-Body Diagram).** As shown on the free-body diagram, Fig. 14–11*b*, the lifting force  $\mathbf{F}$  does positive work, which must be determined by integration since this force is a variable. Also, the weight is constant and will do negative work since the displacement is upward.

**Principles of Work and Energy.**

$$T_1 + \Sigma U_{1-2} = T_2$$

$$\begin{aligned} 0 + \int_0^s (28 + 3s^2)(10^3) ds - (2.50)(10^3)(9.81)s &= \frac{1}{2}(2.50)(10^3)v^2 \\ 28(10^3)s + (10^3)s^3 - 24.525(10^3)s &= 1.25(10^3)v^2 \\ v &= (2.78s + 0.8s^3)^{\frac{1}{2}} \end{aligned} \quad (1)$$

When  $s = 3$  m,

$$v = 5.47 \text{ m/s} \quad \text{Ans.}$$

**Kinematics.** Since we were able to express the velocity as a function of displacement, the time can be determined using  $v = ds/dt$ . In this case,

$$\begin{aligned} (2.78s + 0.8s^3)^{\frac{1}{2}} &= \frac{ds}{dt} \\ t &= \int_0^3 \frac{ds}{(2.78s + 0.8s^3)^{\frac{1}{2}}} \end{aligned}$$

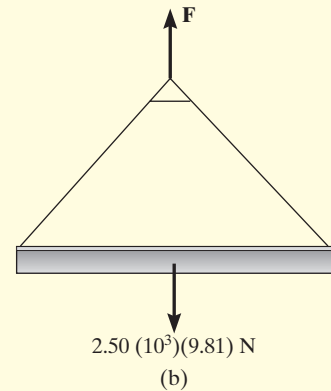
The integration can be performed numerically using a pocket calculator. The result is

$$t = 1.79 \text{ s} \quad \text{Ans.}$$

**NOTE:** The acceleration of the beam can be determined by integrating Eq. (1) using  $v dv = a ds$ , or more directly, by applying the equation of motion,  $\Sigma F = ma$ .



(a)

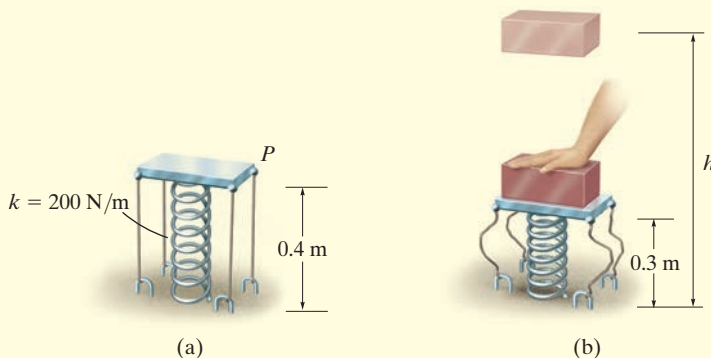


(b)

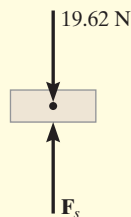
**Fig. 14–11**

**EXAMPLE 14.4**

The platform  $P$ , shown in Fig. 14–12a, has negligible mass and is tied down so that the 0.4-m-long cords keep a 1-m-long spring compressed 0.6 m when *nothing* is on the platform. If a 2-kg block is placed on the platform and released from rest after the platform is pushed down 0.1 m, Fig. 14–12b, determine the maximum height  $h$  the block rises in the air, measured from the ground.

**Fig. 14–12****SOLUTION**

**Work (Free-Body Diagram).** Since the block is released from rest and later reaches its maximum height, the initial and final velocities are zero. The free-body diagram of the block when it is still in contact with the platform is shown in Fig. 14–12c. Note that the weight does negative work and the spring force does positive work. Why? In particular, the *initial compression* in the spring is  $s_1 = 0.6 \text{ m} + 0.1 \text{ m} = 0.7 \text{ m}$ . Due to the cords, the spring's *final compression* is  $s_2 = 0.6 \text{ m}$  (after the block leaves the platform). The bottom of the block rises from a height of  $(0.4 \text{ m} - 0.1 \text{ m}) = 0.3 \text{ m}$  to a final height  $h$ .

**(c)****Principle of Work and Energy.**

$$T_1 + \Sigma U_{1-2} = T_2$$

$$\frac{1}{2}mv_1^2 + \left\{ -\left(\frac{1}{2}ks_2^2 - \frac{1}{2}ks_1^2\right) - W\Delta y \right\} = \frac{1}{2}mv_2^2$$

Note that here  $s_1 = 0.7 \text{ m} > s_2 = 0.6 \text{ m}$  and so the work of the spring as determined from Eq. 14–4 will indeed be positive once the calculation is made. Thus,

$$0 + \left\{ -\left[\frac{1}{2}(200 \text{ N/m})(0.6 \text{ m})^2 - \frac{1}{2}(200 \text{ N/m})(0.7 \text{ m})^2\right] - (19.62 \text{ N})[h - (0.3 \text{ m})] \right\} = 0$$

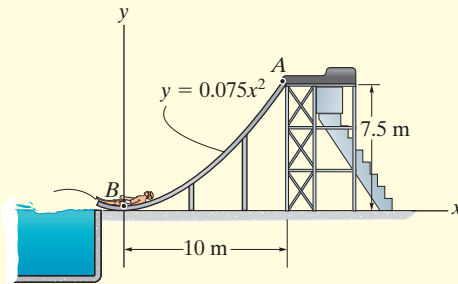
Solving yields

$$h = 0.963 \text{ m}$$

*Ans.*

**EXAMPLE 14.5**

The 40-kg boy in Fig. 14–13*a* slides down the smooth water slide. If he starts from rest at *A*, determine his speed when he reaches *B* and the normal reaction the slide exerts on the boy at this position.



(a)



Please refer to the Companion Website for the animation: *A Skier Showing the Work Done at Different Stages (from Positions 1, 2 to 3)*

**SOLUTION**

**Work (Free-Body Diagram).** As shown on the free-body diagram, Fig. 14–13*b*, there are two forces acting on the boy as he goes down the slide. Note that the normal force does no work.

**Principle of Work and Energy.**

$$T_A + \Sigma U_{A-B} = T_B$$

$$0 + (40(9.81)\text{N})(7.5\text{ m}) = \frac{1}{2}(40\text{ kg})v_B^2$$

$$v_B = 12.13\text{ m/s} = 12.1\text{ m/s}$$

*Ans.*

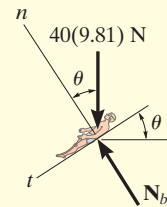
**Equation of Motion.** Referring to the free-body diagram of the boy when he is at *B*, Fig. 14–13*c*, the normal reaction  $N_B$  can now be obtained by applying the equation of motion along the *n* axis. Here the radius of curvature of the path is

$$\rho_B = \frac{\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2}}{|d^2y/dx^2|} = \frac{\left[1 + (0.15x)^2\right]^{3/2}}{|0.15|} \bigg|_{x=0} = 6.667\text{ m}$$

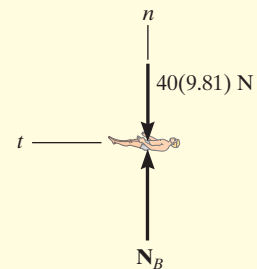
Thus,

$$+\uparrow \Sigma F_n = ma_n; \quad N_B - 40(9.81)\text{ N} = 40\text{ kg} \left( \frac{(12.13\text{ m/s})^2}{6.667\text{ m}} \right)$$

$$N_B = 1275.3\text{ N} = 1.28\text{ kN}$$

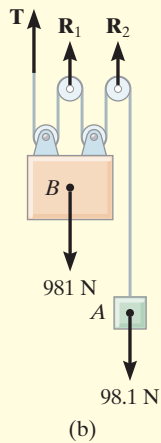
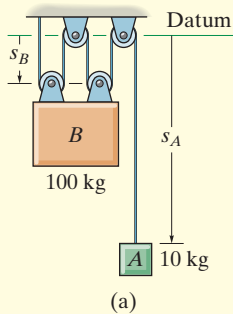
*Ans.*

(b)



(c)

**Fig. 14–13**

**EXAMPLE 14.6****Fig. 14–14**

Blocks *A* and *B* shown in Fig. 14–14*a* have a mass of 10 kg and 100 kg, respectively. Determine the distance *B* travels when it is released from rest to the point where its speed becomes 2 m/s.

**SOLUTION**

This problem may be solved by considering the blocks separately and applying the principle of work and energy to each block. However, the work of the (unknown) cable tension can be eliminated from the analysis by considering blocks *A* and *B* together as a *single system*.

**Work (Free-Body Diagram).** As shown on the free-body diagram of the system, Fig. 14–14*b*, the cable force **T** and reactions **R**<sub>1</sub> and **R**<sub>2</sub> do *no work*, since these forces represent the reactions at the supports and consequently they do not move while the blocks are displaced. The weights both do positive work if we *assume* both move downward, in the positive sense of direction of *s*<sub>A</sub> and *s*<sub>B</sub>.

**Principle of Work and Energy.** Realizing the blocks are released from rest, we have

$$\begin{aligned} \Sigma T_1 + \Sigma U_{1-2} &= \Sigma T_2 \\ \left\{ \frac{1}{2} m_A (v_A)_1^2 + \frac{1}{2} m_B (v_B)_1^2 \right\} + \{ W_A \Delta s_A + W_B \Delta s_B \} &= \left\{ \frac{1}{2} m_A (v_A)_2^2 + \frac{1}{2} m_B (v_B)_2^2 \right\} \\ \{ 0 + 0 \} + \{ 98.1 \text{ N} (\Delta s_A) + 981 \text{ N} (\Delta s_B) \} &= \left\{ \frac{1}{2} (10 \text{ kg}) (v_A)_2^2 + \frac{1}{2} (100 \text{ kg}) (2 \text{ m/s})^2 \right\} \end{aligned} \quad (1)$$

**Kinematics.** Using methods of kinematics, as discussed in Sec. 12.9, it may be seen from Fig. 14–14*a* that the total length *l* of all the vertical segments of cable may be expressed in terms of the position coordinates *s*<sub>A</sub> and *s*<sub>B</sub> as

$$s_A + 4s_B = l$$

Hence, a change in position yields the displacement equation

$$\begin{aligned} \Delta s_A + 4 \Delta s_B &= 0 \\ \Delta s_A &= -4 \Delta s_B \end{aligned}$$

Here we see that a downward displacement of one block produces an upward displacement of the other block. Note that  $\Delta s_A$  and  $\Delta s_B$  must have the *same* sign convention in both Eqs. 1 and 2. Taking the time derivative yields

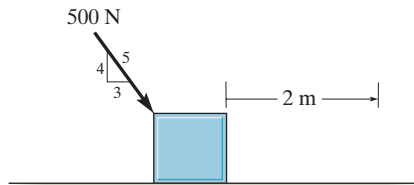
$$v_A = -4v_B = -4(2 \text{ m/s}) = -8 \text{ m/s} \quad (2)$$

Retaining the negative sign in Eq. 2 and substituting into Eq. 1 yields

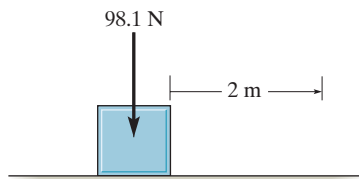
$$\Delta s_B = 0.883 \text{ m} \downarrow \quad \text{Ans.}$$

## PRELIMINARY PROBLEMS

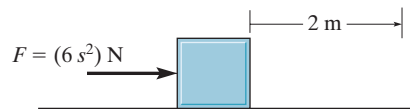
**P14-1.** Determine the work of the force when it displaces 2 m.



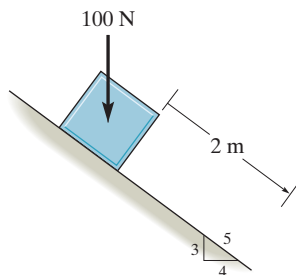
(a)



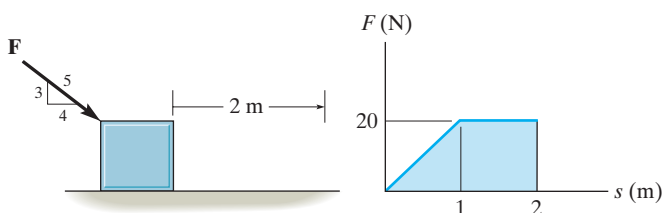
(b)



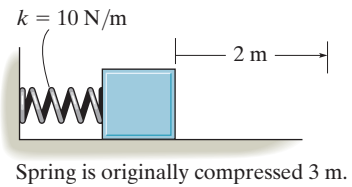
(c)



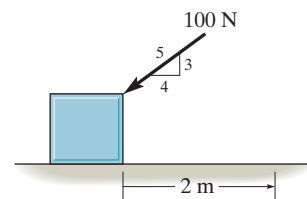
(d)



(e)



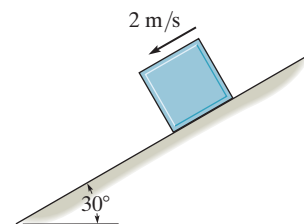
(f)



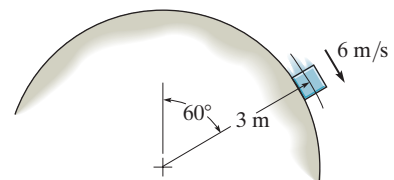
(g)

**Prob. P14-1**

**P14-2.** Determine the kinetic energy of the 10-kg block.



(a)

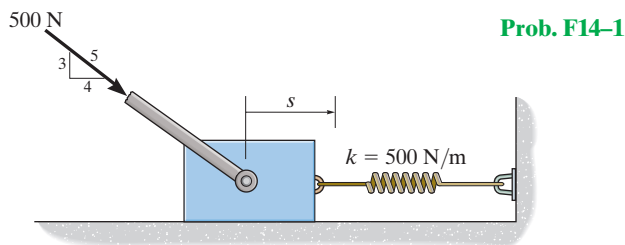


(b)

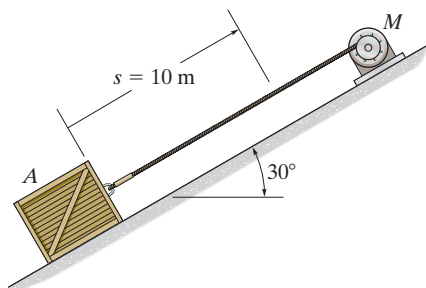
**Prob. P14-2**

## FUNDAMENTAL PROBLEMS

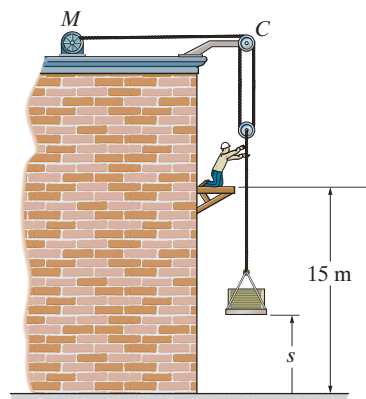
**F14-1.** The spring is placed between the wall and the 10-kg block. If the block is subjected to a force of  $F = 500$  N, determine its velocity when  $s = 0.5$  m. When  $s = 0$ , the block is at rest and the spring is uncompressed. The contact surface is smooth.



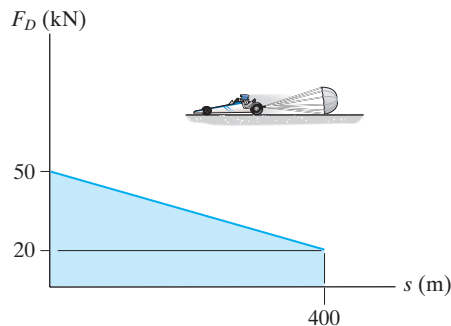
**F14-2.** If the motor exerts a constant force of 300 N on the cable, determine the speed of the 20-kg crate when it travels  $s = 10$  m up the plane, starting from rest. The coefficient of kinetic friction between the crate and the plane is  $\mu_k = 0.3$ .



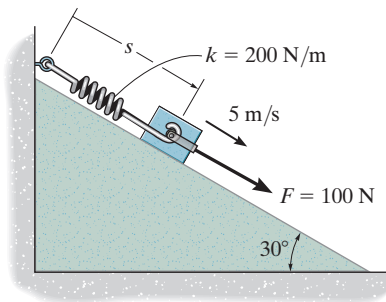
**F14-3.** If the motor exerts a force of  $F = (600 + 2s^2)$  N on the cable, determine the speed of the 100-kg crate when it rises to  $s = 15$  m. The crate is initially at rest on the ground.



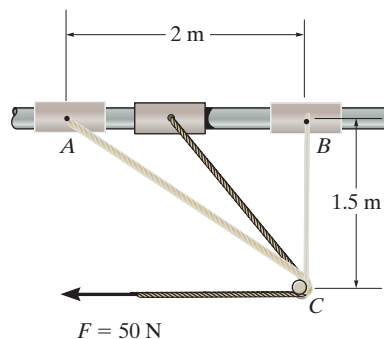
**F14-4.** The 1.8-Mg dragster is traveling at 125 m/s when the engine is shut off and the parachute is released. If the drag force of the parachute can be approximated by the graph, determine the speed of the dragster when it has traveled 400 m.



**F14-5.** When  $s = 0.6$  m, the spring is unstretched and the 10-kg block has a speed of 5 m/s down the smooth plane. Determine the distance  $s$  when the block stops.

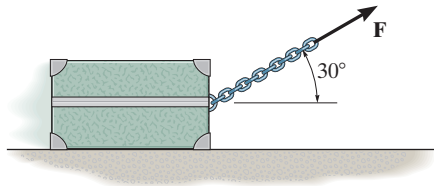


**F14-6.** The 2.5-kg collar is pulled by a cord that passes around a small peg at C. If the cord is subjected to a constant force of  $F = 50$  N, and the collar is at rest when it is at A, determine its speed when it reaches B. Neglect friction.



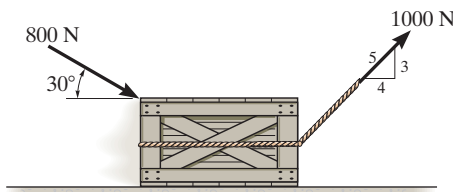
## PROBLEMS

**14-1.** The 20-kg crate is subjected to a force having a constant direction and a magnitude  $F = 100$  N. When  $s = 15$  m, the crate is moving to the right with a speed of 8 m/s. Determine its speed when  $s = 25$  m. The coefficient of kinetic friction between the crate and the ground is  $\mu_k = 0.25$ .



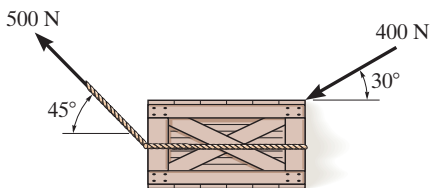
Prob. 14-1

**14-2.** The crate, which has a mass of 100 kg, is subjected to the action of the two forces. If it is originally at rest, determine the distance it slides in order to attain a speed of 6 m/s. The coefficient of kinetic friction between the crate and the surface is  $\mu_k = 0.2$ .



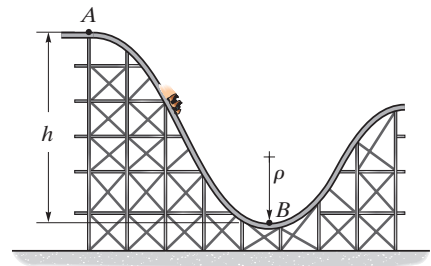
Prob. 14-2

**14-3.** The 100-kg crate is subjected to the forces shown. If it is originally at rest, determine the distance it slides in order to attain a speed of  $v = 8$  m/s. The coefficient of kinetic friction between the crate and the surface is  $\mu_k = 0.2$ .



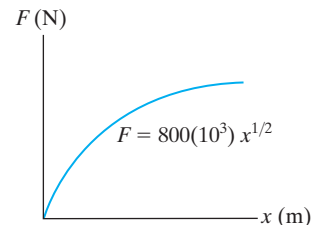
Prob. 14-3

**\*14-4.** Determine the required height  $h$  of the roller coaster so that when it is essentially at rest at the crest of the hill  $A$ , it will reach a speed of 100 km/h when it comes to the bottom  $B$ . Also, what should be the minimum radius of curvature  $\rho$  for the track at  $B$  so that the passengers do not experience a normal force greater than  $4mg = (39.24m)$  N? Neglect the size of the car and passenger.



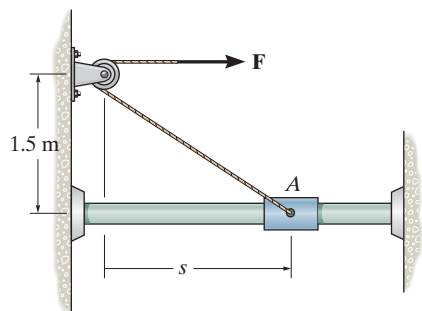
Prob. 14-4

**14-5.** For protection, the barrel barrier is placed in front of the bridge pier. If the relation between the force and deflection of the barrier is  $F = [800(10^3)x^{1/2}]$  N, where  $x$  is in m, determine the car's maximum penetration in the barrier. The car has a mass of 2 Mg and it is traveling with a speed of 20 m/s just before it hits the barrier.



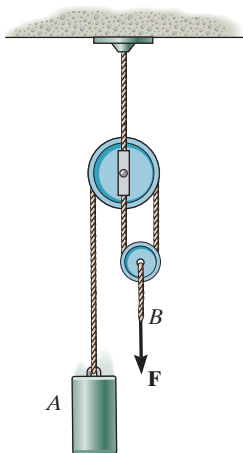
Prob. 14-5

**14-6.** The force of  $F = 50$  N is applied to the cord when  $s = 2$  m. If the 6-kg collar is originally at rest, determine its velocity at  $s = 0$ . Neglect friction.



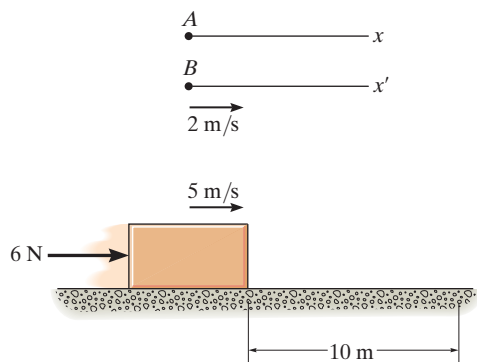
**Prob. 14-6**

**14-7.** A force of  $F = 250$  N is applied to the end at  $B$ . Determine the speed of the 10-kg block when it has moved 1.5 m, starting from rest.



**Prob. 14-7**

**\*14-8.** As indicated by the derivation, the principle of work and energy is valid for observers in *any* inertial reference frame. Show that this is so, by considering the 10-kg block which rests on the smooth surface and is subjected to a horizontal force of 6 N. If observer  $A$  is in a *fixed* frame  $x$ , determine the final speed of the block if it has an initial speed of 5 m/s and travels 10 m, both directed to the right and measured from the fixed frame. Compare the result with that obtained by an observer  $B$ , attached to the  $x'$  axis and moving at a constant velocity of 2 m/s relative to  $A$ . *Hint:* The distance the block travels will first have to be computed for observer  $B$  before applying the principle of work and energy.



**Prob. 14-8**

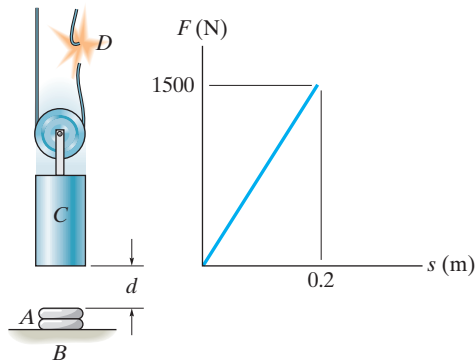
**14-9.** When the driver applies the brakes of a light truck traveling 40 km/h, it skids 3 m before stopping. How far will the truck skid if it is traveling at 80 km/h when the brakes are applied?



**Prob. 14-9**

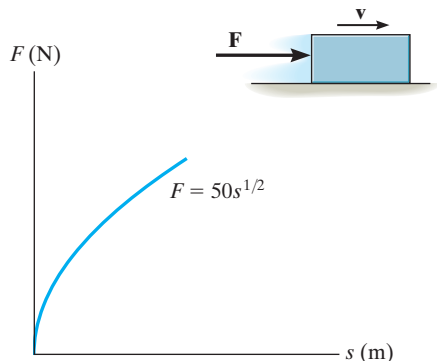


**14-10.** The “air spring”  $A$  is used to protect the support  $B$  and prevent damage to the conveyor-belt tensioning weight  $C$  in the event of a belt failure  $D$ . The force developed by the air spring as a function of its deflection is shown by the graph. If the block has a mass of 20 kg and is suspended a height  $d = 0.4$  m above the top of the spring, determine the maximum deformation of the spring in the event the conveyor belt fails. Neglect the mass of the pulley and belt.



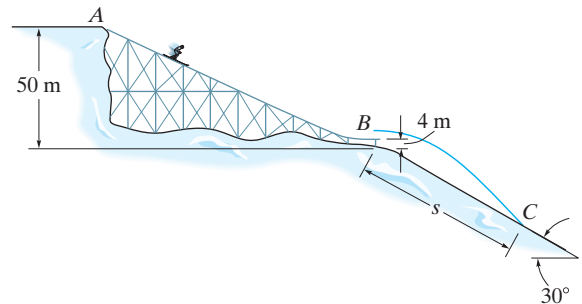
**Prob. 14-10**

**14-11.** The force  $\mathbf{F}$ , acting in a constant direction on the 20-kg block, has a magnitude which varies with the position  $s$  of the block. Determine how far the block must slide before its velocity becomes 15 m/s. When  $s = 0$  the block is moving to the right at  $v = 6$  m/s. The coefficient of kinetic friction between the block and surface is  $\mu_k = 0.3$ .



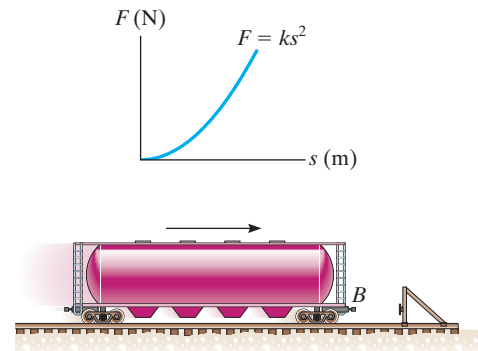
**Prob. 14-11**

**\*14-12.** The skier starts from rest at  $A$  and travels down the ramp. If friction and air resistance can be neglected, determine his speed  $v_B$  when he reaches  $B$ . Also, find the distance  $s$  to where he strikes the ground at  $C$ , if he makes the jump traveling horizontally at  $B$ . Neglect the skier's size. He has a mass of 70 kg.



**Prob. 14-12**

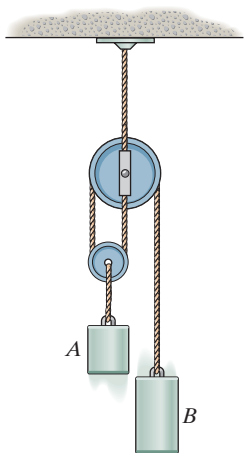
**14-13.** Design considerations for the bumper  $B$  on the 5-Mg train car require use of a nonlinear spring having the load-deflection characteristics shown in the graph. Select the proper value of  $k$  so that the maximum deflection of the spring is limited to 0.2 m when the car, traveling at 4 m/s, strikes the rigid stop. Neglect the mass of the car wheels.



**Prob. 14-13**

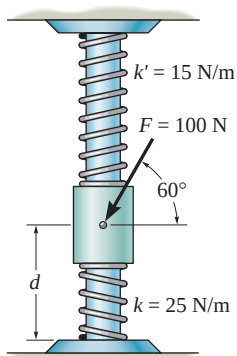
**14-14.** The 8-kg cylinder  $A$  and 3-kg cylinder  $B$  are released from rest. Determine the speed of  $A$  after it has moved 2 m starting from rest. Neglect the mass of the cord and pulleys.

**14-15.** Cylinder  $A$  has a mass of 3 kg and cylinder  $B$  has a mass of 8 kg. Determine the speed of  $A$  after it has moved 2 m starting from rest. Neglect the mass of the cord and pulleys.



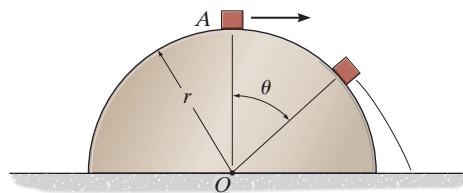
**Probs. 14-14/15**

**\*14-16.** The collar has a mass of 20 kg and is supported on the smooth rod. The attached springs are undeformed when  $d = 0.5$  m. Determine the speed of the collar after the applied force  $F = 100$  N causes it to be displaced so that  $d = 0.3$  m. When  $d = 0.5$  m the collar is at rest.



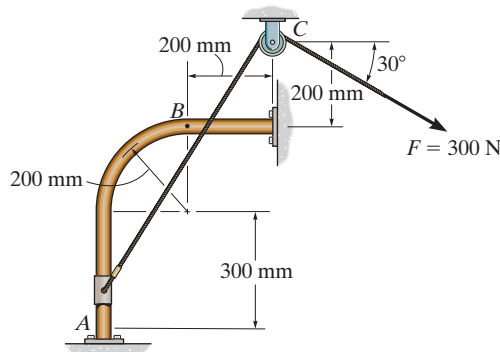
**Prob. 14-16**

**14-17.** A small box of mass  $m$  is given a speed of  $v = \sqrt{\frac{1}{4}gr}$  at the top of the smooth half cylinder. Determine the angle  $\theta$  at which the box leaves the cylinder.



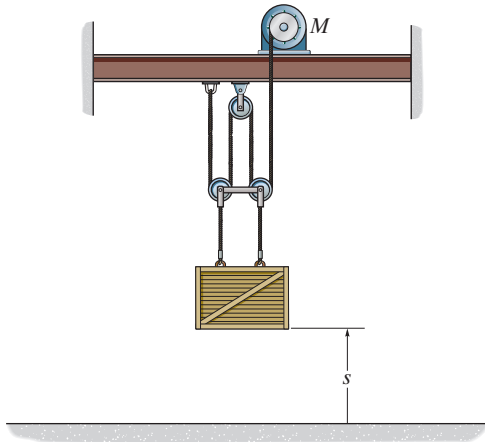
**Prob. 14-17**

**14-18.** If the cord is subjected to a constant force of  $F = 300$  N and the 15-kg smooth collar starts from rest at  $A$ , determine the velocity of the collar when it reaches point  $B$ . Neglect the size of the pulley.



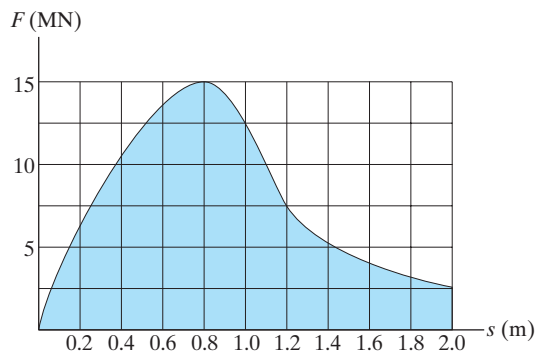
**Prob. 14-18**

**14–19.** If the force exerted by the motor  $M$  on the cable is 250 N, determine the speed of the 100-kg crate when it is hoisted to  $s = 3$  m. The crate is at rest when  $s = 0$ .



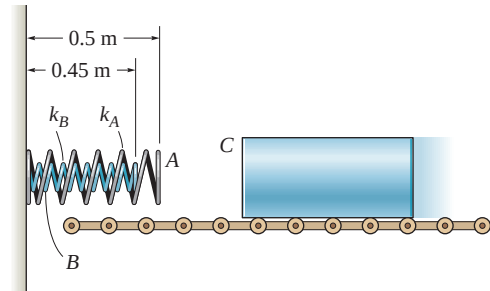
**Prob. 14–19**

**\*14–20.** When a 7-kg projectile is fired from a cannon barrel that has a length of 2 m, the explosive force exerted on the projectile, while it is in the barrel, varies in the manner shown. Determine the approximate muzzle velocity of the projectile at the instant it leaves the barrel. Neglect the effects of friction inside the barrel and assume the barrel is horizontal.



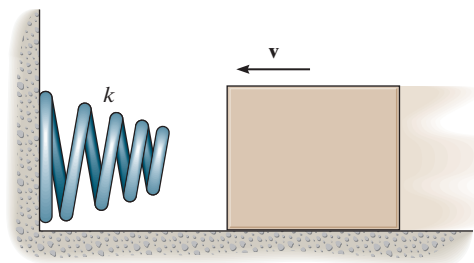
**Prob. 14–20**

**14–21.** The steel ingot has a mass of 1800 kg. It travels along the conveyor at a speed  $v = 0.5$  m/s when it collides with the “nested” spring assembly. If the stiffness of the outer spring is  $k_A = 5$  kN/m, determine the required stiffness  $k_B$  of the inner spring so that the motion of the ingot is stopped at the moment the front,  $C$ , of the ingot is 0.3 m from the wall.



**Prob. 14–21**

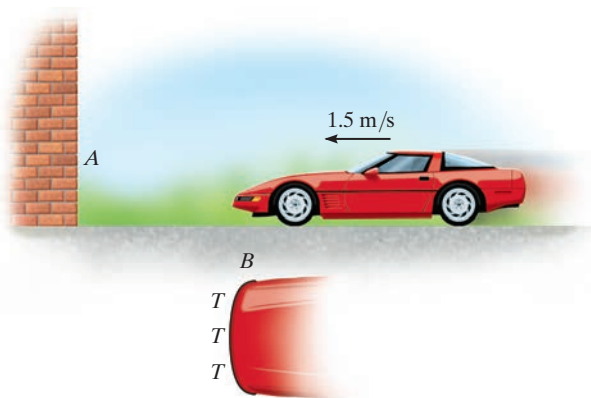
**14–22.** The 1.5-kg block slides along a smooth plane and strikes a *nonlinear spring* with a speed of  $v = 4$  m/s. The spring is termed “nonlinear” because it has a resistance of  $F_s = ks^2$ , where  $k = 900$  N/m<sup>2</sup>. Determine the speed of the block after it has compressed the spring  $s = 0.2$  m.



**Prob. 14–22**

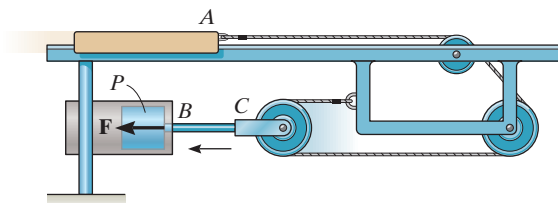
**14-23.** A car is equipped with a bumper  $B$  designed to absorb collisions. The bumper is mounted to the car using pieces of flexible tubing  $T$ . Upon collision with a rigid barrier at  $A$ , a constant horizontal force  $\mathbf{F}$  is developed which causes a car deceleration of  $3g = 29.43 \text{ m/s}^2$  (the highest safe deceleration for a passenger without a seatbelt). If the car and passenger have a total mass of  $1.5 \text{ Mg}$  and the car is initially coasting with a speed of  $1.5 \text{ m/s}$ , determine the magnitude of  $\mathbf{F}$  needed to stop the car and the deformation  $x$  of the bumper tubing.

14



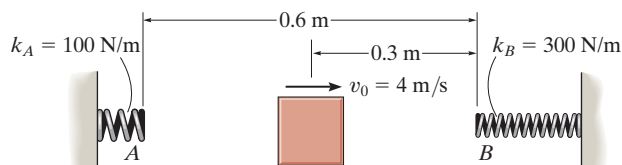
Prob. 14-23

**\*14-24.** The catapulting mechanism is used to propel the  $10\text{-kg}$  slider  $A$  to the right along the smooth track. The propelling action is obtained by drawing the pulley attached to rod  $BC$  rapidly to the left by means of a piston  $P$ . If the piston applies a constant force  $F = 20 \text{ kN}$  to rod  $BC$  such that it moves it  $0.2 \text{ m}$ , determine the speed attained by the slider if it was originally at rest. Neglect the mass of the pulleys, cable, piston, and rod  $BC$ .



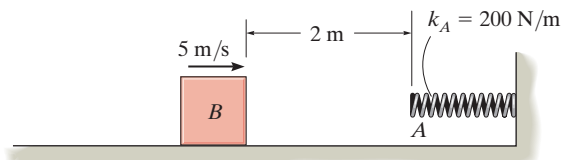
Prob. 14-24

**14-25.** The  $12\text{-kg}$  block has an initial speed of  $v_0 = 4 \text{ m/s}$  when it is midway between springs  $A$  and  $B$ . After striking spring  $B$ , it rebounds and slides across the horizontal plane toward spring  $A$ , etc. If the coefficient of kinetic friction between the plane and the block is  $\mu_k = 0.4$ , determine the total distance traveled by the block before it comes to rest.



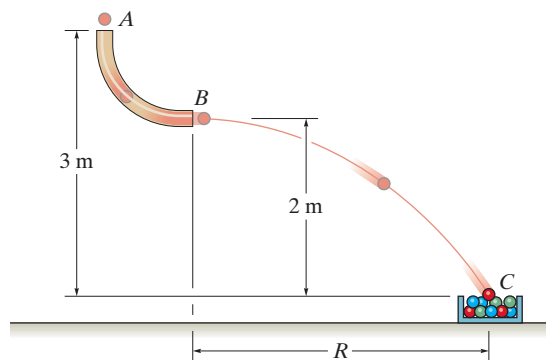
Prob. 14-25

**14-26.** The  $8\text{-kg}$  block is moving with an initial speed of  $5 \text{ m/s}$ . If the coefficient of kinetic friction between the block and plane is  $\mu_k = 0.25$ , determine the compression in the spring when the block momentarily stops.



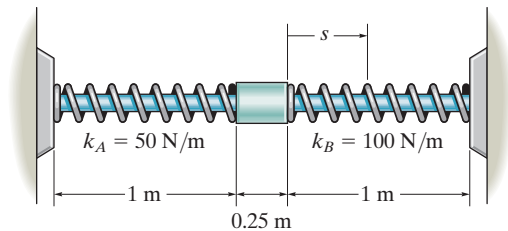
Prob. 14-26

**14-27.** Marbles having a mass of  $5 \text{ g}$  are dropped from rest at  $A$  through the smooth glass tube and accumulate in the can at  $C$ . Determine the placement  $R$  of the can from the end of the tube and the speed at which the marbles fall into the can. Neglect the size of the can.



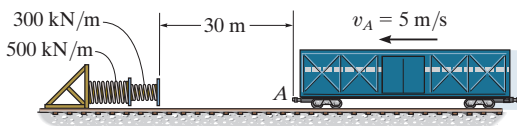
Prob. 14-27

**\*14–28.** The collar has a mass of 20 kg and slides along the smooth rod. Two springs are attached to it and the ends of the rod as shown. If each spring has an uncompressed length of 1 m and the collar has a speed of 2 m/s when  $s = 0$ , determine the maximum compression of each spring due to the back-and-forth (oscillating) motion of the collar.



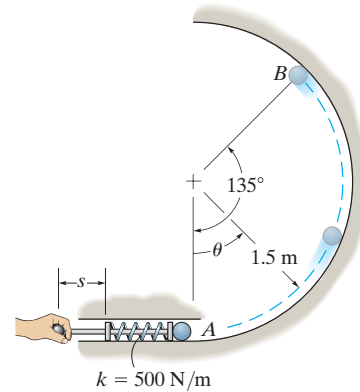
**Prob. 14–28**

**14–29.** The train car has a mass of 10 Mg and is traveling at 5 m/s when it reaches A. If the rolling resistance is 1/100 of the weight of the car, determine the compression of each spring when the car is momentarily brought to rest.



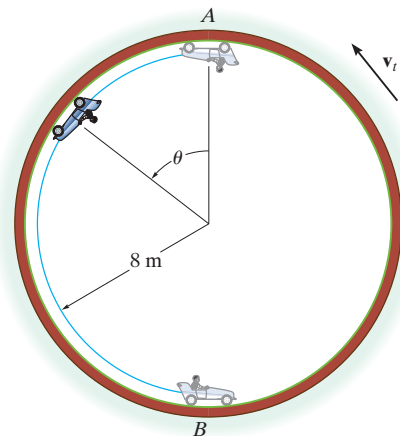
**Prob. 14–29**

**14–30.** The 0.5-kg ball is fired up the smooth vertical circular track using the spring plunger. The plunger keeps the spring compressed 0.08 m when  $s = 0$ . Determine how far  $s$  it must be pulled back and released so that the ball will begin to leave the track when  $\theta = 135^\circ$ .



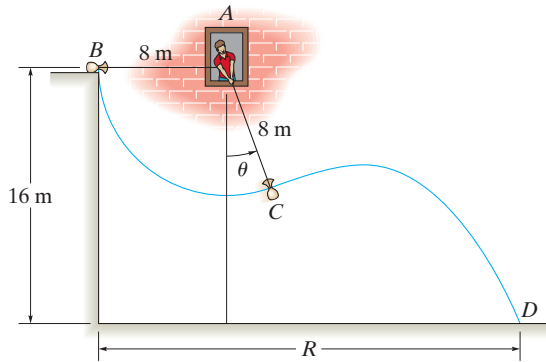
**Prob. 14–30**

**14–31.** The “flying car” is a ride at an amusement park which consists of a car having wheels that roll along a track mounted inside a rotating drum. By design the car cannot fall off the track, however motion of the car is developed by applying the car’s brake, thereby gripping the car to the track and allowing it to move with a constant speed of the track,  $v_t = 3$  m/s. If the rider applies the brake when going from B to A and then releases it at the top of the drum, A, so that the car coasts freely down along the track to B ( $\theta = \pi$  rad), determine the speed of the car at B and the normal reaction which the drum exerts on the car at B. Neglect friction during the motion from A to B. The rider and car have a total mass of 250 kg and the center of mass of the car and rider moves along a circular path having a radius of 8 m.



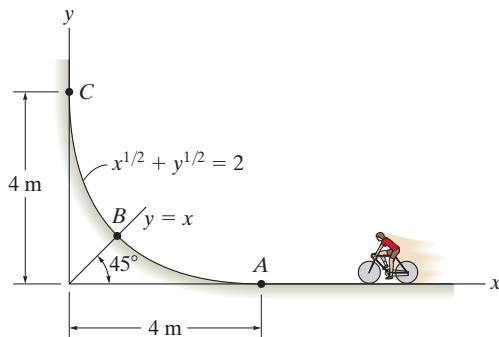
**Prob. 14–31**

**\*14-32.** The man at the window  $A$  wishes to throw the 30-kg sack on the ground. To do this he allows it to swing from rest at  $B$  to point  $C$ , when he releases the cord at  $\theta = 30^\circ$ . Determine the speed at which it strikes the ground and the distance  $R$ .



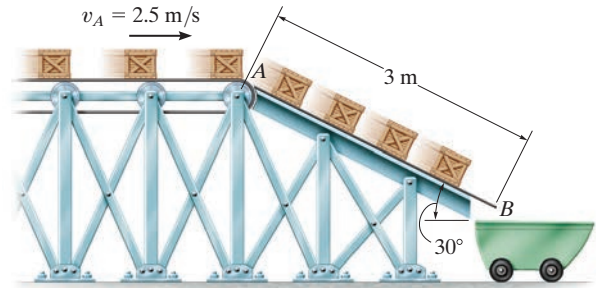
**Prob. 14-32**

**14-33.** The cyclist travels to point  $A$ , pedaling until he reaches a speed  $v_A = 4 \text{ m/s}$ . He then coasts freely up the curved surface. Determine how high he reaches up the surface before he comes to a stop. Also, what are the resultant normal force on the surface at this point and his acceleration? The total mass of the bike and man is 75 kg. Neglect friction, the mass of the wheels, and the size of the bicycle.



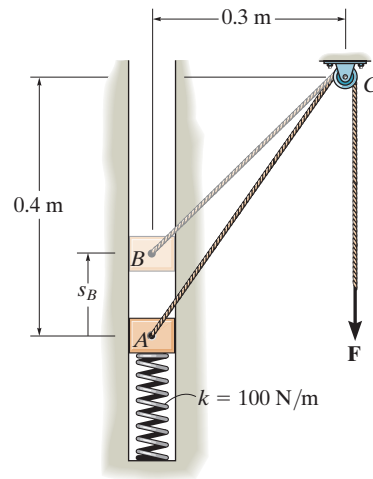
**Prob. 14-33**

**14-34.** The conveyor belt delivers each 12-kg crate to the ramp at  $A$  such that the crate's velocity is  $v_A = 2.5 \text{ m/s}$ , directed down *along* the ramp. If the coefficient of kinetic friction between each crate and the ramp is  $\mu_k = 0.3$ , determine the speed at which each crate slides off the ramp at  $B$ . Assume that no tipping occurs.



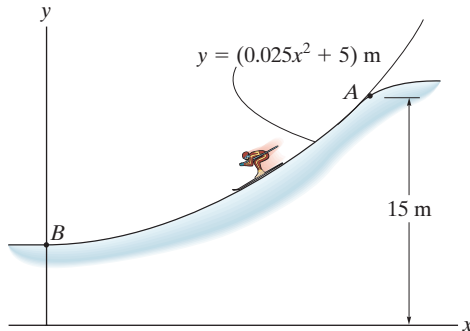
**Prob. 14-34**

**14-35.** The block has a mass of 0.8 kg and moves within the smooth vertical slot. If it starts from rest when the *attached* spring is in the unstretched position at  $A$ , determine the *constant* vertical force  $F$  which must be applied to the cord so that the block attains a speed  $v_B = 2.5 \text{ m/s}$  when it reaches  $B$ ;  $s_B = 0.15 \text{ m}$ . Neglect the size and mass of the pulley. *Hint:* The work of  $\mathbf{F}$  can be determined by finding the difference  $\Delta l$  in cord lengths  $AC$  and  $BC$  and using  $U_F = F \Delta l$ .



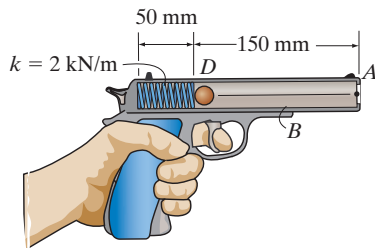
**Prob. 14-35**

**\*14-36.** If the 60-kg skier passes point  $A$  with a speed of 5 m/s, determine his speed when he reaches point  $B$ . Also find the normal force exerted on him by the slope at this point. Neglect friction.



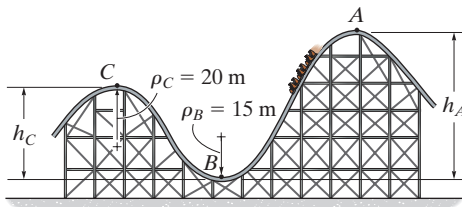
**Prob. 14-36**

**14-37.** The spring in the toy gun has an unstretched length of 100 mm. It is compressed and locked in the position shown. When the trigger is pulled, the spring unstretches 12.5 mm, and the 20-g ball moves along the barrel. Determine the speed of the ball when it leaves the gun. Neglect friction.



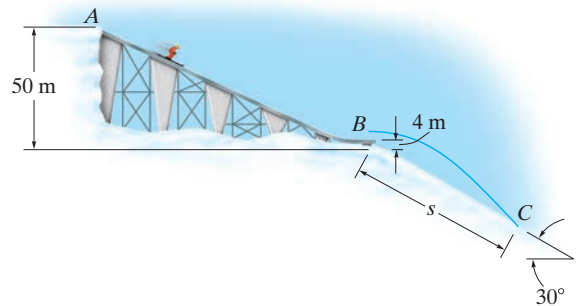
**Prob. 14-37**

**14-38.** If the track is to be designed so that the passengers of the roller coaster do not experience a normal force equal to zero or more than 4 times their weight, determine the limiting heights  $h_A$  and  $h_C$  so that this does not occur. The roller coaster starts from rest at position  $A$ . Neglect friction.



**Prob. 14-38**

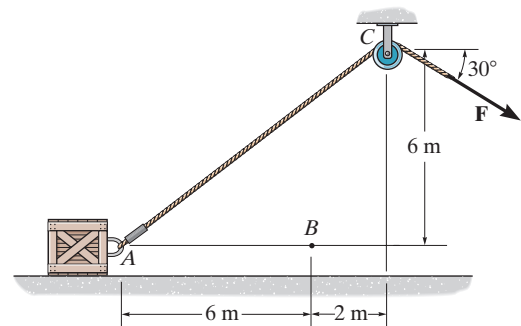
**14-39.** The skier starts from rest at  $A$  and travels down the ramp. If friction and air resistance can be neglected, determine his speed  $v_B$  when he reaches  $B$ . Also, find the distance  $s$  to where he strikes the ground at  $C$ , if he makes the jump traveling horizontally at  $B$ . Neglect the skier's size. He has a mass 75 kg.



**Prob. 14-39**

**\*14-40.** If the 75-kg crate starts from rest at  $A$ , determine its speed when it reaches point  $B$ . The cable is subjected to a constant force of  $F = 300$  N. Neglect friction and the size of the pulley.

**14-41.** If the 75-kg crate starts from rest at  $A$ , and its speed is 6 m/s when it passes point  $B$ , determine the constant force  $F$  exerted on the cable. Neglect friction and the size of the pulley.



**Probs. 14-40/41**

## 14.4 Power and Efficiency

**Power.** The term “power” provides a useful basis for choosing the type of motor or machine which is required to do a certain amount of work in a given time. For example, two pumps may each be able to empty a reservoir if given enough time; however, the pump having the larger power will complete the job sooner.

The *power* generated by a machine or engine that performs an amount of work  $dU$  within the time interval  $dt$  is therefore

$$P = \frac{dU}{dt} \quad (14-9)$$

If the work  $dU$  is expressed as  $dU = \mathbf{F} \cdot d\mathbf{r}$ , then

$$P = \frac{dU}{dt} = \frac{\mathbf{F} \cdot d\mathbf{r}}{dt} = \mathbf{F} \cdot \frac{d\mathbf{r}}{dt}$$

or

$$P = \mathbf{F} \cdot \mathbf{v} \quad (14-10)$$

Hence, power is a *scalar*, where in this formulation  $\mathbf{v}$  represents the velocity of the particle which is acted upon by the force  $\mathbf{F}$ .

The basic unit of power used in the SI system is the watt (W). These units are defined as

$$1 \text{ W} = 1 \text{ J/s} = 1 \text{ N} \cdot \text{m/s}$$



The power output of this locomotive comes from the driving frictional force developed at its wheels. It is this force that overcomes the frictional resistance of the cars in tow and is able to lift the weight of the train up the grade.

**Efficiency.** The *mechanical efficiency* of a machine is defined as the ratio of the output of useful power produced by the machine to the input of power supplied to the machine. Hence,

$$\epsilon = \frac{\text{power output}}{\text{power input}} \quad (14-11)$$



If energy supplied to the machine occurs during the *same time interval* at which it is drawn, then the efficiency may also be expressed in terms of the ratio

$$\varepsilon = \frac{\text{energy output}}{\text{energy input}} \quad (14-12)$$

Since machines consist of a series of moving parts, frictional forces will always be developed within the machine, and as a result, extra energy or power is needed to overcome these forces. Consequently, power output will be less than power input and so *the efficiency of a machine is always less than 1*.

The power supplied to a body can be determined using the following procedure.

### Procedure for Analysis

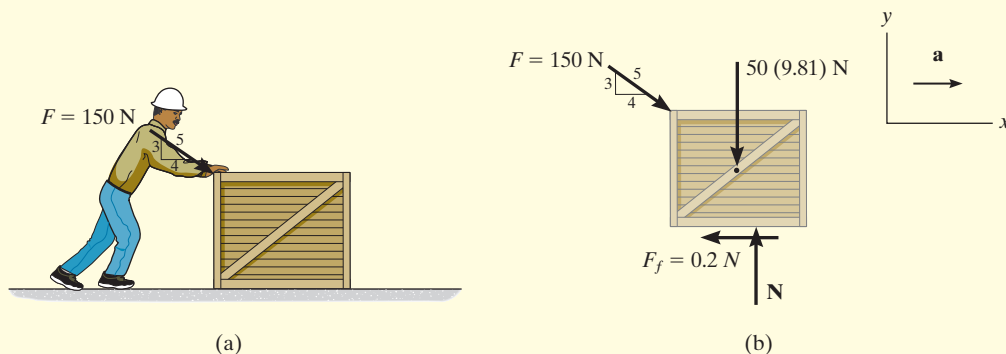
- First determine the external force  $\mathbf{F}$  acting on the body which causes the motion. This force is usually developed by a machine or engine placed either within or external to the body.
- If the body is accelerating, it may be necessary to draw its free-body diagram and apply the equation of motion ( $\Sigma \mathbf{F} = m\mathbf{a}$ ) to determine  $\mathbf{F}$ .
- Once  $\mathbf{F}$  and the velocity  $\mathbf{v}$  of the particle where  $\mathbf{F}$  is applied have been found, the power is determined by multiplying the force magnitude with the component of velocity acting in the direction of  $\mathbf{F}$ , (i.e.,  $P = \mathbf{F} \cdot \mathbf{v} = Fv \cos \theta$ ).
- In some problems the power may be found by calculating the work done by  $\mathbf{F}$  per unit of time ( $P_{\text{avg}} = \Delta U / \Delta t$ ).



The power requirement of this hoist depends upon the vertical force  $\mathbf{F}$  that acts on the elevator and causes it to move upward. If the velocity of the elevator is  $\mathbf{v}$ , then the power output is  $P = \mathbf{F} \cdot \mathbf{v}$ .

**EXAMPLE 14.7**

The man in Fig. 14–15*a* pushes on the 50-kg crate with a force of  $F = 150$  N. Determine the power supplied by the man when  $t = 4$  s. The coefficient of kinetic friction between the floor and the crate is  $\mu_k = 0.2$ . Initially the crate is at rest.

**Fig. 14–15****SOLUTION**

To determine the power developed by the man, the velocity of the 150-N force must be obtained first. The free-body diagram of the crate is shown in Fig. 14–15*b*. Applying the equation of motion,

$$+\uparrow \Sigma F_y = ma_y; \quad N - \left(\frac{3}{5}\right)150 \text{ N} - 50(9.81) \text{ N} = 0$$

$$N = 580.5 \text{ N}$$

$$+\rightarrow \Sigma F_x = ma_x; \quad \left(\frac{4}{5}\right)150 \text{ N} - 0.2(580.5 \text{ N}) = (50 \text{ kg})a$$

$$a = 0.078 \text{ m/s}^2$$

The velocity of the crate when  $t = 4$  s is therefore

$$(\rightarrow) \quad v = v_0 + a_c t$$

$$v = 0 + (0.078 \text{ m/s}^2)(4 \text{ s}) = 0.312 \text{ m/s}$$

The power supplied to the crate by the man when  $t = 4$  s is therefore

$$P = \mathbf{F} \cdot \mathbf{v} = F_x v = \left(\frac{4}{5}\right)(150 \text{ N})(0.312 \text{ m/s})$$

$$= 37.4 \text{ W}$$

*Ans.*

**EXAMPLE 14.8**

The motor  $M$  of the hoist shown in Fig. 14–16a lifts the 35-kg crate  $C$  so that the acceleration of point  $P$  is  $1.2 \text{ m/s}^2$ . Determine the power that must be supplied to the motor at the instant  $P$  has velocity of  $0.6 \text{ m/s}$ . Neglect the mass of the pulley and cable and take  $\varepsilon = 0.85$ .

**SOLUTION**

In order to find the power output of the motor, it is first necessary to determine the tension in the cable since this force is developed by the motor.

From the free-body diagram, Fig. 14–16b, we have

$$+\downarrow \Sigma F_y = ma_y; \quad -2T + 35(9.81) \text{ N} = (35 \text{ kg}) a_c \quad (1)$$

The acceleration of the crate can be obtained by using kinematics to relate it to the known acceleration of point  $P$ , Fig. 14–16a. Using the methods of absolute dependent motion, the coordinates  $s_C$  and  $s_P$  can be related to a constant portion of cable length  $l$  which is changing in the vertical and horizontal directions. We have  $2s_C + s_P = l$ . Taking the second time derivative of this equation yields

$$2a_C = -a_P \quad (2)$$

Since  $a_P = +1.2 \text{ m/s}^2$ , then  $a_C = -(1.2 \text{ m/s}^2)/2 = -0.6 \text{ m/s}^2$ . What does the negative sign indicate? Substituting this result into Eq. 1 and *retaining* the negative sign since the acceleration in *both* Eq. 1 and Eq. 2 was considered positive downward, we have

$$-2T + 35(9.81) \text{ N} = (35 \text{ kg})(-0.6 \text{ m/s}^2)$$

$$T = 182.2 \text{ N}$$

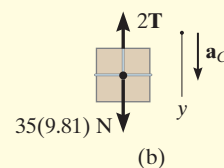
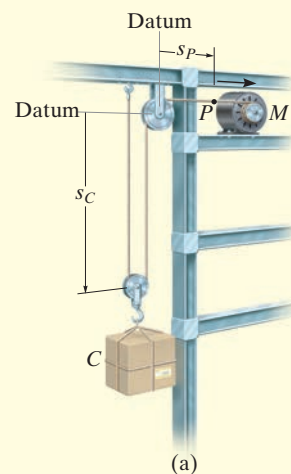
The power output required to draw the cable in at a rate of  $0.6 \text{ m/s}$  is therefore

$$\begin{aligned} P &= \mathbf{T} \cdot \mathbf{v} = (182.2 \text{ N})(0.6 \text{ m/s}) \\ &= 109.3 \text{ W} \end{aligned}$$

This *power output* requires that the motor provide a *power input* of

$$\begin{aligned} \text{power input} &= \frac{1}{\varepsilon}(\text{power output}) \\ &= \frac{1}{0.85}(109.3 \text{ W}) = 129 \text{ W} \end{aligned} \quad \text{Ans.}$$

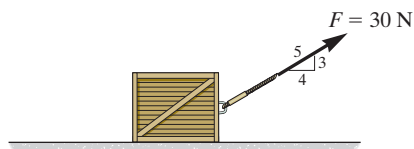
**NOTE:** Since the velocity of the crate is constantly changing, the power requirement is *instantaneous*.



**Fig. 14–16**

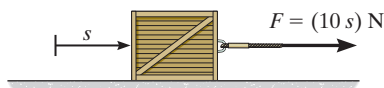
## FUNDAMENTAL PROBLEMS

**F14-7.** If the contact surface between the 20-kg block and the ground is smooth, determine the power of force  $\mathbf{F}$  when  $t = 4$  s. Initially, the block is at rest.



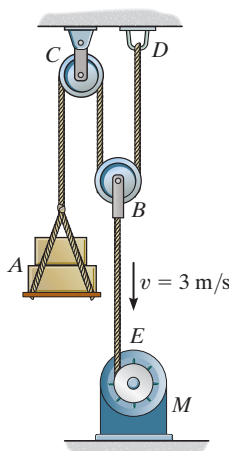
Prob. F14-7

**F14-8.** If  $F = (10s)$  N, where  $s$  is in meters, and the contact surface between the block and the ground is smooth, determine the power of force  $\mathbf{F}$  when  $s = 5$  m. When  $s = 0$ , the 20-kg block is moving at  $v = 1$  m/s.



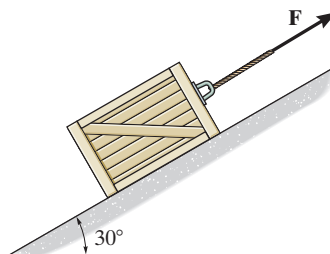
Prob. F14-8

**F14-9.** If the motor winds in the cable with a constant speed of  $v = 3$  m/s, determine the power supplied to the motor. The load weighs 100 N and the efficiency of the motor is  $\epsilon = 0.8$ . Neglect the mass of the pulleys.



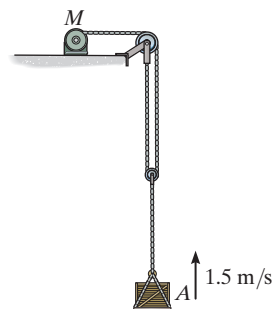
Prob. F14-9

**F14-10.** The coefficient of kinetic friction between the 20-kg block and the inclined plane is  $\mu_k = 0.2$ . If the block is traveling up the inclined plane with a constant velocity  $v = 5$  m/s, determine the power of force  $\mathbf{F}$ .



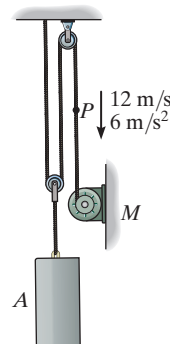
Prob. F14-10

**F14-11.** If the 50-kg load  $A$  is hoisted by motor  $M$  so that the load has a constant velocity of 1.5 m/s, determine the power input to the motor, which operates at an efficiency  $\epsilon = 0.8$ .



Prob. F14-11

**F14-12.** At the instant shown, point  $P$  on the cable has a velocity  $v_P = 12$  m/s, which is increasing at a rate of  $a_P = 6$  m/s<sup>2</sup>. Determine the power input to motor  $M$  at this instant if it operates with an efficiency  $\epsilon = 0.8$ . The mass of block  $A$  is 50 kg.



Prob. F14-12

## PROBLEMS

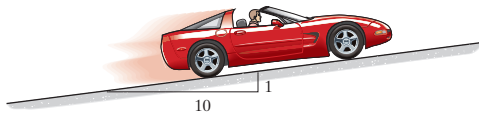
**14-42.** A spring having a stiffness of  $5 \text{ kN/m}$  is compressed  $400 \text{ mm}$ . The stored energy in the spring is used to drive a machine which requires  $90 \text{ W}$  of power. Determine how long the spring can supply energy at the required rate.

**14-43.** To dramatize the loss of energy in an automobile, consider a car having a weight of  $25\,000 \text{ N}$  that is traveling at  $56 \text{ km/h}$ . If the car is brought to a stop, determine how long a  $100\text{-W}$  light bulb must burn to expend the same amount of energy.

**\*14-44.** If the engine of a  $1.5\text{-Mg}$  car generates a constant power of  $15 \text{ kW}$ , determine the speed of the car after it has traveled a distance of  $200 \text{ m}$  on a level road starting from rest. Neglect friction.

**14-45.** If the engine of a  $1.5\text{-Mg}$  car generates a constant power of  $15 \text{ kW}$ , determine the speed of the car after it has traveled a distance of  $200 \text{ m}$  on a level road starting from rest. Neglect friction.

**14-46.** The  $2\text{-Mg}$  car increases its speed uniformly from rest to  $25 \text{ m/s}$  in  $30 \text{ s}$  up the inclined road. Determine the maximum power that must be supplied by the engine, which operates with an efficiency of  $\varepsilon = 0.8$ . Also, find the average power supplied by the engine.



**Prob. 14-46**

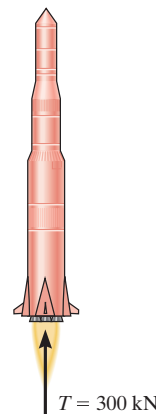
**14-47.** A car has a mass  $m$  and accelerates along a horizontal straight road from rest such that the power is always a constant amount  $P$ . Determine how far it must travel to reach a speed of  $v$ .

**\*14-48.** An automobile having a mass of  $2 \text{ Mg}$  travels up a  $7^\circ$  slope at a constant speed of  $v = 100 \text{ km/h}$ . If mechanical friction and wind resistance are neglected, determine the power developed by the engine if the automobile has an efficiency  $\varepsilon = 0.65$ .



**Prob. 14-48**

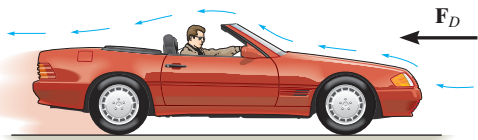
**14-49.** A rocket having a total mass of  $8 \text{ Mg}$  is fired vertically from rest. If the engines provide a constant thrust of  $T = 300 \text{ kN}$ , determine the power output of the engines as a function of time. Neglect the effect of drag resistance and the loss of fuel mass and weight.



**Prob. 14-49**

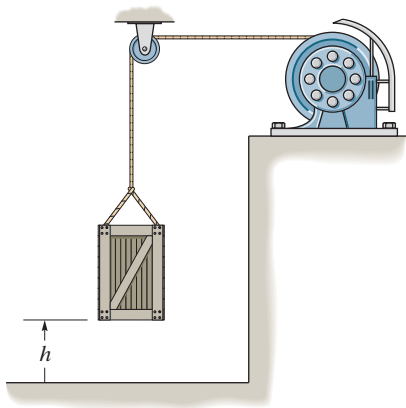
**14-50.** The sports car has a mass of 2.3 Mg, and while it is traveling at 28 m/s the driver causes it to accelerate at  $5 \text{ m/s}^2$ . If the drag resistance on the car due to the wind is  $F_D = (0.3v^2) \text{ N}$ , where  $v$  is the velocity in m/s, determine the power supplied to the engine at this instant. The engine has a running efficiency of  $\varepsilon = 0.68$ .

- 14** **14-51.** The sports car has a mass of 2.3 Mg and accelerates at  $6 \text{ m/s}^2$ , starting from rest. If the drag resistance on the car due to the wind is  $F_D = (10v) \text{ N}$ , where  $v$  is the velocity in m/s, determine the power supplied to the engine when  $t = 5 \text{ s}$ . The engine has a running efficiency of  $\varepsilon = 0.68$ .



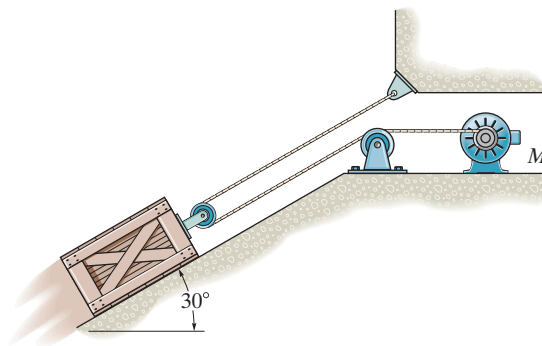
**Probs. 14-50/51**

**\*14-52.** A motor hoists a 60-kg crate at a constant velocity to a height of  $h = 5 \text{ m}$  in 2 s. If the indicated power of the motor is 3.2 kW, determine the motor's efficiency.



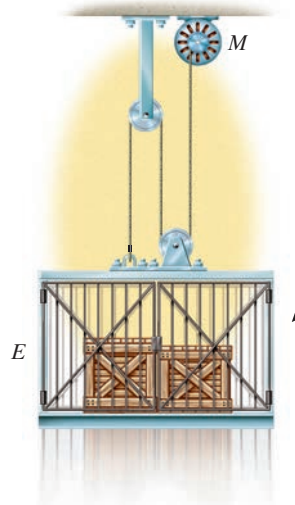
**Prob. 14-52**

**14-53.** The 50-kg crate is hoisted up the  $30^\circ$  incline by the pulley system and motor  $M$ . If the crate starts from rest and, by constant acceleration, attains a speed of 4 m/s after traveling 8 m along the plane, determine the power that must be supplied to the motor at the instant. Neglect friction along the plane. The motor has an efficiency of  $\varepsilon = 0.74$ .



**Prob. 14-53**

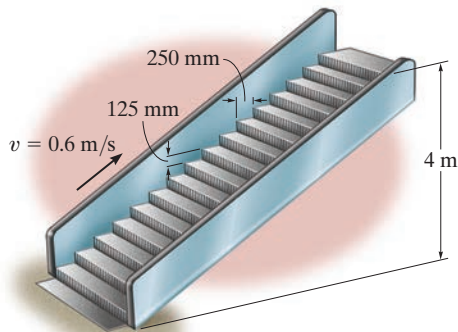
**14-54.** The 500-kg elevator starts from rest and travels upward with a constant acceleration  $a_c = 2 \text{ m/s}^2$ . Determine the power output of the motor  $M$  when  $t = 3 \text{ s}$ . Neglect the mass of the pulleys and cable.



**Prob. 14-54**

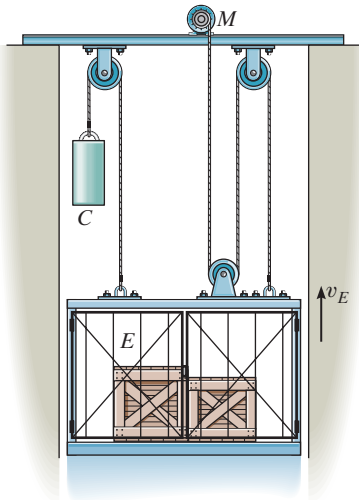
**14-55.** The escalator steps move with a constant speed of  $0.6 \text{ m/s}$ . If the steps are  $125 \text{ mm}$  high and  $250 \text{ mm}$  in length, determine the power of a motor needed to lift an average mass of  $150 \text{ kg}$  per step. There are 32 steps.

**\*14-56.** If the escalator in Prob. 14-55 is not moving, determine the constant speed at which a man having a mass of  $80 \text{ kg}$  must walk up the steps to generate  $100 \text{ W}$  of power—the same amount that is needed to power a standard light bulb.



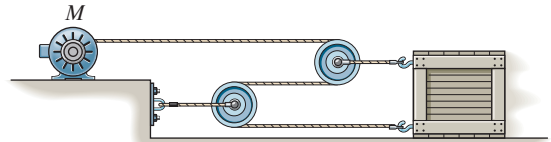
**Probs. 14-55/56**

**14-57.** The elevator  $E$  and its freight have a total mass of  $400 \text{ kg}$ . Hoisting is provided by the motor  $M$  and the  $60\text{-kg}$  block  $C$ . If the motor has an efficiency of  $\varepsilon = 0.6$ , determine the power that must be supplied to the motor when the elevator is hoisted upward at a constant speed of  $v_E = 4 \text{ m/s}$ .



**Prob. 14-57**

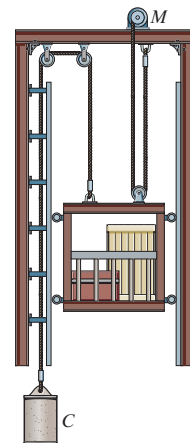
**14-58.** The crate has a mass of  $150 \text{ kg}$  and rests on a surface for which the coefficients of static and kinetic friction are  $\mu_s = 0.3$  and  $\mu_k = 0.2$ , respectively. If the motor  $M$  supplies a cable force of  $F = (8t^2 + 20) \text{ N}$ , where  $t$  is in seconds, determine the power output developed by the motor when  $t = 5 \text{ s}$ .



**Prob. 14-58**

**14-59.** The material hoist and the load have a total mass of  $800 \text{ kg}$  and the counterweight  $C$  has a mass of  $150 \text{ kg}$ . At a given instant, the hoist has an upward velocity of  $2 \text{ m/s}$  and an acceleration of  $1.5 \text{ m/s}^2$ . Determine the power generated by the motor  $M$  at this instant if it operates with an efficiency of  $\varepsilon = 0.8$ .

**\*14-60.** The material hoist and the load have a total mass of  $800 \text{ kg}$  and the counterweight  $C$  has a mass of  $150 \text{ kg}$ . If the upward speed of the hoist increases uniformly from  $0.5 \text{ m/s}$  to  $1.5 \text{ m/s}$  in  $1.5 \text{ s}$ , determine the average power generated by the motor  $M$  during this time. The motor operates with an efficiency of  $\varepsilon = 0.8$ .

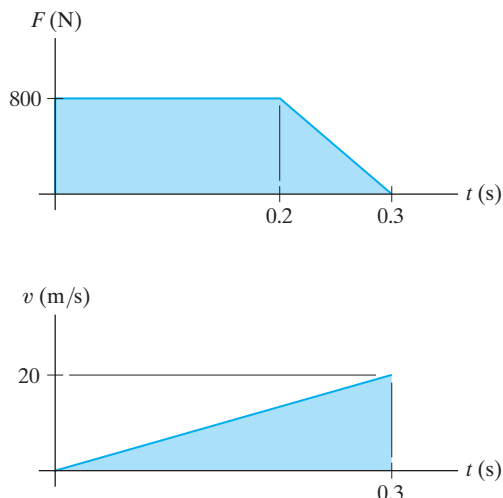


**Probs. 14-59/60**

**14–61.** An athlete pushes against an exercise machine with a force that varies with time as shown in the first graph. Also, the velocity of the athlete's arm acting in the same direction as the force varies with time as shown in the second graph. Determine the power applied as a function of time and the work done in  $t = 0.3$  s.

**14–62.** An athlete pushes against an exercise machine with a force that varies with time as shown in the first graph. Also, the velocity of the athlete's arm acting in the same direction as the force varies with time as shown in the second graph. Determine the maximum power developed during the 0.3-second time period.

14



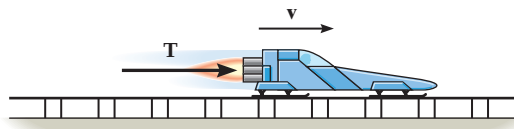
**Probs. 14–61/62**

**14–63.** If the jet on the dragster supplies a constant thrust of  $T = 20$  kN, determine the power generated by the jet as a function of time. Neglect drag and rolling resistance, and the loss of fuel. The dragster has a mass of 1 Mg and starts from rest.



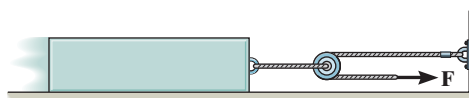
**Prob. 14–63**

**\*14–64.** The rocket sled has a mass of 4 Mg and travels from rest along the horizontal track for which the coefficient of kinetic friction is  $\mu_k = 0.20$ . If the engine provides a constant thrust  $T = 150$  kN, determine the power output of the engine as a function of time. Neglect the loss of fuel mass and air resistance.



**Prob. 14–64**

**14–65.** The block has a mass of 150 kg and rests on a surface for which the coefficients of static and kinetic friction are  $\mu_s = 0.5$  and  $\mu_k = 0.4$ , respectively. If a force  $F = (60t^2)$  N, where  $t$  is in seconds, is applied to the cable, determine the power developed by the force when  $t = 5$  s. *Hint:* First determine the time needed for the force to cause motion.



**Prob. 14–65**



## 14.5 Conservative Forces and Potential Energy

**Conservative Force.** If the work of a force is *independent of the path* and depends only on the force's initial and final positions on the path, then we can classify this force as a *conservative force*. Examples of conservative forces are the weight of a particle and the force developed by a spring. The work done by the weight depends *only* on the *vertical displacement* of the weight, and the work done by a spring force depends *only* on the spring's *elongation* or *compression*.

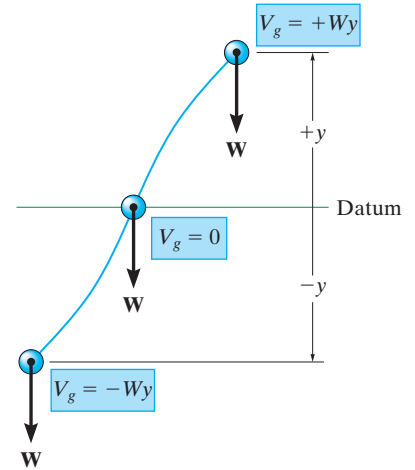
In contrast to a conservative force, consider the force of friction exerted on a sliding object by a fixed surface. The work done by the frictional force *depends on the path*—the longer the path, the greater the work. Consequently, *frictional forces are nonconservative*. The work is dissipated from the body in the form of heat.

**Energy.** Energy is defined as the capacity for doing work. For example, if a particle is originally at rest, then the principle of work and energy states that  $\Sigma U_{1 \rightarrow 2} = T_2$ . In other words, the kinetic energy is equal to the work that must be done on the particle to bring it from a state of rest to a speed  $v$ . Thus, the *kinetic energy* is a measure of the particle's *capacity to do work*, which is associated with the *motion* of the particle. When energy comes from the *position* of the particle, measured from a fixed datum or reference plane, it is called potential energy. Thus, *potential energy* is a measure of the amount of work a conservative force will do when it moves from a given position to the datum. In mechanics, the potential energy created by gravity (weight) and an elastic spring is important.

**Gravitational Potential Energy.** If a particle is located a distance  $y$  *above* an arbitrarily selected datum, as shown in Fig. 14–17, the particle's weight  $\mathbf{W}$  has positive *gravitational potential energy*,  $V_g$ , since  $\mathbf{W}$  has the capacity of doing positive work when the particle is moved back down to the datum. Likewise, if the particle is located a distance  $y$  *below* the datum,  $V_g$  is negative since the weight does negative work when the particle is moved back up to the datum. At the datum  $V_g = 0$ .

In general, if  $y$  is *positive upward*, the gravitational potential energy of the particle of weight  $\mathbf{W}$  is\*

$$V_g = Wy \quad (14-13)$$



Gravitational potential energy

Fig. 14–17



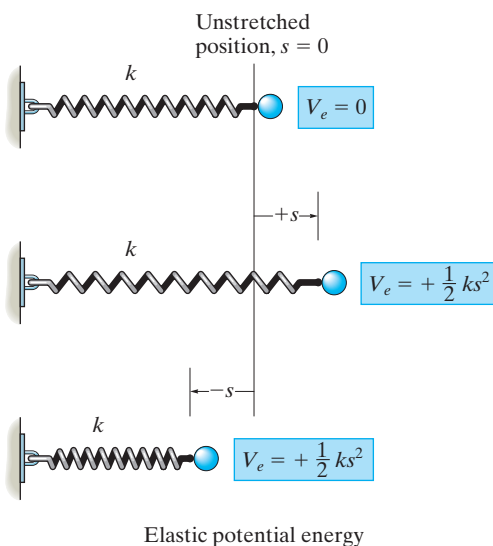
Gravitational potential energy of this weight is increased as it is hoisted upward.

\*Here the weight is assumed to be *constant*. This assumption is suitable for small differences in elevation  $\Delta y$ . If the elevation change is significant, however, a variation of weight with elevation must be taken into account (see Prob. 14–81).

**Elastic Potential Energy.** When an elastic spring is elongated or compressed a distance  $s$  from its unstretched position, elastic potential energy  $V_e$  can be stored in the spring. This energy is

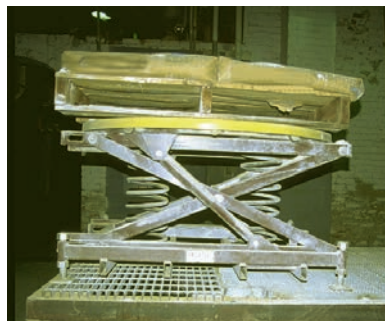
$$V_e = +\frac{1}{2}ks^2 \quad (14-14)$$

Here  $V_e$  is *always positive* since, in the deformed position, the force of the spring has the *capacity* or “potential” for always doing positive work on the particle when the spring is returned to its unstretched position, Fig. 14–18.



**Fig. 14–18**

The weight of the sacks resting on this platform causes potential energy to be stored in the supporting springs. As each sack is removed, the platform will *rise* slightly since some of the potential energy within the springs will be transformed into an increase in gravitational potential energy of the remaining sacks. Such a device is useful for removing the sacks without having to bend over to pick them up as they are unloaded.



**Potential Function.** In the general case, if a particle is subjected to both gravitational and elastic forces, the particle's potential energy can be expressed as a *potential function*, which is the algebraic sum

$$V = V_g + V_e \quad (14-15)$$

Measurement of  $V$  depends on the location of the particle with respect to a selected datum in accordance with Eqs. 14-13 and 14-14.

The work done by a conservative force in moving the particle from one point to another point is measured by the *difference* of this function, i.e.,

$$U_{1-2} = V_1 - V_2 \quad (14-16)$$

For example, the potential function for a particle of weight  $W$  suspended from a spring can be expressed in terms of its position,  $s$ , measured from a datum located at the unstretched length of the spring, Fig. 14-19. We have

$$\begin{aligned} V &= V_g + V_e \\ &= -Ws + \frac{1}{2}ks^2 \end{aligned}$$

If the particle moves from  $s_1$  to a lower position  $s_2$ , then applying Eq. 14-16 it can be seen that the work of  $\mathbf{W}$  and  $\mathbf{F}_s$  is

$$\begin{aligned} U_{1-2} &= V_1 - V_2 = \left(-Ws_1 + \frac{1}{2}ks_1^2\right) - \left(-Ws_2 + \frac{1}{2}ks_2^2\right) \\ &= W(s_2 - s_1) - \left(\frac{1}{2}ks_2^2 - \frac{1}{2}ks_1^2\right) \end{aligned}$$

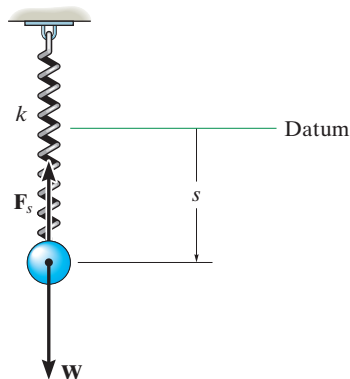


Fig. 14-19

When the displacement along the path is infinitesimal, i.e., from point  $(x, y, z)$  to  $(x + dx, y + dy, z + dz)$ , Eq. 14-16 becomes

$$\begin{aligned} dU &= V(x, y, z) - V(x + dx, y + dy, z + dz) \\ &= -dV(x, y, z) \end{aligned} \quad (14-17)$$

If we represent both the force and its displacement as Cartesian vectors, then the work can also be expressed as

$$\begin{aligned} dU &= \mathbf{F} \cdot d\mathbf{r} = (F_x \mathbf{i} + F_y \mathbf{j} + F_z \mathbf{k}) \cdot (dx \mathbf{i} + dy \mathbf{j} + dz \mathbf{k}) \\ &= F_x dx + F_y dy + F_z dz \end{aligned}$$

Substituting this result into Eq. 14-17 and expressing the differential  $dV(x, y, z)$  in terms of its partial derivatives yields

$$F_x dx + F_y dy + F_z dz = - \left( \frac{\partial V}{\partial x} dx + \frac{\partial V}{\partial y} dy + \frac{\partial V}{\partial z} dz \right)$$

Since changes in  $x$ ,  $y$ , and  $z$  are all independent of one another, this equation is satisfied provided

$$F_x = -\frac{\partial V}{\partial x}, \quad F_y = -\frac{\partial V}{\partial y}, \quad F_z = -\frac{\partial V}{\partial z} \quad (14-18)$$

Thus,

$$\begin{aligned} \mathbf{F} &= -\frac{\partial V}{\partial x} \mathbf{i} - \frac{\partial V}{\partial y} \mathbf{j} - \frac{\partial V}{\partial z} \mathbf{k} \\ &= - \left( \frac{\partial}{\partial x} \mathbf{i} + \frac{\partial}{\partial y} \mathbf{j} + \frac{\partial}{\partial z} \mathbf{k} \right) V \end{aligned}$$

or

$$\mathbf{F} = -\nabla V \quad (14-19)$$

where  $\nabla$  (del) represents the vector operator

$$\nabla = \left( \frac{\partial}{\partial x} \right) \mathbf{i} + \left( \frac{\partial}{\partial y} \right) \mathbf{j} + \left( \frac{\partial}{\partial z} \right) \mathbf{k}.$$

Equation 14-19 relates a force  $\mathbf{F}$  to its potential function  $V$  and thereby provides a mathematical criterion for proving that  $\mathbf{F}$  is conservative. For example, the gravitational potential function for a weight located a distance  $y$  above a datum is  $V_g = Wy$ . To prove that  $\mathbf{W}$  is conservative, it is necessary to show that it satisfies Eq. 14-18 (or Eq. 14-19), in which case

$$F_y = -\frac{\partial V}{\partial y}; \quad F_y = -\frac{\partial}{\partial y}(Wy) = -W$$

The negative sign indicates that  $\mathbf{W}$  acts downward, opposite to positive  $y$ , which is upward.

## 14.6 Conservation of Energy

When a particle is acted upon by a system of *both* conservative and nonconservative forces, the portion of the work done by the *conservative forces* can be written in terms of the difference in their potential energies using Eq. 14-16, i.e.,  $(\Sigma U_{1-2})_{\text{cons.}} = V_1 - V_2$ . As a result, the principle of work and energy can be written as

$$T_1 + V_1 + (\Sigma U_{1-2})_{\text{noncons.}} = T_2 + V_2 \quad (14-20)$$

Here  $(\Sigma U_{1-2})_{\text{noncons.}}$  represents the work of the nonconservative forces acting on the particle. If *only conservative forces* do work then we have

$$T_1 + V_1 = T_2 + V_2 \quad (14-21)$$

This equation is referred to as the *conservation of mechanical energy* or simply the *conservation of energy*. It states that during the motion the sum of the particle's kinetic and potential energies remains *constant*. For this to occur, kinetic energy must be transformed into potential energy, and vice versa. For example, if a ball of weight **W** is dropped from a height  $h$  above the ground (datum), Fig. 14-20, the potential energy of the ball is maximum before it is dropped, at which time its kinetic energy is zero. The total mechanical energy of the ball in its initial position is thus

$$E = T_1 + V_1 = 0 + Wh = Wh$$

When the ball has fallen a distance  $h/2$ , its speed can be determined by using  $v^2 = v_0^2 + 2a_c(y - y_0)$ , which yields  $v = \sqrt{2g(h/2)} = \sqrt{gh}$ . The energy of the ball at the mid-height position is therefore

$$E = T_2 + V_2 = \frac{1}{2} \frac{W}{g} (\sqrt{gh})^2 + W \left( \frac{h}{2} \right) = Wh$$

Just before the ball strikes the ground, its potential energy is zero and its speed is  $v = \sqrt{2gh}$ . Here, again, the total energy of the ball is

$$E = T_3 + V_3 = \frac{1}{2} \frac{W}{g} (\sqrt{2gh})^2 + 0 = Wh$$

Note that when the ball comes in contact with the ground, it deforms somewhat, and provided the ground is hard enough, the ball will rebound off the surface, reaching a new height  $h'$ , which will be *less* than the height  $h$  from which it was first released. Neglecting air friction, the difference in height accounts for an energy loss,  $E_l = W(h - h')$ , which occurs during the collision. Portions of this loss produce noise, localized deformation of the ball and ground, and heat.

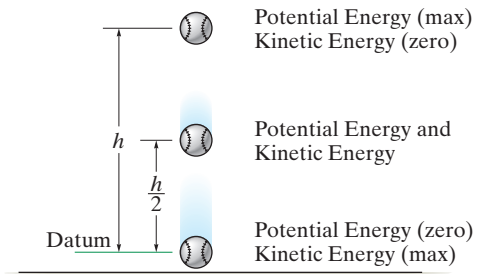


Fig. 14-20

**System of Particles.** If a system of particles is *subjected only to conservative forces*, then an equation similar to Eq. 14–21 can be written for the particles. Applying the ideas of the preceding discussion, Eq. 14–8 ( $\Sigma T_1 + \Sigma U_{1-2} = \Sigma T_2$ ) becomes

$$\Sigma T_1 + \Sigma V_1 = \Sigma T_2 + \Sigma V_2 \quad (14-22)$$

Here, the sum of the system's initial kinetic and potential energies is equal to the sum of the system's final kinetic and potential energies. In other words,  $\Sigma T + \Sigma V = \text{const.}$

## Procedure for Analysis

The conservation of energy equation can be used to solve problems involving *velocity*, *displacement*, and *conservative force systems*. It is generally *easier to apply* than the principle of work and energy because this equation requires specifying the particle's kinetic and potential energies at only *two points* along the path, rather than determining the work when the particle moves through a *displacement*. For application it is suggested that the following procedure be used.

### Potential Energy.

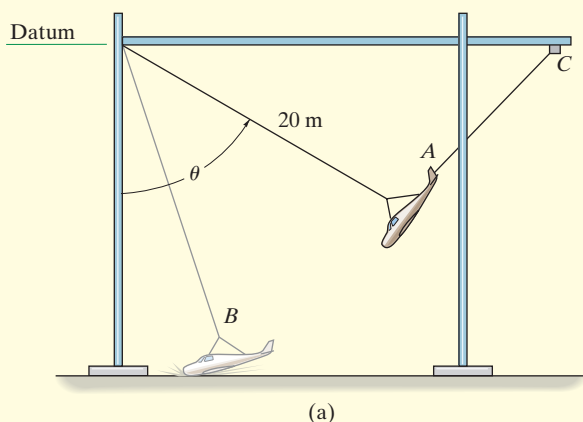
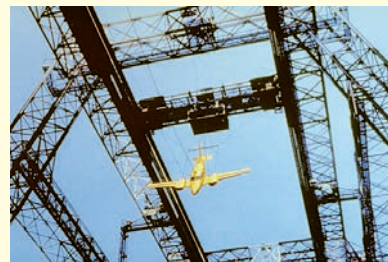
- Draw two diagrams showing the particle located at its initial and final points along the path.
- If the particle is subjected to a vertical displacement, establish the fixed horizontal datum from which to measure the particle's gravitational potential energy  $V_g$ .
- Data pertaining to the elevation  $y$  of the particle from the datum and the stretch or compression  $s$  of any connecting springs can be determined from the geometry associated with the two diagrams.
- Recall  $V_g = Wy$ , where  $y$  is positive upward from the datum and negative downward from the datum; also for a spring,  $V_e = \frac{1}{2}ks^2$ , which is *always positive*.

### Conservation of Energy.

- Apply the equation  $T_1 + V_1 = T_2 + V_2$ .
- When determining the kinetic energy,  $T = \frac{1}{2}mv^2$ , remember that the particle's speed  $v$  must be measured from an inertial reference frame.

**EXAMPLE 14.9**

The gantry structure in the photo is used to test the response of an airplane during a crash. As shown in Fig. 14–21*a*, the plane, having a mass of 8 Mg, is hoisted back until  $\theta = 60^\circ$ , and then the pull-back cable *AC* is released when the plane is at rest. Determine the speed of the plane just before it crashes into the ground,  $\theta = 15^\circ$ . Also, what is the maximum tension developed in the supporting cable during the motion? Neglect the size of the airplane and the effect of lift caused by the wings during the motion.

**SOLUTION**

Since the force of the cable does *no work* on the plane, it must be obtained using the equation of motion. First, however, we must determine the plane's speed at *B*.

**Potential Energy.** For convenience, the datum has been established at the top of the gantry, Fig. 14–21*a*.

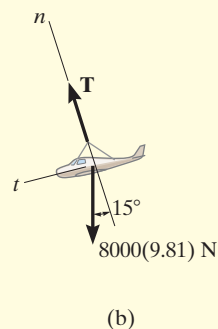
**Conservation of Energy.**

$$\begin{aligned}
 T_A + V_A &= T_B + V_B \\
 0 - 8000 \text{ kg} (9.81 \text{ m/s}^2)(20 \cos 60^\circ \text{ m}) &= \\
 \frac{1}{2}(8000 \text{ kg})v_B^2 - 8000 \text{ kg} (9.81 \text{ m/s}^2)(20 \cos 15^\circ \text{ m}) &= \\
 v_B &= 13.52 \text{ m/s} = 13.5 \text{ m/s} \quad \text{Ans.}
 \end{aligned}$$

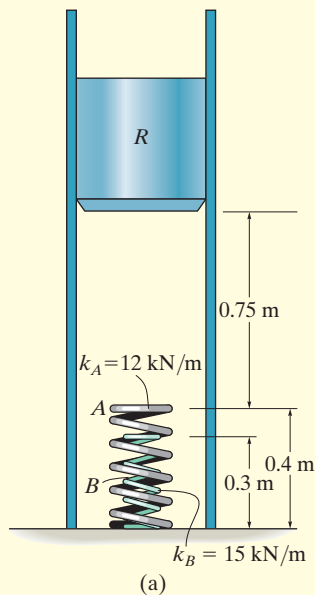
**Equation of Motion.** From the free-body diagram when the plane is at *B*, Fig. 14–21*b*, we have

$$+\curvearrowleft \quad \Sigma F_n = ma_n;$$

$$\begin{aligned}
 T - (8000(9.81) \text{ N}) \cos 15^\circ &= (8000 \text{ kg}) \frac{(13.52 \text{ m/s})^2}{20 \text{ m}} \\
 T &= 149 \text{ kN} \quad \text{Ans.}
 \end{aligned}$$

**Fig. 14–21**

## EXAMPLE 14.10



The ram  $R$  shown in Fig. 14–22*a* has a mass of 100 kg and is released from rest 0.75 m from the top of a spring,  $A$ , that has a stiffness  $k_A = 12$  kN/m. If a second spring  $B$ , having a stiffness  $k_B = 15$  kN/m, is “nested” in  $A$ , determine the maximum displacement of  $A$  needed to stop the downward motion of the ram. The unstretched length of each spring is indicated in the figure. Neglect the mass of the springs.

## SOLUTION

**Potential Energy.** We will *assume* that the ram compresses *both* springs at the instant it comes to rest. The datum is located through the center of gravity of the ram at its initial position, Fig. 14–22*b*. When the kinetic energy is reduced to zero ( $v_2 = 0$ ),  $A$  is compressed a distance  $s_A$  and  $B$  compresses  $s_B = s_A - 0.1$  m.

## Conservation of Energy.

$$T_1 + V_1 = T_2 + V_2$$

$$0 + 0 = 0 + \left\{ \frac{1}{2}k_A s_A^2 + \frac{1}{2}k_B (s_A - 0.1)^2 - Wh \right\}$$

$$0 + 0 = 0 + \left\{ \frac{1}{2}(12\,000 \text{ N/m})s_A^2 + \frac{1}{2}(15\,000 \text{ N/m})(s_A - 0.1 \text{ m})^2 - 981 \text{ N}(0.75 \text{ m} + s_A) \right\}$$

Rearranging the terms,

$$13\,500s_A^2 - 2481s_A - 660.75 = 0$$

Using the quadratic formula and solving for the positive root, we have

$$s_A = 0.331 \text{ m} \quad \text{Ans.}$$

Since  $s_B = 0.331 \text{ m} - 0.1 \text{ m} = 0.231 \text{ m}$ , which is positive, the assumption that *both* springs are compressed by the ram is correct.

**NOTE:** The second root,  $s_A = -0.148 \text{ m}$ , does not represent the physical situation. Since positive  $s$  is measured downward, the negative sign indicates that spring  $A$  would have to be “extended” by an amount of 0.148 m to stop the ram.

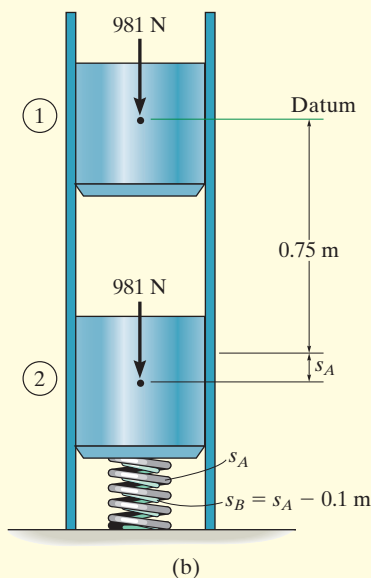


Fig. 14–22



**EXAMPLE 14.11**

A smooth 2-kg collar, shown in Fig. 14–23a, fits loosely on the vertical shaft. If the spring is unstretched when the collar is in the position *A*, determine the speed at which the collar is moving when  $y = 1$  m, if (a) it is released from rest at *A*, and (b) it is released at *A* with an upward velocity  $v_A = 2$  m/s.

**SOLUTION**

**Part (a) Potential Energy.** For convenience, the datum is established through *AB*, Fig. 14–23b. When the collar is at *C*, the gravitational potential energy is  $-(mg)y$ , since the collar is *below* the datum, and the elastic potential energy is  $\frac{1}{2}ks_{CB}^2$ . Here  $s_{CB} = 0.5$  m, which represents the *stretch* in the spring as shown in the figure.

**Conservation of Energy.**

$$T_A + V_A = T_C + V_C$$

$$0 + 0 = \frac{1}{2}mv_C^2 + \left\{ \frac{1}{2}ks_{CB}^2 - mgy \right\}$$

$$0 + 0 = \left\{ \frac{1}{2}(2 \text{ kg})v_C^2 \right\} + \left\{ \frac{1}{2}(3 \text{ N/m})(0.5 \text{ m})^2 - 2(9.81) \text{ N}(1 \text{ m}) \right\}$$

$$v_C = 4.39 \text{ m/s} \downarrow$$

*Ans.*

This problem can also be solved by using the equation of motion or the principle of work and energy. Note that for *both* of these methods the variation of the magnitude and direction of the spring force must be taken into account (see Example 13.4). Here, however, the above solution is clearly advantageous since the calculations depend *only* on data calculated at the initial and final points of the path.

**Part (b) Conservation of Energy.** If  $v_A = 2$  m/s, using the data in Fig. 14–23b, we have

$$T_A + V_A = T_C + V_C$$

$$\frac{1}{2}mv_A^2 + 0 = \frac{1}{2}mv_C^2 + \left\{ \frac{1}{2}ks_{CB}^2 - mgy \right\}$$

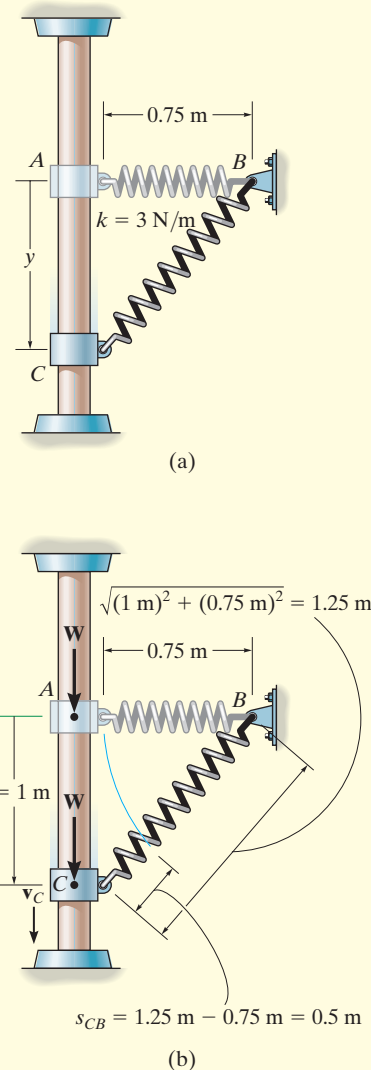
$$\frac{1}{2}(2 \text{ kg})(2 \text{ m/s})^2 + 0 = \frac{1}{2}(2 \text{ kg})v_C^2 + \left\{ \frac{1}{2}(3 \text{ N/m})(0.5 \text{ m})^2 \right.$$

$$\left. - 2(9.81) \text{ N}(1 \text{ m}) \right\}$$

$$v_C = 4.82 \text{ m/s} \downarrow$$

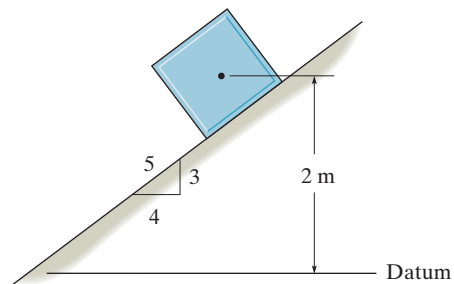
*Ans.*

**NOTE:** The kinetic energy of the collar depends only on the *magnitude* of velocity, and therefore it is immaterial if the collar is moving up or down at 2 m/s when released at *A*.

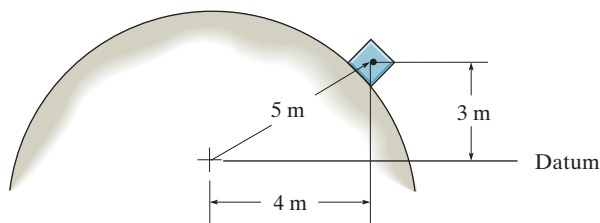
**Fig. 14–23**

## PRELIMINARY PROBLEMS

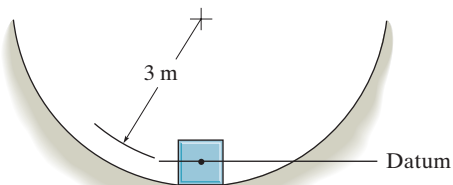
**P14-3.** Determine the potential energy of the block that has a weight of 100 N.



(a)



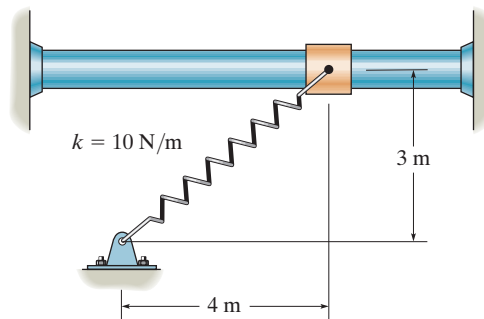
(b)



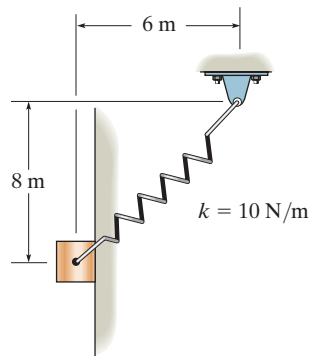
(c)

**Prob. P14-3**

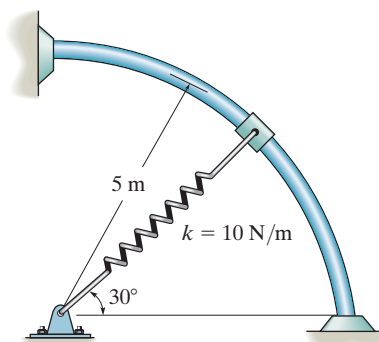
**P14-4.** Determine the potential energy in the spring that has an unstretched length of 4 m.



(a)



(b)

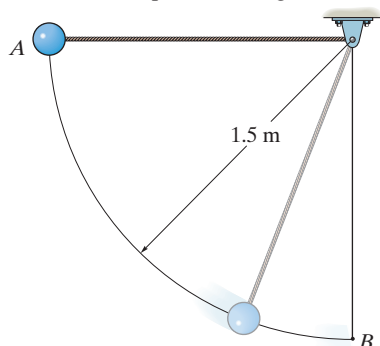


(c)

**Prob. P14-4**

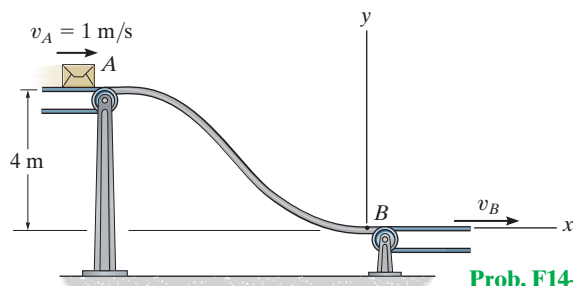
## FUNDAMENTAL PROBLEMS

**F14-13.** The 2-kg pendulum bob is released from rest when it is at  $A$ . Determine the speed of the bob and the tension in the cord when the bob passes through its lowest position,  $B$ .



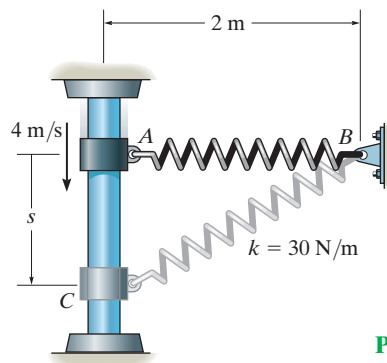
Prob. F14-13

**F14-14.** The 2-kg package leaves the conveyor belt at  $A$  with a speed of  $v_A = 1$  m/s and slides down the smooth ramp. Determine the required speed of the conveyor belt at  $B$  so that the package can be delivered without slipping on the belt. Also, find the normal reaction the curved portion of the ramp exerts on the package at  $B$  if  $\rho_B = 2$  m.



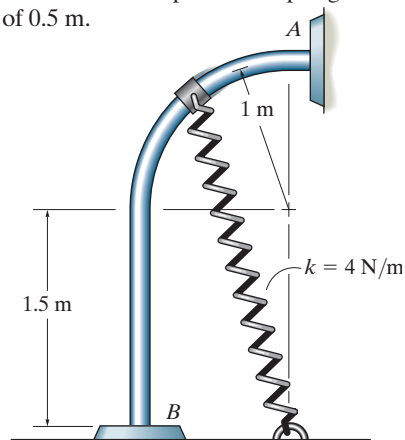
Prob. F14-14

**F14-15.** The 2-kg collar is given a downward velocity of 4 m/s when it is at  $A$ . If the spring has an unstretched length of 1 m and a stiffness of  $k = 30$  N/m, determine the velocity of the collar at  $s = 1$  m.



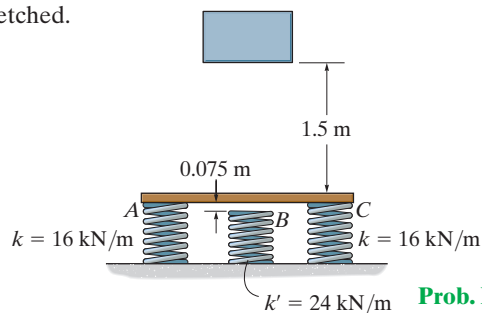
Prob. F14-15

**F14-16.** The 5-kg collar is released from rest at  $A$  and travels along the frictionless guide. Determine the speed of the collar when it strikes the stop  $B$ . The spring has an unstretched length of 0.5 m.



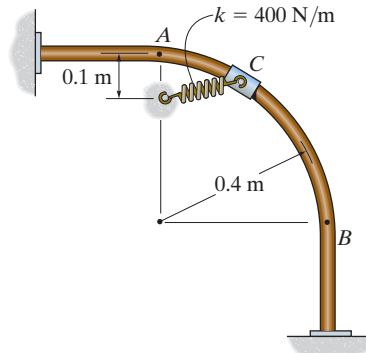
Prob. F14-16

**F14-17.** The 35-kg block is released from rest 1.5 m above the plate. Determine the compression of each spring when the block momentarily comes to rest after striking the plate. Neglect the mass of the plate. The springs are initially unstretched.



Prob. F14-17

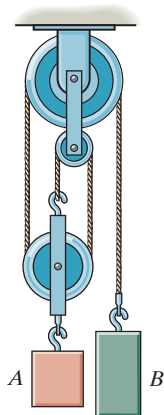
**F14-18.** The 4-kg collar  $C$  has a velocity of  $v_A = 2$  m/s when it is at  $A$ . If the guide rod is smooth, determine the speed of the collar when it is at  $B$ . The spring has an unstretched length of  $l_0 = 0.2$  m.



Prob. F14-18

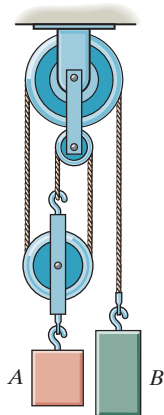
## PROBLEMS

**14–66.** The assembly consists of two blocks  $A$  and  $B$ , which have a mass of 20 kg and 30 kg, respectively. Determine the distance  $B$  must descend in order for  $A$  to achieve a speed of 3 m/s starting from rest.



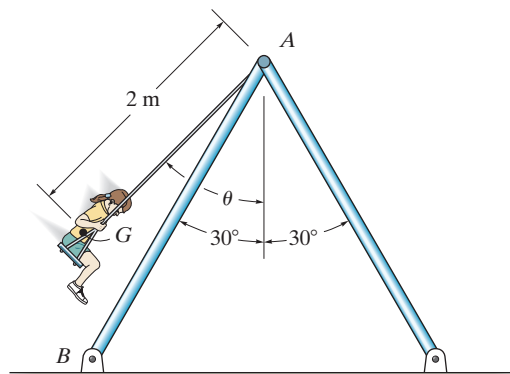
Prob. 14–66

**14–67.** The assembly consists of two blocks  $A$  and  $B$  which have a mass of 20 kg and 30 kg, respectively. Determine the speed of each block when  $B$  descends 1.5 m. The blocks are released from rest. Neglect the mass of the pulleys and cords.



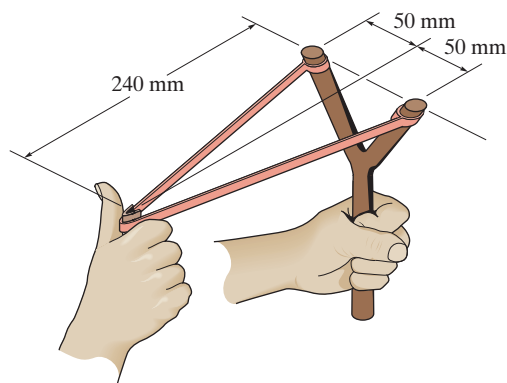
Prob. 14–67

**\*14–68.** The girl has a mass of 40 kg and center of mass at  $G$ . If she is swinging to a maximum height defined by  $\theta = 60^\circ$ , determine the force developed along each of the four supporting posts such as  $AB$  at the instant  $\theta = 0^\circ$ . The swing is centrally located between the posts.



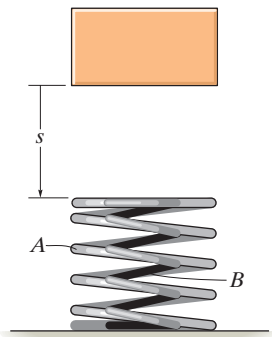
Prob. 14–68

**14–69.** Each of the two elastic rubber bands of the slingshot has an unstretched length of 180 mm. If they are pulled back to the position shown and released from rest, determine the maximum height the 30-g pellet will reach if it is fired vertically upward. Neglect the mass of the rubber bands and the change in elevation of the pellet while it is constrained by the rubber bands. Each rubber band has a stiffness  $k = 80 \text{ N/m}$ .



Prob. 14–69

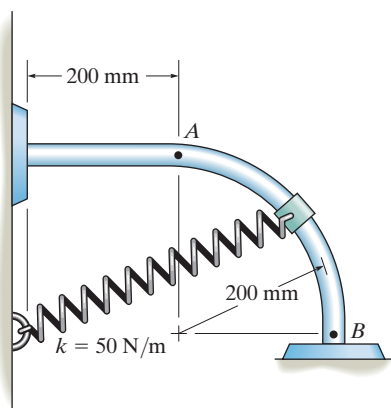
**14-70.** Two equal-length springs are “nested” together in order to form a shock absorber. If it is designed to arrest the motion of a 2-kg mass that is dropped  $s = 0.5$  m above the top of the springs from an at-rest position, and the maximum compression of the springs is to be 0.2 m, determine the required stiffness of the inner spring,  $k_B$ , if the outer spring has a stiffness  $k_A = 400$  N/m.



**Prob. 14-70**

**14-71.** The 5-kg collar has a velocity of 5 m/s to the right when it is at *A*. It then travels down along the smooth guide. Determine the speed of the collar when it reaches point *B*, which is located just before the end of the curved portion of the rod. The spring has an unstretched length of 100 mm and *B* is located just before the end of the curved portion of the rod.

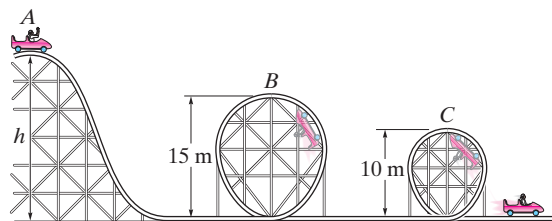
**\*14-72.** The 5-kg collar has a velocity of 5 m/s to the right when it is at *A*. It then travels along the smooth guide. Determine its speed when its center reaches point *B* and the normal force it exerts on the rod at this point. The spring has an unstretched length of 100 mm and *B* is located just before the end of the curved portion of the rod.



**Probs. 14-71/72**

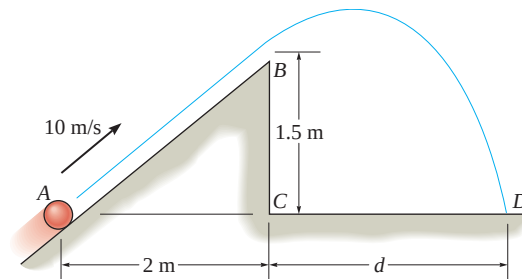
**14-73.** The roller coaster car has a mass of 700 kg, including its passenger. If it starts from the top of the hill *A* with a speed  $v_A = 3$  m/s, determine the minimum height  $h$  of the hill crest so that the car travels around the inside loops without leaving the track. Neglect friction, the mass of the wheels, and the size of the car. What is the normal reaction on the car when the car is at *B* and when it is at *C*? Take  $\rho_B = 7.5$  m and  $\rho_C = 5$  m.

**14-74.** The roller coaster car has a mass of 700 kg, including its passenger. If it is released from rest at the top of the hill *A*, determine the minimum height  $h$  of the hill crest so that the car travels around both inside loops without leaving the track. Neglect friction, the mass of the wheels, and the size of the car. What is the normal reaction on the car when the car is at *B* and when it is at *C*? Take  $\rho_B = 7.5$  m and  $\rho_C = 5$  m.



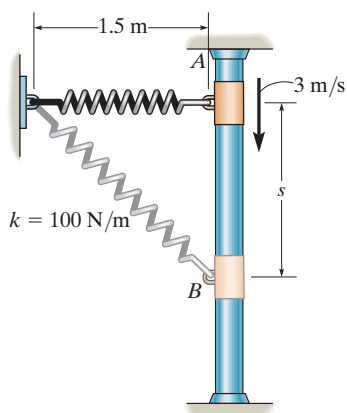
**Probs. 14-73/74**

**14-75.** The 2-kg ball of negligible size is fired from point *A* with an initial velocity of 10 m/s up the smooth inclined plane. Determine the distance from point *C* to where it hits the horizontal surface at *D*. Also, what is its velocity when it strikes the surface?



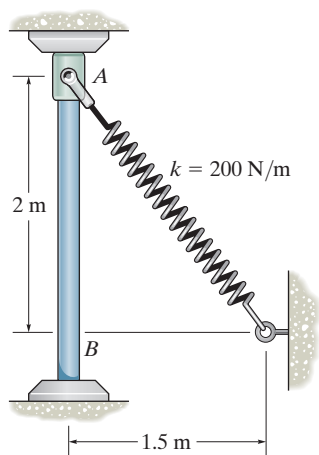
**Prob. 14-75**

**\*14-76.** The 4-kg smooth collar has a speed of 3 m/s when it is at  $s = 0$ . Determine the maximum distance  $s$  it travels before it stops momentarily. The spring has an unstretched length of 1 m.



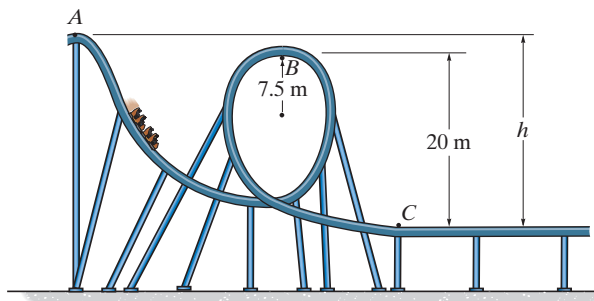
**Prob. 14-76**

**14-77.** The spring has a stiffness  $k = 200$  N/m and an unstretched length of 0.5 m. If it is attached to the 3-kg smooth collar and the collar is released from rest at  $A$ , determine the speed of the collar when it reaches  $B$ . Neglect the size of the collar.



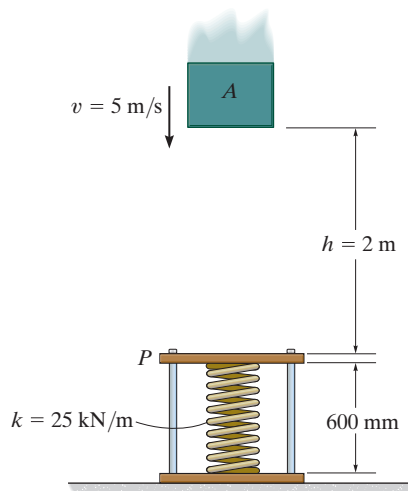
**Prob. 14-77**

**14-78.** The roller coaster car having a mass  $m$  is released from rest at point  $A$ . If the track is to be designed so that the car does not leave it at  $B$ , determine the required height  $h$ . Also, find the speed of the car when it reaches point  $C$ . Neglect friction.



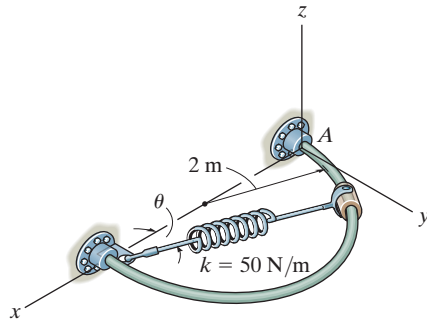
**Prob. 14-78**

**14-79.** A 750-mm-long spring is compressed and confined by the plate  $P$ , which can slide freely along the vertical 600-mm-long rods. The 40-kg block is given a speed of  $v = 5$  m/s when it is  $h = 2$  m above the plate. Determine how far the plate moves downwards when the block momentarily stops after striking it. Neglect the mass of the plate.



**Prob. 14-79**

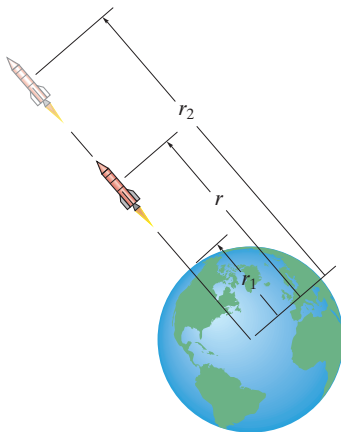
**\*14–80.** The spring has a stiffness  $k = 50 \text{ N/m}$  and an unstretched length of  $0.3 \text{ m}$ . If it is attached to the  $2\text{-kg}$  smooth collar and the collar is released from rest at  $A$  ( $\theta = 0^\circ$ ), determine the speed of the collar when  $\theta = 60^\circ$ . The motion occurs in the horizontal plane. Neglect the size of the collar.



**Prob. 14–80**

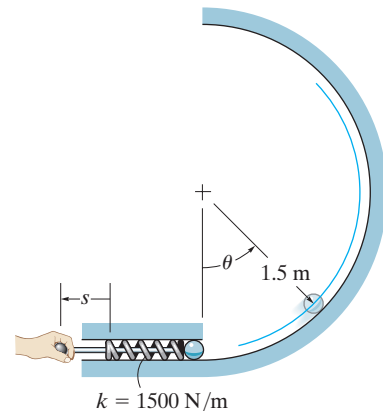
**14–81.** If the mass of the earth is  $M_e$ , show that the gravitational potential energy of a body of mass  $m$  located a distance  $r$  from the center of the earth is  $V_g = -GM_em/r$ . Recall that the gravitational force acting between the earth and the body is  $F = G(M_em/r^2)$ , Eq. 13–1. For the calculation, locate the datum at  $r \rightarrow \infty$ . Also, prove that  $F$  is a conservative force.

**14–82.** A rocket of mass  $m$  is fired vertically from the surface of the earth, i.e., at  $r = r_1$ . Assuming that no mass is lost as it travels upward, determine the work it must do against gravity to reach a distance  $r_2$ . The force of gravity is  $F = GM_em/r^2$  (Eq. 13–1), where  $M_e$  is the mass of the earth and  $r$  the distance between the rocket and the center of the earth.



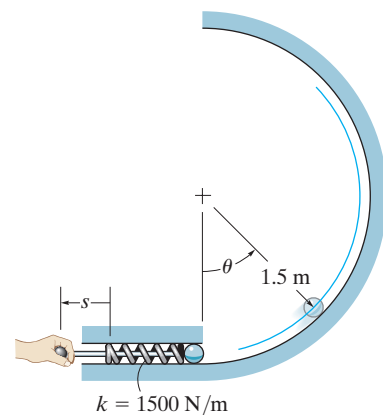
**Probs. 14–81/82**

**14–83.** When  $s = 0$ , the spring on the firing mechanism is unstretched. If the arm is pulled back such that  $s = 100 \text{ mm}$  and released, determine the speed of the  $0.3\text{-kg}$  ball and the normal reaction of the circular track on the ball when  $\theta = 60^\circ$ . Assume all surfaces of contact to be smooth. Neglect the mass of the spring and the size of the ball.



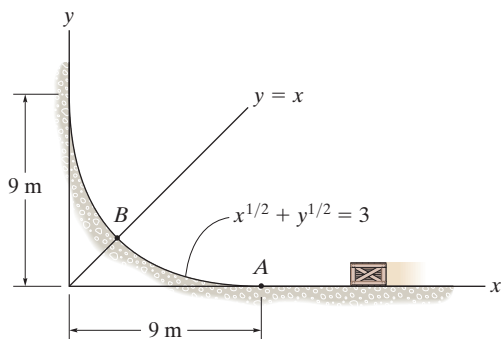
**Prob. 14–83**

**\*14–84.** When  $s = 0$ , the spring on the firing mechanism is unstretched. If the arm is pulled back such that  $s = 100 \text{ mm}$  and released, determine the maximum angle  $\theta$  the ball will travel without leaving the circular track. Assume all surfaces of contact to be smooth. Neglect the mass of the spring and the size of the ball.



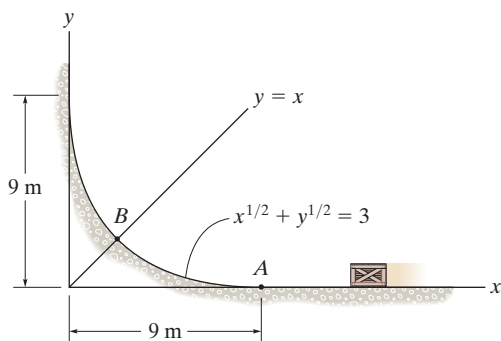
**Prob. 14–84**

**14-85.** When the 5-kg box reaches point  $A$  it has a speed  $v_A = 10$  m/s. Determine the normal force the box exerts on the surface when it reaches point  $B$ . Neglect friction and the size of the box.



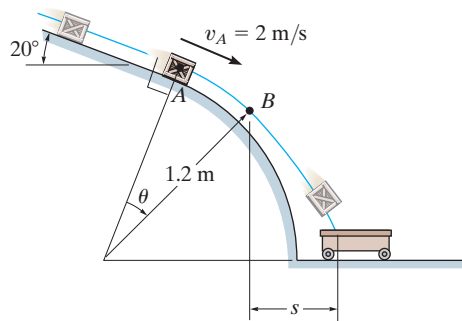
**Prob. 14-85**

**14-86.** When the 5-kg box reaches point  $A$  it has a speed  $v_A = 10$  m/s. Determine how high the box reaches up the surface before it comes to a stop. Also, what is the resultant normal force on the surface at this point and the acceleration? Neglect friction and the size of the box.



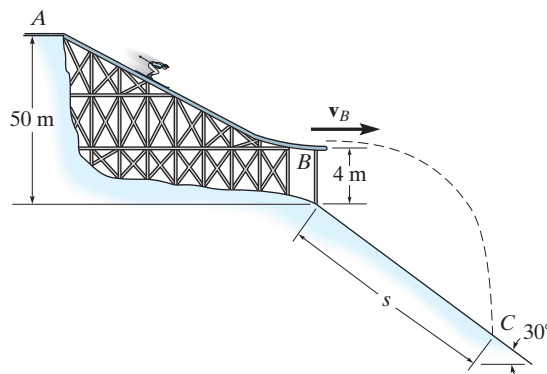
**Prob. 14-86**

**14-87.** When the 6-kg box reaches point  $A$  it has a speed of  $v_A = 2$  m/s. Determine the angle  $\theta$  at which it leaves the smooth circular ramp and the distance  $s$  to where it falls into the cart. Neglect friction.



**Prob. 14-87**

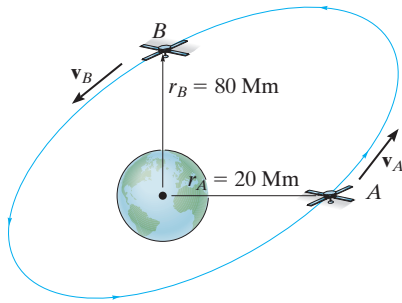
**\*14-88.** The skier starts from rest at  $A$  and travels down the ramp. If friction and air resistance can be neglected, determine his speed  $v_B$  when he reaches  $B$ . Also, compute the distance  $s$  to where he strikes the ground at  $C$ , if he makes the jump traveling horizontally at  $B$ . Neglect the skier's size. He has a mass of 70 kg.



**Prob. 14-88**

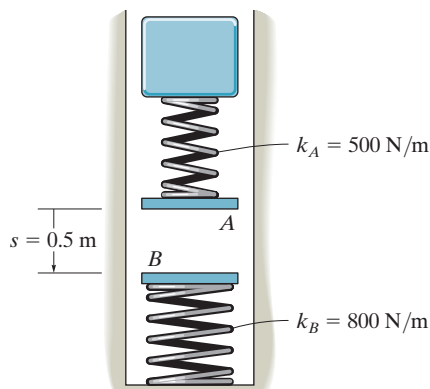


**14-89.** A 60-kg satellite travels in free flight along an elliptical orbit such that at  $A$ , where  $r_A = 20$  Mm, it has a speed  $v_A = 40$  Mm/h. What is the speed of the satellite when it reaches point  $B$ , where  $r_B = 80$  Mm? *Hint:* See Prob. 14-81, where  $M_e = 5.976(10^{24})$  kg and  $G = 66.73(10^{-12})$  m<sup>3</sup>/(kg · s<sup>2</sup>).



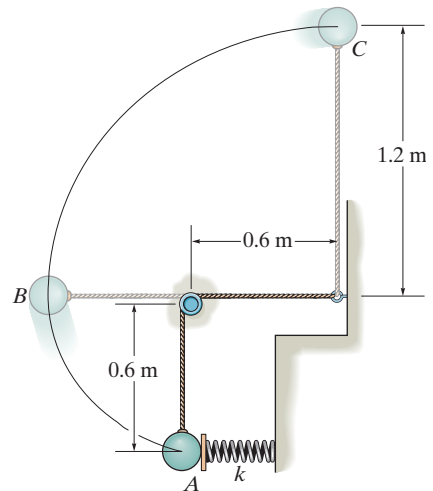
**Prob. 14-89**

**14-90.** The block has a mass of 20 kg and is released from rest when  $s = 0.5$  m. If the mass of the bumpers  $A$  and  $B$  can be neglected, determine the maximum deformation of each spring due to the collision.



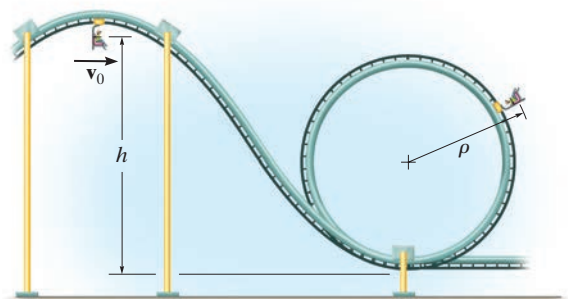
**Prob. 14-90**

**14-91.** The 0.75-kg bob of a pendulum is fired from rest at position  $A$  by a spring which has a stiffness  $k = 6$  kN/m and is compressed 125 mm. Determine the speed of the bob and the tension in the cord when the bob is at positions  $B$  and  $C$ . Point  $B$  is located on the path where the radius of curvature is still 0.6 m, i.e., just before the cord becomes horizontal.



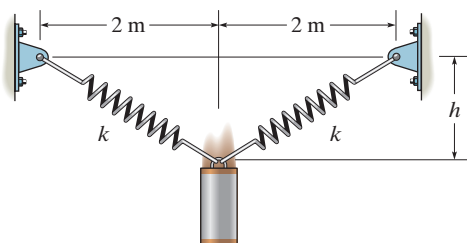
**Prob. 14-91**

**\*14-92.** The Raptor is an outside loop roller coaster in which riders are belted into seats resembling ski-lift chairs. If the cars travel at  $v_0 = 4$  m/s when they are at the top of the hill, determine their speed when they are at the top of the loop and the reaction of the 70-kg passenger on his seat at this instant. The car has a mass of 50 kg. Take  $h = 12$  m,  $\rho = 5$  m. Neglect friction and the size of the car and passenger.



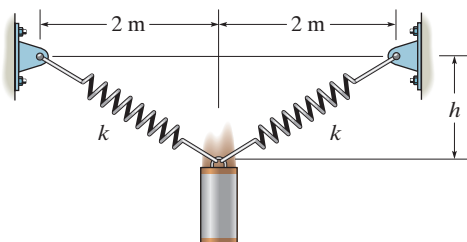
**Prob. 14-92**

**14-93.** If the 20-kg cylinder is released from rest at  $h = 0$ , determine the required stiffness  $k$  of each spring so that its motion is arrested or stops when  $h = 0.5$  m. Each spring has an unstretched length of 1 m.



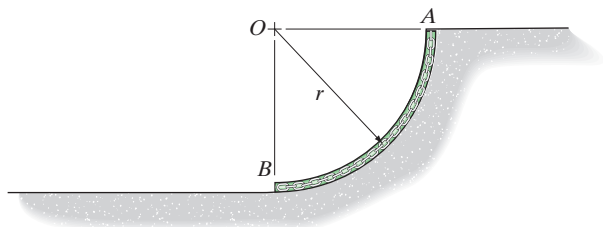
**Prob. 14-93**

**14-94.** The cylinder has a mass of 20 kg and is released from rest when  $h = 0$ . Determine its speed when  $h = 3$  m. Each spring has a stiffness  $k = 40$  N/m and an unstretched length of 2 m.



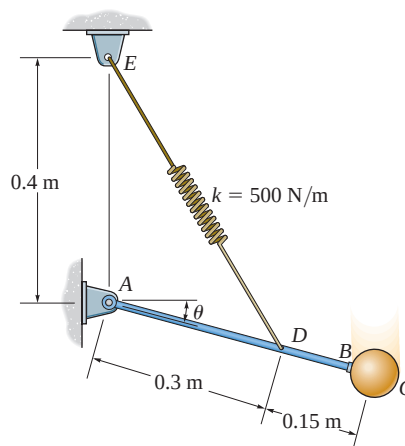
**Prob. 14-94**

**14-95.** A quarter-circular tube  $AB$  of mean radius  $r$  contains a smooth chain that has a mass per unit length of  $m_0$ . If the chain is released from rest from the position shown, determine its speed when it emerges completely from the tube.



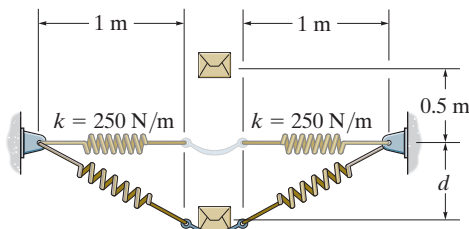
**Prob. 14-95**

**\*14-96.** The 10-kg sphere  $C$  is released from rest when  $\theta = 0^\circ$  and the tension in the spring is 100 N. Determine the speed of the sphere at the instant  $\theta = 90^\circ$ . Neglect the mass of rod  $AB$  and the size of the sphere.



**Prob. 14-96**

**14-97.** A pan of negligible mass is attached to two identical springs of stiffness  $k = 250$  N/m. If a 10-kg box is dropped from a height of 0.5 m above the pan, determine the maximum vertical displacement  $d$ . Initially each spring has a tension of 50 N.



**Prob. 14-97**

## CONCEPTUAL PROBLEMS

**C14-1.** The roller coaster is momentarily at rest at *A*. Determine the approximate normal force it exerts on the track at *B*. Also determine its approximate acceleration at this point. Use numerical data, and take scaled measurements from the photo with a known height at *A*.



**Prob. C14-1**

**C14-2.** As the large ring rotates, the operator can apply a breaking mechanism that binds the cars to the ring, which then allows the cars to rotate with the ring. Assuming the passengers are not belted into the cars, determine the smallest speed of the ring (cars) so that no passenger will fall out. When should the operator release the brake so that the cars can achieve their greatest speed as they slide freely on the ring? Estimate the greatest normal force of the seat on a passenger when this speed is reached. Use numerical values to explain your answer.



**Prob. C14-2**

**C14-3.** The woman pulls the water balloon launcher back, stretching each of the four elastic cords. Estimate the maximum height and the maximum range of a ball placed within the container if it is released from the position shown. Use numerical values and any necessary measurements from the photo. Assume the unstretched length and stiffness of each cord is known.



**Prob. C14-3**

**C14-4.** The girl is momentarily at rest in the position shown. If the unstretched length and stiffness of each of the two elastic cords is known, determine approximately how far the girl descends before she again becomes momentarily at rest. Use numerical values and take any necessary measurements from the photo.



**Prob. C14-4**

## CHAPTER REVIEW

### Work of a Force

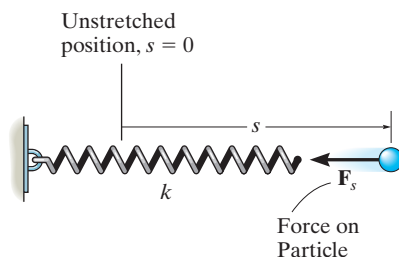
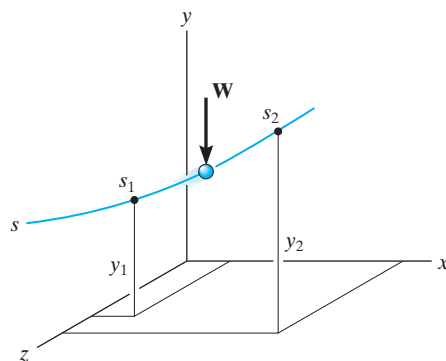
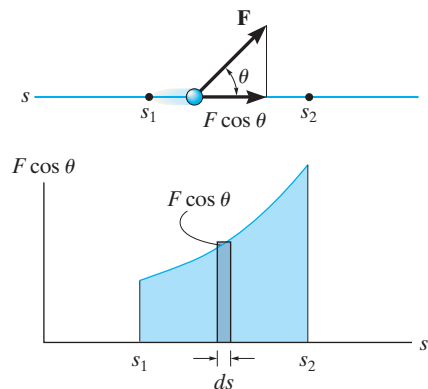
A force does work when it undergoes a displacement along its line of action. If the force varies with the displacement, then the work is  $U = \int F \cos \theta \, ds$ .

Graphically, this represents the area under the  $F$ - $s$  diagram.

If the force is constant, then for a displacement  $\Delta s$  in the direction of the force,  $U = F_c \Delta s$ . A typical example of this case is the work of a weight,  $U = -W \Delta y$ . Here,  $\Delta y$  is the vertical displacement.

The work done by a spring force,  $F = ks$ , depends upon the stretch or compression  $s$  of the spring.

$$U = -\left(\frac{1}{2}ks_2^2 - \frac{1}{2}ks_1^2\right)$$



### The Principle of Work and Energy

If the equation of motion in the tangential direction,  $\Sigma F_t = ma_t$ , is combined with the kinematic equation,  $a_t ds = v \, dv$ , we obtain the principle of work and energy. This equation states that the initial kinetic energy  $T$ , plus the work done  $\Sigma U_{1-2}$  is equal to the final kinetic energy.

$$T_1 + \Sigma U_{1-2} = T_2$$

The principle of work and energy is useful for solving problems that involve force, velocity, and displacement. For application, the free-body diagram of the particle should be drawn in order to identify the forces that do work.

### Power and Efficiency

Power is the time rate of doing work. For application, the force  $\mathbf{F}$  creating the power and its velocity  $\mathbf{v}$  must be specified.

Efficiency represents the ratio of power output to power input. Due to frictional losses, it is always less than one.

$$P = \frac{dU}{dt}$$

$$P = \mathbf{F} \cdot \mathbf{v}$$

$$\varepsilon = \frac{\text{power output}}{\text{power input}}$$

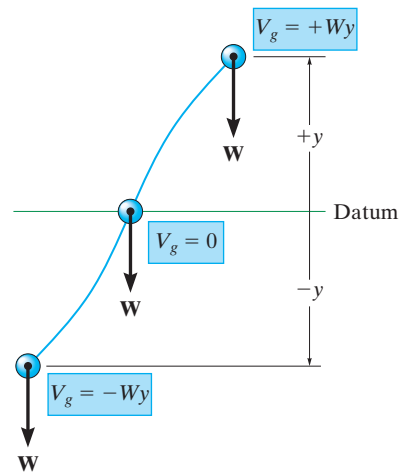
### Conservation of Energy

A conservative force does work that is independent of its path. Two examples are the weight of a particle and the spring force.

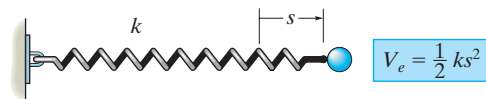
Friction is a nonconservative force since the work depends upon the length of the path. The longer the path, the more work done.

The work done by a conservative force depends upon its position relative to a datum. When this work is referenced from a datum, it is called potential energy. For a weight, it is  $V_g = \pm Wy$ , and for a spring it is  $V_e = +\frac{1}{2}ks^2$ .

Mechanical energy consists of kinetic energy  $T$  and gravitational and elastic potential energies  $V$ . According to the conservation of energy, this sum is constant and has the same value at any position on the path. If only gravitational and spring forces cause motion of the particle, then the conservation-of-energy equation can be used to solve problems involving these conservative forces, displacement, and velocity.



Gravitational potential energy

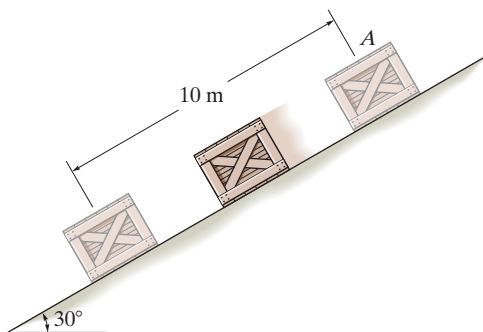


Elastic potential energy

$$T_1 + V_1 = T_2 + V_2$$

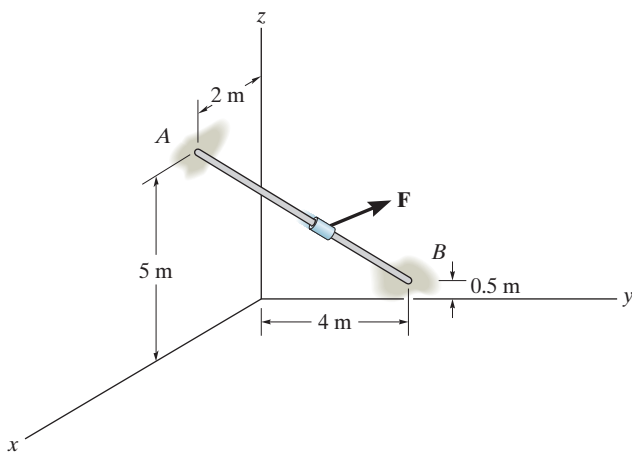
## REVIEW PROBLEMS

**R14-1.** If a 50-kg crate is released from rest at  $A$ , determine its speed after it slides 10 m down the plane. The coefficient of kinetic friction between the crate and plane is  $\mu_k = 0.3$ .



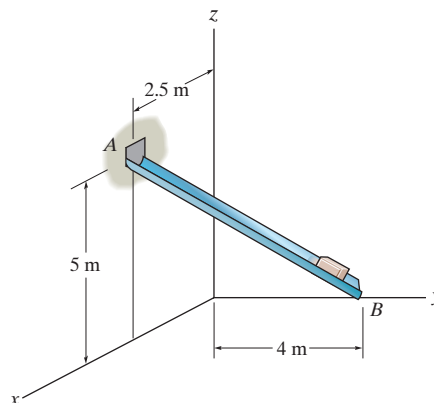
Prob. R14-1

**R14-2.** The small 1-kg collar starting from rest at  $A$  slides down along the smooth rod. During the motion, the collar is acted upon by a force  $\mathbf{F} = \{50\mathbf{i} + 30y\mathbf{j} + 10z\mathbf{k}\}$  N, where  $x, y, z$  are in meters. Determine the collar's speed when it strikes the wall at  $B$ .



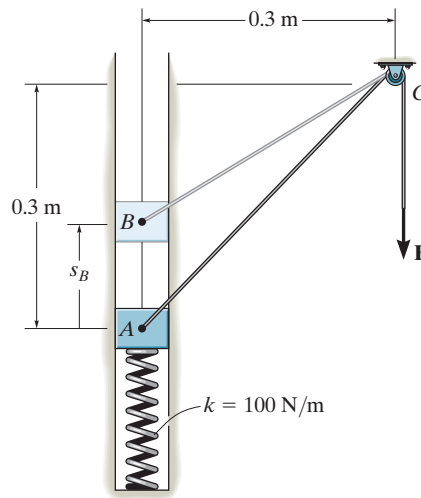
Prob. R14-2

**R14-3.** The block has a mass of 0.75 kg and slides along the smooth chute  $AB$ . It is released from rest at  $A$ , which has coordinates of  $A(2.5 \text{ m}, 0, 5 \text{ m})$ . Determine the speed at which it slides off at  $B$ , which has coordinates of  $B(0, 4 \text{ m}, 0)$ .



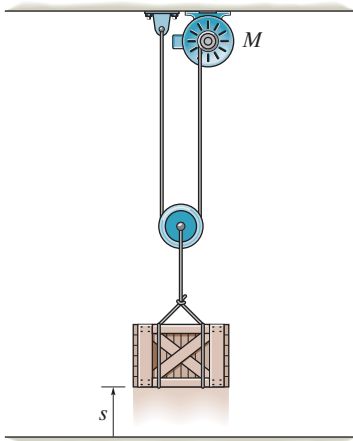
Prob. R14-3

**R14-4.** The block has a mass of 0.5 kg and moves within the smooth vertical slot. If the block starts from rest when the attached spring is in the unstretched position at  $A$ , determine the constant vertical force  $F$  which must be applied to the cord so that the block attains a speed  $v_B = 2.5 \text{ m/s}$  when it reaches  $B$ ;  $s_B = 0.15 \text{ m}$ . Neglect the mass of the cord and pulley.



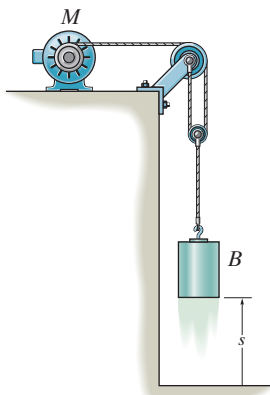
Prob. R14-4

**R14-5.** The crate, having a mass of 25 kg, is hoisted by the pulley system and motor  $M$ . If the crate starts from rest and, by constant acceleration, attains a speed of 6 m/s after rising 3 m, determine the power that must be supplied to the motor at the instant  $s = 3$  m. The motor has an efficiency  $\varepsilon = 0.74$ .



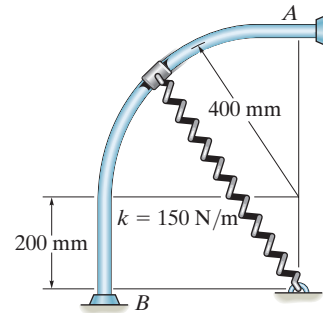
**Prob. R14-5**

**R14-6.** The 25-kg load is hoisted by the pulley system and motor  $M$ . If the motor exerts a constant force of 150 N on the cable, determine the power that must be supplied to the motor if the load has been hoisted  $s = 3$  m starting from rest. The motor has an efficiency of  $\varepsilon = 0.76$ .



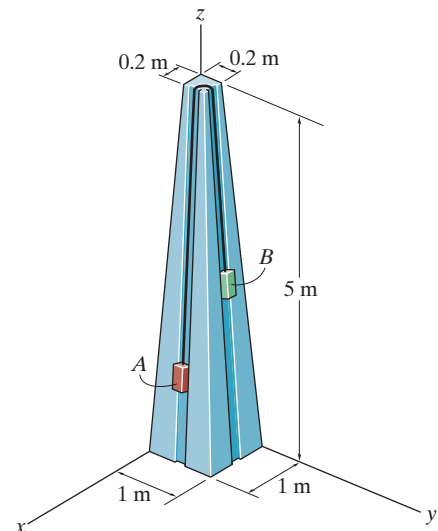
**Prob. R14-6**

**R14-7.** The collar of negligible size has a mass of 0.25 kg and is attached to a spring having an unstretched length of 100 mm. If the collar is released from rest at  $A$  and travels along the smooth guide, determine its speed just before it strikes  $B$ .



**Prob. R14-7**

**R14-8.** The blocks  $A$  and  $B$  have the mass of 5 kg and 15 kg, respectively. They are connected together by a light cord and ride in the frictionless grooves. Determine the speed of each block after block  $A$  moves 2 m up along the plane. The blocks are released from rest.



**Prob. R14-8**



# Chapter 15



(© David J. Green/Alamy)

The design of the bumper cars used for this amusement park ride requires knowledge of the principles of impulse and momentum.



# Kinetics of a Particle: Impulse and Momentum

## CHAPTER OBJECTIVES

- To develop the principle of linear impulse and momentum for a particle and apply it to solve problems that involve force, velocity, and time.
- To study the conservation of linear momentum for particles.
- To analyze the mechanics of impact.
- To introduce the concept of angular impulse and momentum.
- To solve problems involving steady fluid streams and propulsion with variable mass.



Video Solutions are available for selected questions in this chapter.

## 15.1 Principle of Linear Impulse and Momentum

In this section we will integrate the equation of motion with respect to time and thereby obtain the principle of impulse and momentum. The resulting equation will be useful for solving problems involving force, velocity, and time.

Using kinematics, the equation of motion for a particle of mass  $m$  can be written as

$$\Sigma \mathbf{F} = m\mathbf{a} = m \frac{d\mathbf{v}}{dt} \quad (15-1)$$

where  $\mathbf{a}$  and  $\mathbf{v}$  are both measured from an inertial frame of reference. Rearranging the terms and integrating between the limits  $\mathbf{v} = \mathbf{v}_1$  at  $t = t_1$  and  $\mathbf{v} = \mathbf{v}_2$  at  $t = t_2$ , we have

$$\Sigma \int_{t_1}^{t_2} \mathbf{F} dt = m \int_{\mathbf{v}_1}^{\mathbf{v}_2} d\mathbf{v}$$



The impulse tool is used to remove the dent in the trailer fender. To do so its end is first screwed into a hole drilled in the fender, then the weight is gripped and jerked upwards, striking the stop ring. The impulse developed is transferred along the shaft of the tool and pulls suddenly on the dent.

or

$$\Sigma \int_{t_1}^{t_2} \mathbf{F} dt = m\mathbf{v}_2 - m\mathbf{v}_1 \quad (15-2)$$

This equation is referred to as the *principle of linear impulse and momentum*. From the derivation it can be seen that it is simply a time integration of the equation of motion. It provides a *direct means* of obtaining the particle's final velocity  $\mathbf{v}_2$  after a specified time period when the particle's initial velocity is known and the forces acting on the particle are either constant or can be expressed as functions of time. By comparison, if  $\mathbf{v}_2$  was determined using the equation of motion, a two-step process would be necessary; i.e., apply  $\Sigma \mathbf{F} = m\mathbf{a}$  to obtain  $\mathbf{a}$ , then integrate  $\mathbf{a} = d\mathbf{v}/dt$  to obtain  $\mathbf{v}_2$ .

**Linear Momentum.** Each of the two vectors of the form  $\mathbf{L} = m\mathbf{v}$  in Eq. 15-2 is referred to as the particle's linear momentum. Since  $m$  is a positive scalar, the linear-momentum vector has the same direction as  $\mathbf{v}$ , and its magnitude  $mv$  has units of mass times velocity, e.g.,  $\text{kg} \cdot \text{m/s}$ .

**Linear Impulse.** The integral  $\mathbf{I} = \int \mathbf{F} dt$  in Eq. 15-2 is referred to as the *linear impulse*. This term is a vector quantity which measures the effect of a force during the time the force acts. Since time is a positive scalar, the impulse acts in the same direction as the force, and its magnitude has units of force times time, e.g.,  $\text{N} \cdot \text{s}$ .\*

If the force is expressed as a function of time, the impulse can be determined by direct evaluation of the integral. In particular, if the force is constant in both magnitude and direction, the resulting impulse becomes

$$\mathbf{I} = \int_{t_1}^{t_2} \mathbf{F}_c dt = \mathbf{F}_c(t_2 - t_1).$$

Graphically the magnitude of the impulse can be represented by the shaded area under the curve of force versus time, Fig. 15-1. A constant force creates the shaded rectangular area shown in Fig. 15-2.

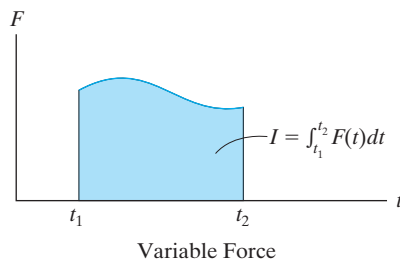


Fig. 15-1

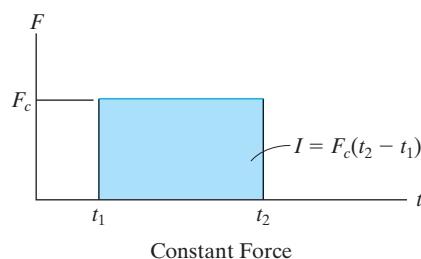


Fig. 15-2

\*Although the units for impulse and momentum are defined differently, it can be shown that Eq. 15-2 is dimensionally homogeneous.

**Principle of Linear Impulse and Momentum.** For problem solving, Eq. 15-2 will be rewritten in the form

$$m\mathbf{v}_1 + \Sigma \int_{t_1}^{t_2} \mathbf{F} dt = m\mathbf{v}_2 \quad (15-3)$$

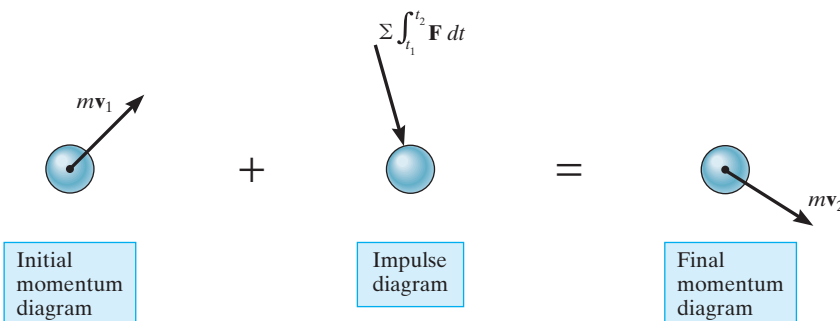
which states that the initial momentum of the particle at time  $t_1$  plus the sum of all the impulses applied to the particle from  $t_1$  to  $t_2$  is equivalent to the final momentum of the particle at time  $t_2$ . These three terms are illustrated graphically on the *impulse and momentum diagrams* shown in Fig. 15-3. The two *momentum diagrams* are simply outlined shapes of the particle which indicate the direction and magnitude of the particle's initial and final momenta,  $m\mathbf{v}_1$  and  $m\mathbf{v}_2$ . Similar to the free-body diagram, the *impulse diagram* is an outlined shape of the particle showing all the impulses that act on the particle when it is located at some intermediate point along its path.

If each of the vectors in Eq. 15-3 is resolved into its  $x$ ,  $y$ ,  $z$  components, we can write the following three scalar equations of linear impulse and momentum.

$$\begin{aligned} m(v_x)_1 + \Sigma \int_{t_1}^{t_2} F_x dt &= m(v_x)_2 \\ m(v_y)_1 + \Sigma \int_{t_1}^{t_2} F_y dt &= m(v_y)_2 \\ m(v_z)_1 + \Sigma \int_{t_1}^{t_2} F_z dt &= m(v_z)_2 \end{aligned} \quad (15-4)$$



The study of many types of sports, such as golf, requires application of the principle of linear impulse and momentum.



**Fig. 15-3**

## 15.2 Principle of Linear Impulse and Momentum for a System of Particles

The principle of linear impulse and momentum for a system of particles moving relative to an inertial reference, Fig. 15–4, is obtained from the equation of motion applied to all the particles in the system, i.e.,

$$\Sigma \mathbf{F}_i = \Sigma m_i \frac{d\mathbf{v}_i}{dt} \quad (15-5)$$

The term on the left side represents only the sum of the *external forces* acting on the particles. Recall that the internal forces  $\mathbf{f}_i$  acting between particles do not appear with this summation, since by Newton's third law they occur in equal but opposite collinear pairs and therefore cancel out. Multiplying both sides of Eq. 15–5 by  $dt$  and integrating between the limits  $t = t_1$ ,  $\mathbf{v}_i = (\mathbf{v}_i)_1$  and  $t = t_2$ ,  $\mathbf{v}_i = (\mathbf{v}_i)_2$  yields

$$\Sigma m_i (\mathbf{v}_i)_1 + \Sigma \int_{t_1}^{t_2} \mathbf{F}_i dt = \Sigma m_i (\mathbf{v}_i)_2 \quad (15-6)$$

This equation states that the initial linear momentum of the system plus the impulses of all the *external forces* acting on the system from  $t_1$  to  $t_2$  is equal to the system's final linear momentum.

Since the location of the mass center  $G$  of the system is determined from  $m\mathbf{r}_G = \Sigma m_i \mathbf{r}_i$ , where  $m = \Sigma m_i$  is the total mass of all the particles, Fig. 15–4, then taking the time derivative, we have

$$m\mathbf{v}_G = \Sigma m_i \mathbf{v}_i$$

which states that the total linear momentum of the system of particles is equivalent to the linear momentum of a “fictitious” aggregate particle of mass  $m = \Sigma m_i$  moving with the velocity of the mass center of the system. Substituting into Eq. 15–6 yields

$$m(\mathbf{v}_G)_1 + \Sigma \int_{t_1}^{t_2} \mathbf{F}_i dt = m(\mathbf{v}_G)_2 \quad (15-7)$$

Here the initial linear momentum of the aggregate particle plus the external impulses acting on the system of particles from  $t_1$  to  $t_2$  is equal to the aggregate particle's final linear momentum. As a result, the above equation justifies application of the principle of linear impulse and momentum to a system of particles that compose a rigid body.

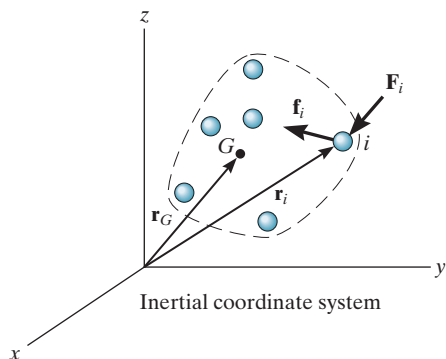


Fig. 15–4

## Procedure for Analysis

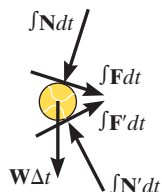
The principle of linear impulse and momentum is used to solve problems involving *force*, *time*, and *velocity*, since these terms are involved in the formulation. For application it is suggested that the following procedure be used.\*

### Free-Body Diagram.

- Establish the  $x$ ,  $y$ ,  $z$  inertial frame of reference and draw the particle's free-body diagram in order to account for all the forces that produce impulses on the particle.
- The direction and sense of the particle's initial and final velocities should be established.
- If a vector is unknown, assume that the sense of its components is in the direction of the positive inertial coordinate(s).
- As an alternative procedure, draw the impulse and momentum diagrams for the particle as discussed in reference to Fig. 15–3.

### Principle of Impulse and Momentum.

- In accordance with the established coordinate system, apply the principle of linear impulse and momentum,  $m\mathbf{v}_1 + \sum \int_{t_1}^{t_2} \mathbf{F} dt = m\mathbf{v}_2$ . If motion occurs in the  $x$ - $y$  plane, the two scalar component equations can be formulated by either resolving the vector components of  $\mathbf{F}$  from the free-body diagram, or by using the data on the impulse and momentum diagrams.
- Realize that every force acting on the particle's free-body diagram will create an impulse, even though some of these forces will do no work.
- Forces that are functions of time must be integrated to obtain the impulse. Graphically, the impulse is equal to the area under the force–time curve.

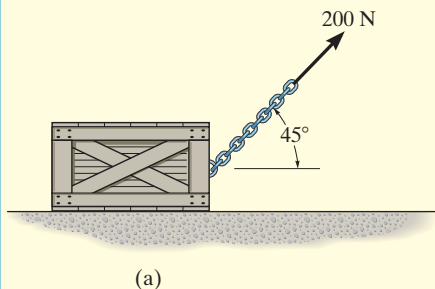


As the wheels of the pitching machine rotate, they apply frictional impulses to the ball, thereby giving it a linear momentum. These impulses are shown on the impulse diagram. Here both the frictional and normal impulses vary with time. By comparison, the weight impulse is constant and is very small since the time  $\Delta t$  the ball is in contact with the wheels is very small.



\*This procedure will be followed when developing the proofs and theory in the text.

## EXAMPLE 15.1



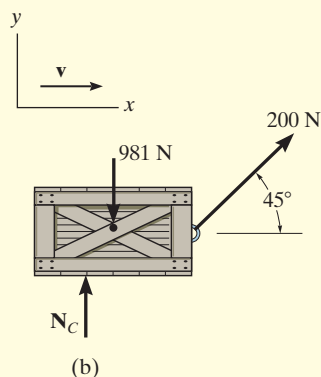
The 100-kg crate shown in Fig. 15–5a is originally at rest on the smooth horizontal surface. If a towing force of 200 N, acting at an angle of  $45^\circ$ , is applied for 10 s, determine the final velocity and the normal force which the surface exerts on the crate during this time interval.

## SOLUTION

This problem can be solved using the principle of impulse and momentum since it involves force, velocity, and time.

**Free-Body Diagram.** See Fig. 15–5b. Since all the forces acting are *constant*, the impulses are simply the product of the force magnitude and 10 s [ $\mathbf{I} = \mathbf{F}_c(t_2 - t_1)$ ]. Note the alternative procedure of drawing the crate's impulse and momentum diagrams, Fig. 15–5c.

**Principle of Impulse and Momentum.** Applying Eqs. 15–4 yields



$$\begin{aligned}
 (\rightarrow) \quad m(v_x)_1 + \sum \int_{t_1}^{t_2} F_x dt &= m(v_x)_2 \\
 0 + 200 \text{ N} \cos 45^\circ (10 \text{ s}) &= (100 \text{ kg})v_2 \\
 v_2 &= 14.1 \text{ m/s} \quad \text{Ans.}
 \end{aligned}$$

$$\begin{aligned}
 (+\uparrow) \quad m(v_y)_1 + \sum \int_{t_1}^{t_2} F_y dt &= m(v_y)_2 \\
 0 + N_C(10 \text{ s}) - 981 \text{ N}(10 \text{ s}) + 200 \text{ N} \sin 45^\circ (10 \text{ s}) &= 0 \\
 N_C &= 840 \text{ N} \quad \text{Ans.}
 \end{aligned}$$

**NOTE:** Since no motion occurs in the  $y$  direction, direct application of the equilibrium equation  $\Sigma F_y = 0$  gives the same result for  $N_C$ . Try to solve the problem by first applying  $\Sigma F_x = ma_x$ , then  $v = v_0 + a_c t$ .

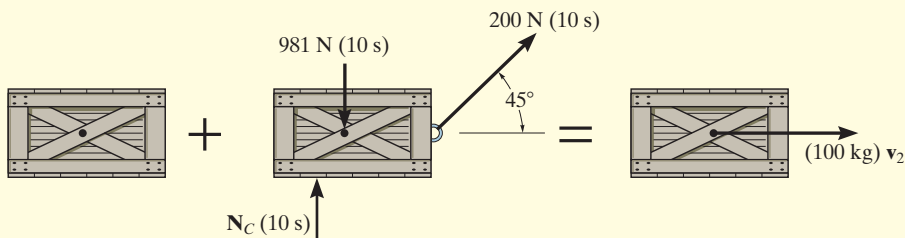


Fig.15–5

**EXAMPLE 15.2**

The 25-kg crate shown in Fig. 15–6a is acted upon by a force having a variable magnitude  $P = (100t)$  N, where  $t$  is in seconds. Determine the crate's velocity 2 s after  $\mathbf{P}$  has been applied. The initial velocity is  $v_1 = 1$  m/s down the plane, and the coefficient of kinetic friction between the crate and the plane is  $\mu_k = 0.3$ .

**SOLUTION**

**Free-Body Diagram.** See Fig. 15–6b. Since the magnitude of force  $P = 100t$  varies with time, the impulse it creates must be determined by integrating over the 2-s time interval.

**Principle of Impulse and Momentum.** Applying Eqs. 15–4 in the  $x$  direction, we have

$$\begin{aligned}
 (+\curvearrowleft) \quad m(v_x)_1 + \Sigma \int_{t_1}^{t_2} F_x dt &= m(v_x)_2 \\
 (25 \text{ kg})(1 \text{ m/s}) + \int_0^{2 \text{ s}} 100t dt - 0.3N_C(2 \text{ s}) + (25 \text{ kg})(9.81 \text{ m/s}^2) \sin 30^\circ(2 \text{ s}) &= (25 \text{ kg})v_2 \\
 25 + 200 - 0.6N_C + 245.25 &= 25v_2
 \end{aligned}$$

The equation of equilibrium can be applied in the  $y$  direction. Why?

$$+\nearrow \Sigma F_y = 0; \quad N_C - 25(9.81) \cos 30^\circ \text{ N} = 0$$

Solving,

$$\begin{aligned}
 N_C &= 212.39 \text{ N} \\
 v_2 &= 13.7 \text{ m/s } \curvearrowleft
 \end{aligned}$$

*Ans.*

**NOTE:** We can also solve this problem using the equation of motion. From Fig. 15–6b,

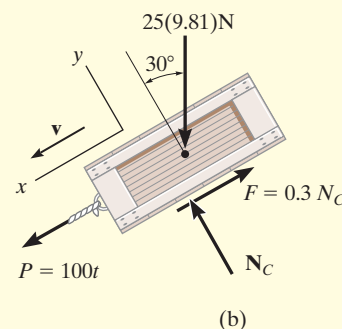
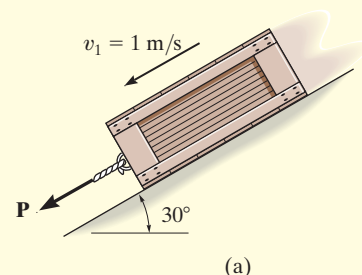
$$\begin{aligned}
 +\curvearrowleft \Sigma F_x &= ma_x; \quad 100t - 0.3(212.39) + 25(9.81) \sin 30^\circ = 25a \\
 a &= 4t + 2.356
 \end{aligned}$$

Using kinematics

$$\begin{aligned}
 +\curvearrowleft dv &= a dt; \quad \int_{1 \text{ m/s}}^v dv = \int_0^{2 \text{ s}} (4t + 2.356) dt \\
 v &= 13.7 \text{ m/s}
 \end{aligned}$$

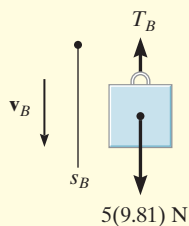
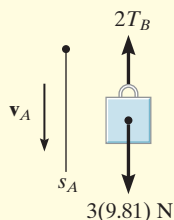
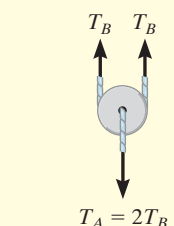
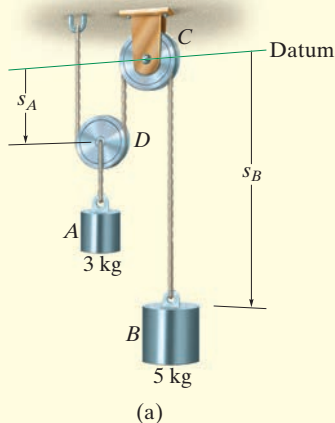
*Ans.*

By comparison, application of the principle of impulse and momentum eliminates the need for using kinematics ( $a = dv/dt$ ) and thereby yields an easier method for solution.



**Fig. 15–6**

## EXAMPLE 15.3



(b)

Fig. 15-7

Blocks *A* and *B* shown in Fig. 15-7*a* have a mass of 3 kg and 5 kg, respectively. If the system is released from rest, determine the velocity of block *B* in 6 s. Neglect the mass of the pulleys and cord.

## SOLUTION

**Free-Body Diagram.** See Fig. 15-7*b*. Since the weight of each block is constant, the cord tensions will also be constant. Furthermore, since the mass of pulley *D* is neglected, the cord tension  $T_A = 2T_B$ . Note that the blocks are both assumed to be moving downward in the positive coordinate directions,  $s_A$  and  $s_B$ .

## Principle of Impulse and Momentum.

Block *A*:

$$\begin{aligned}
 (+\downarrow) \quad m(v_A)_1 + \Sigma \int_{t_1}^{t_2} F_y dt &= m(v_A)_2 \\
 0 - 2T_B(6 \text{ s}) + 3(9.81) \text{ N}(6 \text{ s}) &= (3 \text{ kg})(v_A)_2
 \end{aligned} \quad (1)$$

Block *B*:

$$\begin{aligned}
 (+\downarrow) \quad m(v_B)_1 + \Sigma \int_{t_1}^{t_2} F_y dt &= m(v_B)_2 \\
 0 + 5(9.81) \text{ N}(6 \text{ s}) - T_B(6 \text{ s}) &= (5 \text{ kg})(v_B)_2
 \end{aligned} \quad (2)$$

**Kinematics.** Since the blocks are subjected to dependent motion, the velocity of *A* can be related to that of *B* by using the kinematic analysis discussed in Sec. 12.9. A horizontal datum is established through the fixed point at *C*, Fig. 15-7*a*, and the position coordinates,  $s_A$  and  $s_B$ , are related to the constant total length  $l$  of the vertical segments of the cord by the equation

$$2s_A + s_B = l$$

Taking the time derivative yields

$$2v_A = -v_B \quad (3)$$

As indicated by the negative sign, when *B* moves downward *A* moves upward. Substituting this result into Eq. 1 and solving Eqs. 1 and 2 yields

$$\begin{aligned}
 (v_B)_2 &= 35.8 \text{ m/s} \downarrow \\
 T_B &= 19.2 \text{ N}
 \end{aligned} \quad \text{Ans.}$$

**NOTE:** Realize that the *positive* (downward) direction for  $\mathbf{v}_A$  and  $\mathbf{v}_B$  is *consistent* in Figs. 15-7*a* and 15-7*b* and in Eqs. 1 to 3. This is important since we are seeking a simultaneous solution of equations.